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MASTER'S DEGREE IN BIOINFORMATICS

Learning predictive models of time-to-conversion from MCI A+ to AD using CSF proteomics and clinical data

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Abstract

Alzheimer’s disease (AD) is the most common cause of dementia worldwide and, when diagnosed, irreversible brain damage has often already occurred. Conventional cerebrospinal fluid (CSF) biomarkers such as amyloid- β , phosphorylated tau, and total tau provide diagnostic information but have limited prognostic value for predicting the timing of conversion from Mild Cognitive Impairment (MCI) to AD dementia. CSF proteomics may offer a complementary substrate, potentially capturing upstream molecular processes preceding clinical decline and enabling therapeutic interventions within the most effective time window.

An extended ensemble feature selection framework (FSE-StabPred) was applied to two independent cohorts: the Lisbon-Coimbra (LC) cohort ($n = 56$, LC-SWATH-MS, predominantly A+T+) and the ADNI-A+ cohort ($n = 187$, SOMAscan 7K, entirely A+T−), evaluating binary one-vs-all classification, multiclass classification, and regression using Random Forest and Support Vector Machine, and comparing proteomic features against conventional CSF biomarkers and clinical variables (MMSE, age, gender) within a nested cross-validation framework.

Meaningful predictive performance was limited to the 2-year OVA window in LC (MCC 0.448 ± 0.072), where proteomic features provided a statistically supported improvement over conventional biomarkers, and the 5+ year OVA window in ADNI-A+ (MCC 0.328 ± 0.018), driven predominantly by conventional CSF biomarkers with only a marginal proteomic contribution. Downstream analysis on the LC cohort identified ten candidate proteins with convergent support, with haptoglobin (HP) and complement C1q subunit B (C1QB) emerging as the most robustly supported. Cross-cohort comparison revealed largely non-overlapping discriminative signatures, which may reflect biological and technical differences between cohorts.

These findings provide a proof of concept that CSF proteomics may contribute prognostic information for short-term MCI-to-AD conversion, underscoring the need for independent replication in larger, harmonised cohorts.

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1. Introduction

1.1 Context and motivation

Clinical Motivation. Alzheimer’s disease (AD) is a neurodegenerative disorder that frequently manifests with overt clinical symptoms only when substantial and largely irreversible brain damage has already occurred [34]. Early identification of patients at risk of conversion to dementia is therefore of critical importance, as it could enable the application of therapeutic interventions within the most effective time window, before neurodegeneration becomes widespread.

Biological complexity of AD. Predicting not only whether a patient will convert to dementia, but when this transition will occur, remains an unresolved clinical and scientific challenge [34]. This difficulty stems from the remarkable biological complexity of the disease, which involves multiple interacting molecular pathways and proteins, and which continues to be the subject of extensive research [40]. In addition, the existence of distinct disease subtypes, each characterised by different underlying biological processes, has been proposed and is increasingly supported by emerging evidence [29], further complicating the development of reliable prognostic models.

To further complicate this landscape, individuals with Mild Cognitive Impairment (MCI) and abnormal amyloid pathology (A+) represent a population at considerably increased risk of progressing to dementia, yet they exhibit highly heterogeneous rates of cognitive decline and conversion. While tau status has been identified as one factor associated with a steeper cognitive decline in MCI A+ individuals, it explains only part of the observed variability, and the biological mechanisms underlying this heterogeneity remain largely unclear [52].

Recent work has provided compelling evidence that CSF proteomic profiles carry biologically meaningful prognostic information in this population. Vromen et al. [52] demonstrated, using an untargeted mass spectrometry approach in A+ MCI individuals from the Amsterdam Dementia Cohort (ADC) with external replication in Alzheimer’s Disease Neuroimaging Initiative cohort (ADNI), that baseline CSF protein levels were significantly associated with both the rate of longitudinal cognitive decline on the MMSE and time to clinical progression to dementia. Crucially, distinct proteomic signatures emerged depending on tau status, with proteins related to

synaptic plasticity, blood-brain barrier integrity, and immune processes differentially contributing to progression in A+T+ and A+T− subgroups. However, while that study employed repeated longitudinal measures to model cognitive trajectories over time using Cox proportional hazard models and linear mixed models, the predictive power of these proteomic biomarkers at a single baseline timepoint has not been systematically evaluated within a machine learning framework designed for prognostic classification. In other words, it remains unclear whether baseline CSF proteomic concentrations alone are sufficient to stratify patients by their expected conversion horizon, independently of subsequent longitudinal measurements.

Previous Predictive Strategy. Predictive studies of this kind have so far been conducted primarily using neuropsychological test data, owing to the relative accessibility of such measures compared to proteomic data, and partly due to the biological complexity introduced by proteomic heterogeneity. Pereira et al. [35] proposed a rigorous heterogeneous ensemble feature selection framework, *FSE-StabPred*, that combines the output of multiple filter and embedded feature selection methods, optimising the trade-off between feature stability, measured via the Kuncheva index [23], and predictive performance across repeated cross-validation folds.

Applied to neuropsychological data from two independent cohorts, ADNI and the Portuguese Cognitive Complaints Cohort (CCC), this approach consistently identified stable and discriminative predictor subsets that outperformed individual feature selection methods, while removing the need to choose a single selection algorithm a priori. The application of an analogous framework to CSF proteomic data represents a natural and important extension of this line of research, which to date has not been undertaken.

Aims of the Study. The overall aim of this study is therefore to apply the ensemble feature selection pipeline of Pereira et al. [35] to CSF proteomic features, in order to evaluate their predictive efficacy for time-to-conversion from MCI to Alzheimer’s dementia, using a cohort from the Lisbon-Coimbra clinical centres (LC) as the primary dataset, with a comparative analysis on the Alzheimer’s Disease Neuroimaging Initiative (ADNI) cohort as an independent dataset evaluated under the same methodology. Specifically, this study addresses four interconnected research questions:

1. **Do CSF proteomic features improve prognostic performance beyond conventional biomarkers?** In particular, whether proteomic features improve prognostic performance beyond what can be achieved using conventional CSF biomarkers and clinical features alone, namely A β 42, t-tau, p-tau, age, and gender, and whether combining all feature sets yields further gains.

2. **Which modelling configuration is best suited to time-to-conversion prediction?** Specifically, whether a classification or regression framework is more appropriate for predicting time-to-conversion, and, within a classification setting, whether a one-vs-all (OVA) strategy applied independently to each time window outperforms a single multiclass model.
3. **Which proteomic biomarkers emerge as the most informative and what is their biological relevance?** This question shifts from predictive performance to biological interpretability, asking which proteomic biomarkers emerge as the most informative across the analyses and how they relate to known Alzheimer’s disease pathological mechanisms, as assessed through downstream analyses including Gini importance, Cox proportional hazards survival analysis, and pathway enrichment analysis.
4. **To what extent do dataset size and panel composition influence generalisability?** This question addresses whether the methodology developed on the LC cohort yields comparable results when applied independently to the larger ADNI-A+ dataset, and how dataset size and proteomic panel composition affect the robustness and replicability of the identified signatures.

Together, these questions define a study that sits at the intersection of clinical proteomics and machine learning, and whose findings are intended to contribute both to the methodological development of robust prognostic pipelines for AD research, and to the biological understanding of the molecular processes that govern the temporal dynamics of MCI-to-AD conversion ultimately aiming to support the identification of patients within the critical time window in which therapeutic intervention may still be effective, before neurodegeneration becomes too advanced.

1.2 Contributions

The present thesis makes four main contributions to the field of computational proteomics and machine learning for Alzheimer’s disease research:

Methodological.

1. The first contribution is the adaptation and extension of the FSE-StabPred ensemble feature selection framework of Pereira et al. [35] to a high-dimensional CSF proteomic setting. This extension includes the replacement of the Chi-squared ranking method with the ANOVA F-test to accommodate standardised continuous features, the development of an analogous ensemble of regression-adapted ranking methods for the continuous time-to-conversion target, and the

formulation of a dedicated predictability metric for regression within the RPT- β framework. The resulting pipeline provides a principled and reproducible methodology for stable proteomic biomarker selection under the severe sample size constraints typical of clinical proteomic studies.

2. The second contribution is the cross-cohort methodological consistency analysis on the overlapping panel of 40 proteins shared between the LC and ADNI proteomic panels, which characterises the degree of cross-platform and cross-biological comparability achievable under the present experimental constraints.

Clinical.

3. The third contribution consists of the predictive models for time-to-conversion from MCI to Alzheimer’s dementia developed on the LC and ADNI cohorts. The analysis systematically evaluates the relative informativeness of four feature sets: clinical variables, conventional CSF biomarkers, proteomic features, and their combination; every feature set is evaluated across multiple modelling approaches, including one-vs-all and multiclass classification and regression, and two predictive models, Random Forest and Support Vector Machine, providing a comprehensive characterisation of the conditions under which CSF proteomics adds predictive value beyond standard clinical and biochemical measures.

Biological.

4. The fourth contribution is the identification and biological characterisation of a set of strong candidate proteomic biomarkers for MCI-to-AD conversion, supported by convergent evidence from ensemble feature selection, Gini importance analysis, and Cox proportional hazards survival analysis.

1.3 Thesis Structure

This thesis is organised as follows. Chapter 2 (*Background*) provides the biological and methodological background necessary to contextualise the study, covering the neuropathology of Alzheimer’s disease and mild cognitive impairment, the proteomic platforms used in the two cohorts, and a part that described the supervised learning methods used and the feature selection methods employed in the pipeline, and related works that brought to this thesis study. Chapter 3 (*Materials and Methods*) describes the study cohorts, the preprocessing procedures applied to each dataset, the exploratory data analysis, and the machine learning pipeline in detail, including the nested cross-validation design, the extended FSE-StabPred framework, and the downstream biomarker analyses. Chapter 4 (*Results*) presents the results in terms

of predictive performance evaluation, feature selection of the meaningful configurations based on the performances obtained, and biomarker analyses across these configurations, focusing on the biological meaning and pathway enrichment analysis. Chapter 5 (*Discussion*) discusses the main findings in light of the methodological choices made, the limitations of the data, and the broader context of the field, and provides a detailed biological interpretation of the candidate biomarkers identified. Finally, Chapter 6 (*Conclusions and Future Directions*) summarises the principal conclusions of the study and outlines directions for future work.

2. Background

2.1 Alzheimer’s Disease

Alzheimer’s disease (AD) is the most common cause of dementia and one of the leading neurodegenerative disorders worldwide, affecting approximately 50 million people globally, a number projected to rise to over 130 million by 2050 [27].

The disease was first characterised by Alois Alzheimer in 1906, who described the histological features of senile plaques and neurofibrillary tangles in the post-mortem brain of a patient with progressive cognitive decline.

Its neuropathological hallmarks include the accumulation of insoluble amyloid- β ($A\beta$) in the form of parenchymal plaques and vascular deposits, intraneuronal neurofibrillary tangles (NFTs) composed of misfolded and hyperphosphorylated tau protein, activated astrocytes and microglia, and widespread loss of synapses and neuronal cell bodies.

Despite decades of research, neither effective preventive measures nor disease-modifying treatments are currently available, making the identification of reliable prognostic biomarkers a central goal of the field.

Aetiology and genetic risk factors. The majority of AD cases are sporadic, with risk predominantly determined by a combination of genetic and environmental factors. The greatest single risk factor is advancing age: the likelihood of developing AD is approximately 3% at age 65 and rises to over 30% at age 85 [1]. Among genetic determinants, the apolipoprotein E ϵ 4 allele (APOE4) is the most common genetic risk factor for late-onset AD, increasing risk in an allele dose-dependent fashion, while the APOE ϵ 2 allele exerts a protective effect [24]. Early-onset familial AD, accounting for approximately 2–5% of all cases, is caused by autosomal dominant mutations in the amyloid precursor protein gene (APP), presenilin-1 (PSEN1), or presenilin-2 (PSEN2), all of which alter $A\beta$ production and lead to early amyloid accumulation [10]. Beyond these major loci, genome-wide association studies have identified more than 20 susceptibility loci, including the rare variant TREM2 (triggering receptor expressed on myeloid cells), which has been implicated in neuroinflammatory pathways and microglial function [19]. Sporadic AD, by contrast, typically results from the cumulative interaction of multiple genetic variants and environmental risk factors, including cerebral hypoperfusion, systemic inflammation, diabetes, and

traumatic brain injury.

Amyloid- β pathology and APP processing. At the molecular level, A β is generated through the amyloidogenic processing of APP, in which sequential cleavage by β - and γ -secretases releases A β peptides into the extracellular space. In the alternative non-amyloidogenic pathway, cleavage by α -secretase precludes A β generation (Fig.2.1). Senile plaques are primarily composed of 36–43 residue A β peptides that undergo fibrillisation into β -sheet structures resistant to degradation, and activate microglia and astrocytes in surrounding tissue [46]. More recently, attention has shifted towards soluble A β oligomers as potentially more relevant correlates of cognitive decline than mature plaques: oligomers accumulate early in pathogenesis, cluster near synapses, inhibit long-term potentiation in vivo, and induce synapse loss and neuronal dysfunction [49]. The relationship between A β deposition and cognitive decline is not fully resolved, as cognitively unimpaired elderly individuals can also harbour significant A β deposits, suggesting that amyloid pathology alone may be insufficient to account for the full clinical heterogeneity of the disease.

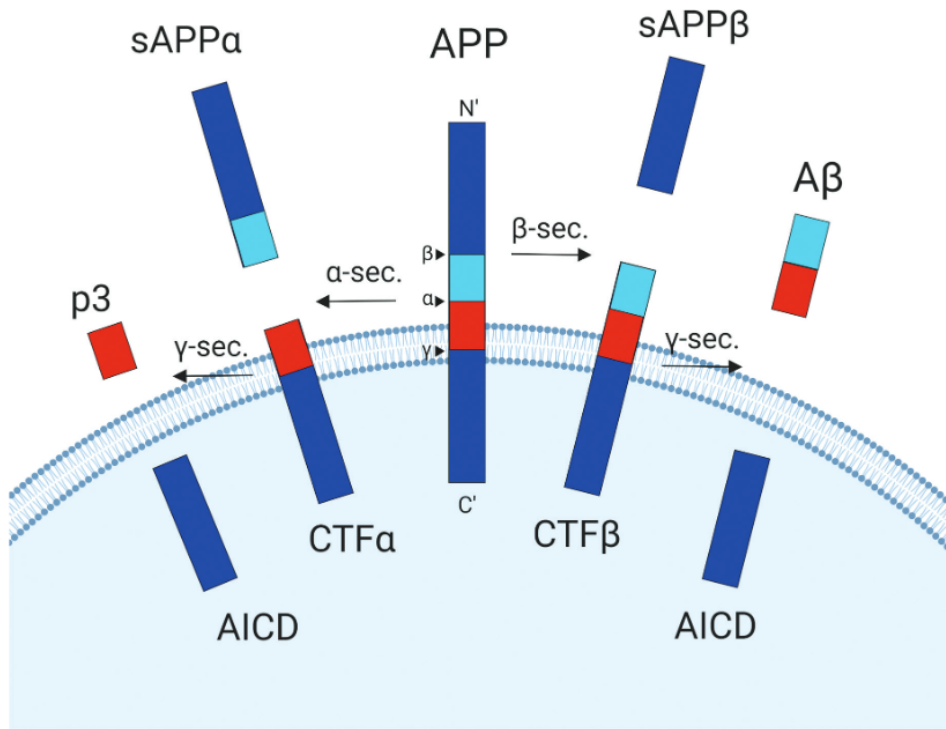


Figure 2.1: Post-translational processing of A β precursor protein (APP) is thought to occur at the cell surface or within endosomes. It includes cleavage by either α - then γ -secretase (non-amyloidogenic), or β - then γ -secretase (amyloidogenic pathway) [40].

Tau pathology and neurofibrillary tangles. NFTs are intraneuronal aggregates of hyperphosphorylated tau protein, encoded by the microtubule-associated protein tau (MAPT) gene, and composed of paired helical filaments approximately 20 nm in

diameter [4] (Fig.2.2). Under physiological conditions, tau stabilises microtubules through interaction with tubulin; in AD, abnormal hyperphosphorylation disrupts this function, promotes tau aggregation, and impairs axonal transport. The spatial progression of NFTs through the brain follows a stereotyped pattern described by Braak staging, beginning in the trans-entorhinal cortex (stages I–II), spreading to the limbic system (stages III–IV), and eventually reaching the neocortex (stages V–VI) [6] (Fig.2.3). While both amyloid plaques and NFTs are associated with neuronal loss and toxicity, synaptic loss is considered the pathological change most strongly correlated with cognitive decline in AD [45]. $A\beta$ and tau appear to act synergistically: $A\beta$ -induced toxicity and impairment of long-term potentiation have been shown to require the presence of endogenous tau, and their interaction may result in transcriptional deficits and synaptic alterations that drive disease progression [56].

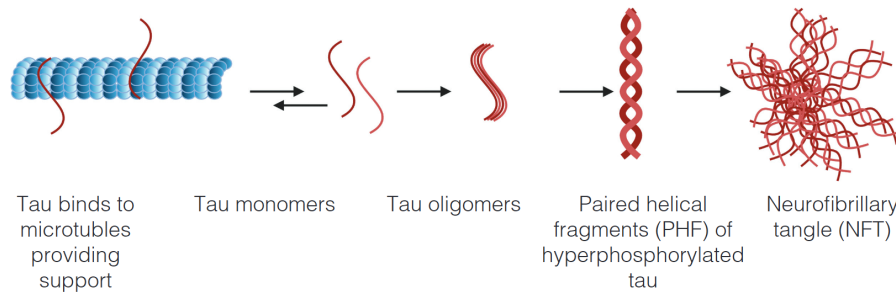


Figure 2.2: Microtubule-associated protein Tau (MAPT) aggregation results in the accumulation of neurofibrillary tau tangles (NFTs) [40].

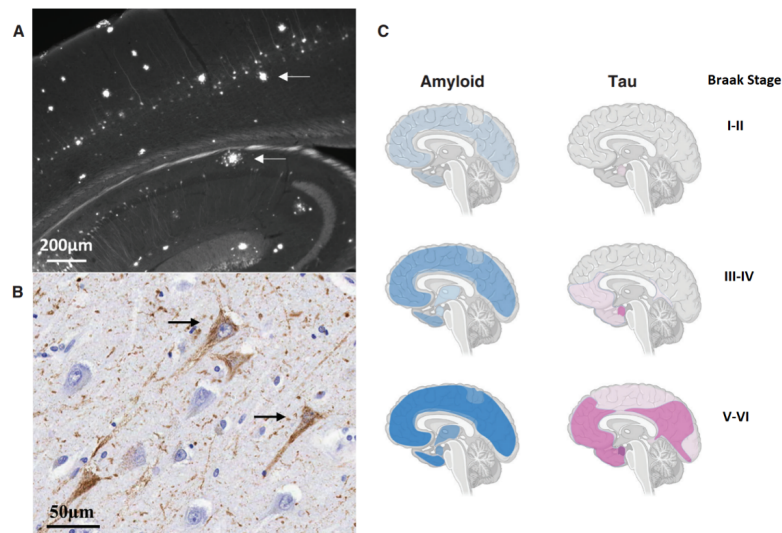


Figure 2.3: Amyloid and tau pathology. (A) Thioflavin S staining of $A\beta$ plaques in the cortex of a CRND8 APP transgenic mouse. (B) AT8 staining of neurofibrillary tau tangles (NFTs) within an aged human CA1 region of the hippocampus. (C) The spread of amyloid and tau pathology throughout the brain during AD Braak Stages (adapted from Braak and Braak [6]).

Additional pathological processes. Beyond amyloid and tau pathology, several interconnected biological processes contribute to AD progression. Mitochondrial dysfunction is among the most prominent: alterations in mitochondrial morphology, reduced cytochrome oxidase activity, and increased production of reactive oxygen species (ROS) at synapses have all been documented in AD brains [43]. Neuroinflammation represents a second major contributor: while acute microglial and astrocytic activation is neuroprotective, chronic activation leads to a feed-forward loop of inflammatory damage that increases $A\beta$ burden and accelerates neurodegeneration [20]. Insulin resistance and reduced glucose metabolism, observed in the brains of AD patients, contribute to energetic deficits and oxidative stress, and have been proposed as links between systemic metabolic disease and AD risk [12]. The ubiquitin-proteasome system (UPS) and the autophagy-lysosomal pathway, which are responsible for the clearance of misfolded proteins, are progressively impaired by the accumulation of amyloid and tau aggregates, creating a vicious cycle of proteostatic failure [31]. These processes interact in a complex and heterogeneous manner, and their relative contributions likely vary across disease stages, patient subtypes, and genetic backgrounds.

2.2 Mild Cognitive Impairment

Mild cognitive impairment (MCI) describes a clinical state intermediate between normal cognitive ageing and dementia, characterised by objective cognitive deficits that exceed what would be expected for a patient’s age and education, without fulfilling the diagnostic criteria for dementia and with largely preserved activities of daily living [36]. MCI was formally characterised by Petersen et al. [36], and its diagnostic criteria have since been refined to include deficits in any cognitive domain, not exclusively memory, and to incorporate biomarker evidence of underlying AD pathology [37].

Individuals with MCI and abnormal amyloid pathology (A+) represent a population at considerably increased risk of progression to Alzheimer’s dementia, yet they exhibit highly heterogeneous rates of cognitive decline and conversion, making reliable prognostic stratification a central and unresolved challenge.

Subtypes and progression. MCI is broadly classified into amnesic MCI (aMCI), characterised by predominant memory impairment and higher risk of progression to AD, and non-amnesic MCI (naMCI), presenting with deficits in other cognitive domains and more commonly associated with Lewy body disease or frontotemporal dementia [39]. Both forms can be further categorised by the number of domains affected.

The Mini-Mental State Examination (MMSE) is among the most widely used screening instruments for MCI and cognitive staging, and is employed as a clinical variable in the present thesis. Tau status has been identified as a key stratification variable for MCI patients: individuals with an A+T+ profile show a steeper rate of cognitive decline than A+T− individuals [38], and as demonstrated by Vromen et al. [52], the two subgroups are characterised by substantially different CSF proteomic signatures, with synaptic plasticity processes predominantly associated with decline in T+ patients and blood-brain barrier dysfunction predominating in T− patients. This biological heterogeneity fundamentally complicates the development of generalizable prognostic models and motivates the cohort-specific analyses performed in the present thesis.

2.3 Proteomic Techniques

The two cohorts analysed in this thesis were profiled using two distinct CSF proteomic platforms, reflecting different methodological philosophies in high-throughput protein quantification: a mass-spectrometry-based, semi-targeted approach (Short-GeLC LC-SWATH-MS) for the LC cohort, and an aptamer-based, highly multiplexed targeted approach (SomaScan) for the ADNI cohort. This section introduces the general principles underlying each technique, independently of the specific samples and procedures applied in this thesis, which are instead described together with each cohort in the Materials and Methods chapter (Sections 3.1.1 and 3.1.2).

Short-GeLC LC-SWATH-MS Short-GeLC LC-SWATH-MS [29] combines gel-based protein fractionation with liquid-chromatography tandem mass spectrometry operated in a data-independent acquisition (DIA) mode known as SWATH-MS (Sequential Window Acquisition of all Theoretical fragment ion spectra). In the gel-based fractionation step (GeLC), proteins are first separated by molecular weight via SDS-PAGE and the resulting gel lane is cut into a small number of slices, each of which is processed and digested independently; this short, coarse fractionation step reduces sample complexity ahead of mass spectrometry, improving the depth and reproducibility of protein identification relative to direct injection of an undigested, unfractionated sample, while remaining considerably faster than extensive multi-fraction proteomic workflows.

Following digestion, peptides are analysed by liquid chromatography coupled to tandem mass spectrometry. Unlike data-dependent acquisition (DDA), in which the instrument selects, at each scan, a limited number of precursor ions for fragmentation based on their intensity, SWATH-MS operates in a data-independent fashion: the full mass-to-charge (m/z) range of interest is divided into a series of overlapping

precursor isolation windows, and the instrument cycles through all of these windows systematically, fragmenting and recording the resulting spectra for every precursor ion that falls within each window, regardless of its intensity. This exhaustive, window-by-window acquisition produces a comprehensive and highly reproducible digital record of the sample’s fragment-ion spectra, at the cost of more complex, multiplexed spectra that require a previously generated, sample-specific spectral library (typically obtained from complementary DDA runs) for confident peptide identification and quantification. Once such a library is available, SWATH-MS allows relative protein quantification across large sample cohorts with good reproducibility, since every analyte is, in principle, captured in every run, in contrast to the more stochastic precursor selection of DDA, which can lead to the same peptide being identified in some runs and missed in others.

SomaScan SomaScan [14] is an aptamer-based, multiplexed proteomic platform that enables the simultaneous, targeted quantification of several thousand pre-specified protein analytes from a single small-volume biological sample. The technology relies on Slow Off-rate Modified Aptamers (SOMAmers), chemically modified single-stranded DNA sequences engineered to fold into three-dimensional structures that bind a specific protein target with high affinity and specificity, in a manner functionally analogous to an antibody-antigen interaction, but with the advantage that DNA-based aptamers can be synthesised, quantified, and multiplexed at very large scale using standard nucleic acid technologies. Each SOMAmer is associated with a single protein target, and the abundance of the corresponding protein in a sample is inferred indirectly, from the amount of bound SOMAmer recovered after a series of selection and washing steps, which is subsequently quantified via hybridisation to a DNA microarray and expressed in Relative Fluorescence Units (RFU).

Because each protein is queried by a dedicated, pre-validated aptamer, SomaScan is, unlike SWATH-MS, a *targeted* platform in the sense that only the protein targets for which a SOMAmer has been developed can be measured, rather than allowing the unrestricted detection of whatever peptides happen to be detectable in a given run. This design trades the hypothesis-free discovery potential of mass-spectrometry approaches for substantially higher throughput, lower per-sample cost, and a fixed, highly reproducible analyte panel.

2.4 Supervised Learning

This section introduces the two predictive models evaluated across all experimental configurations: Support Vector Machines (Subsection 2.4.1) and Random Forests

(Subsection 2.4.2), describing their theoretical foundations and their application to both classification and regression tasks.

2.4.1 Support Vector Machines

Support Vector Machines (SVMs) are supervised learning models that identify the optimal separating hyperplane between classes in a high-dimensional feature space by maximising the margin between the decision boundary and the nearest training examples of each class, known as support vectors [9]. The core principle of margin maximisation confers SVMs a strong regularisation property: by explicitly penalising decision boundaries that lie too close to the training data, the model is encouraged to generalise to unseen examples rather than memorising the training set, an advantage that is particularly valuable in high-dimensional, low-sample-size settings such as those encountered in proteomic biomarker studies.

SVM for classification. In the classification setting, the SVM seeks a hyperplane $w \cdot x + b = 0$ that separates two classes with maximum margin. When the classes are not linearly separable in the original feature space, SVMs employ the kernel trick: input features are implicitly mapped to a higher-dimensional space through a kernel function $K(\mathbf{x}_i, \mathbf{x}_j)$, within which a linear separator may exist. The radial basis function (RBF) kernel, defined as $K(\mathbf{x}_i, \mathbf{x}_j) = \exp(-\gamma \|\mathbf{x}_i - \mathbf{x}_j\|^2)$, is among the most widely used, as it implicitly projects data into an infinite-dimensional space and can model complex non-linear decision boundaries. The regularisation parameter C controls the trade-off between maximising the margin and minimising the classification error on the training set: small values of C favour a wider margin at the cost of more misclassifications, while large values enforce a stricter fit to the training data. In the present thesis, SVMs are applied to both multiclass and one-vs-all binary classification tasks, using a balanced class weighting scheme to account for the pronounced class imbalance across time windows.

SVM for regression. The regression counterpart of the SVM, the Support Vector Regressor (SVR), extends the margin maximisation principle to a continuous target. Rather than separating classes, the SVR seeks a function $f(\mathbf{x}) = w \cdot x + b$ that lies within an ε -large tolerance band around the true target values: predictions within ε of the true value are not penalized, while deviations beyond this band are penalised linearly. This formulation makes the SVR robust to small perturbations in the target variable and explicitly controls the width of the acceptable prediction range through the ε parameter. As for classification, the RBF kernel is used to model non-linear relationships, and the regularisation parameter C governs the trade-off between flatness of the regression function and tolerance of training errors. In the

present thesis, the SVR is employed for the regression task targeting the continuous time-to-conversion variable.

2.4.2 Random Forest

Random Forest (RF) is an ensemble learning method that constructs a large number of decision trees on bootstrap samples of the training data, aggregating their predictions to produce a final output that is more stable and accurate than any individual tree [7]. Each tree is built using a random subset of features at each split, a mechanism known as feature bagging, which decorrelates the individual trees and reduces the variance of the ensemble relative to a single decision tree. The combination of bootstrap aggregation (bagging) and random feature subsampling gives the Random Forest its characteristic robustness to overfitting, even in high-dimensional settings where the number of features substantially exceeds the number of samples.

Random Forest for classification. In the classification setting, each tree in the ensemble produces a class prediction for a given input, and the final prediction is determined by majority vote across all trees. Node splits within each tree are chosen to maximise the reduction in Gini impurity, defined as $G = 1 - \sum_k p_k^2$, where p_k is the proportion of samples of class k at a given node. A natural product of this splitting criterion is the Gini importance score for each feature, computed as the total weighted reduction in Gini impurity contributed by that feature across all splits in all trees of the ensemble; this score provides a model-intrinsic feature ranking that is used in the present thesis as a complementary analysis to the FSE-StabPred ensemble selection (Section 4.3). To account for the marked class imbalance across time windows, a balanced class weighting scheme is applied.

Random Forest for regression. In the regression setting, the Random Forest Regressor replaces majority voting with averaging of the continuous predictions produced by each individual tree, which themselves are fitted to minimise the mean squared error at each node. The ensemble average reduces the variance of the prediction relative to a single regression tree, while preserving its ability to model non-linear and interaction effects in the feature space. The Gini importance score is replaced by the impurity-based feature importance computed from the reduction in mean squared error at each split.

2.5 Model Evaluation and Feature Selection

This section describes the strategies adopted to evaluate model performance and select informative features under the data constraints of the present thesis. Subsection 2.5.1 introduces the nested cross-validation strategy used to obtain unbiased performance estimates under limited sample sizes. The remaining subsections describe the components of the Feature Selection Ensemble combining Stability and Predictability (FSE-StabPred) framework by Pereira et al. [34]: Subsection 2.5.2 presents the ensemble ranking methods used to aggregate feature relevance across multiple criteria; Subsection 2.5.3 describes the stability criterion and the Kuncheva index used to quantify the consistency of feature selections across data partitions; Subsection 2.5.4 introduces the predictability criterion and the performance metrics used to assess model fit; and Subsection 2.5.5 presents the Robustness-Performance Trade-off (RPT- β) score that integrates stability and predictability to identify the optimal feature subset size.

2.5.1 Nested Cross-Validation

When the available sample size is limited, a single training/test split produces performance estimates that are highly sensitive to the particular partition chosen, and risks conflating the optimisation of the model with its evaluation. *Nested cross-fold validation* addresses this problem by strictly separating the phases of feature selection, hyperparameter tuning, and final performance evaluation, so that the data used to choose a model (or a feature subset) are never the same data used to assess how well that choice generalises.

The strategy is organised across two nested levels. An *outer loop* partitions the dataset into a small number of folds: at each iteration, one fold is held out entirely as a *benchmark* set, while the remaining folds form a *development* set. This rotation is repeated until every partition has been used exactly once as benchmark fold, yielding an unbiased and generalized estimate of performance once results are aggregated across outer iterations. Crucially, the benchmark fold of each outer iteration is kept completely isolated from every step performed inside that iteration. An *inner loop*, operating exclusively on the development set, is then responsible for the steps that require features selection: ranking and selecting the most relevant features through the *FSE-StabPred* strategy described in Section 2.5.2. Only once these choices have been finalised within the inner loop the hyperparameter are tuned and the final model is evaluated, for the first and only time, on the benchmark fold of the corresponding outer iteration.

This two-level architecture is particularly important when working with limited

samples. With few available patients, every individual partition of the data into training and test subsets carries a significant risk of being unrepresentative, so that both the estimated performance and the subset of features selected as most relevant could end up reflecting the specific characteristics of that exact split rather than a genuine, generalisable signal. By rotating the outer benchmark fold across all patients and by confining every model-selection decision to the inner loop, nested cross-validation makes it possible to extract as much information as possible from a small dataset while keeping the bias introduced by any single partition under control. The specific configuration adopted for the outer and inner loops in this thesis, including the number of folds, the number of repetitions, and the stratification scheme, is detailed in the Materials and Methods chapter (Section 3.3.1).

2.5.2 Ensemble Ranking Methods

The feature selection strategy adopted in this thesis is built upon the *Feature Selection Ensemble combining Stability and Predictability* framework (FSE-StabPred) introduced by Pereira et al. [35]. The framework rests on a simple but consequential observation: no single feature ranking method is universally optimal, since different methods rely on different criteria to quantify the relevance of a feature, and therefore tend to sometimes substantially disagree on which features matter most for a given prediction task. Rather than committing to one such method and accepting its particular biases, Pereira et al. [35] proposed to combine the rankings produced by several methodologically diverse feature selection algorithms into a single, aggregated ranking, and subsequently to select the size of the final feature subset by jointly considering two properties: how *stable* that subset is across repeated applications of the procedure to different partitions of the data, and how *predictive* it is of the outcome of interest. This combination of diversity in the ranking stage and a principled criterion for subset selection is what the present thesis inherits and extends, adapting it to a setting that, unlike the original neuropsychological application, must accommodate both a classification and a regression target.

The implementation details of each base ranking method described below follow the respective software documentation: the scikit-learn library [33] for Mutual Information, the Chi2, LR-L1, and SVM-RFE; the `skrebate` library [48] for ReliefF; and the `mrmr` library [13] for mRMR. For the classification task, six ranking methods are combined into the ensemble, following the same six methods employed by Pereira et al. [35]:

1. **Mutual Information** (MI) quantifies the statistical dependence between each feature and the class label, capturing both linear and non-linear relationships: a mutual information of zero indicates that the feature and the class are entirely

independent.

2. **Chi-squared test** (χ^2) evaluates whether the observed distribution of a feature differs significantly from what would be expected under independence from the class label, providing a measure of class separability commonly used for non-negative, count-like features.
3. **Logistic Regression with L1 regularisation** (LR-L1), fitted under a one-vs-rest scheme, is an embedded method in which the L1 penalty shrinks the coefficients of uninformative features to exactly zero.
4. **Recursive Feature Elimination with a linear SVM** (SVM-RFE), a wrapper method that iteratively trains a linear support vector classifier (described in detail in Section 2.4.1), ranks features by the magnitude of their weight vector, and removes the least informative feature at each step, producing a ranking based on the order of elimination; this approach is particularly suited to capturing interactions between features that purely univariate methods may miss.
5. **ReliefF** ranks features according to their ability to distinguish between nearest-neighbour samples belonging to different classes, making it well suited to capture also the finest differences between different time-window targets.
6. **Minimum Redundancy Maximum Relevance** (mRMR) selects features that are simultaneously maximally correlated with the target and minimally correlated with one another, explicitly penalising redundancy within the selected subset, a property that complements the other five methods, because none of them account for inter-feature dependencies.

For the regression task, since the original FSE-StabPred framework of Pereira et al. [35] does not address regression settings, an analogous ensemble was developed as part of this thesis by adapting the same criteria described above to a continuous time-to-conversion target. The specific ranking methods composing this regression ensemble, and how each was adapted from its classification counterpart, are described in detail in the Materials and Methods chapter (Section 3.3.2).

Within each fold of the inner cross-validation loop (Section 2.5.1), every base method produces an independent ranking of all candidate features. These rankings are combined into a single ensemble ranking by averaging, feature by feature, the positions assigned by each method: features that are consistently ranked highly across methods with different underlying criteria are pushed towards the top of the aggregated ranking, while features favoured by only one or two methods, and potentially reflecting specificity and ungeneralizability dependent on that particular

criterion, and are penalised. This aggregated ranking is what the stability and predictability criteria, described below, are subsequently applied to.

2.5.3 Stability

A ranking that changes substantially every time that is recomputed on a slightly different sample of patients possess a limited clinical value, even if it achieves good predictive performance on average, since it would offer no consistent guidance as to which biomarkers are genuinely informative. *Stability* quantifies the degree of agreement between the feature subsets selected across the repeated partitions of the inner cross-validation loop. Following Pereira et al. [35], stability is measured using the index proposed by Kuncheva [23], which evaluates the similarity between two feature subsets of equal size k , drawn from a pool of m total features, where r denotes the number of features common to both subsets, by modelling their intersection as a hypergeometric distribution and correcting explicitly for the overlap expected by randomness alone:

$$\text{Sim}(k) = \frac{r \cdot m - k^2}{k \cdot (m - k)} \quad (2.1)$$

In this formulation, $r \cdot m$ scales the observed overlap by the total number of available features, while k^2 represents the overlap expected by chance alone; the numerator $r \cdot m - k^2$ is therefore a scaled measure of the excess overlap beyond what randomness would produce, normalised by the denominator $k(m - k)$, which corresponds to the maximum value the numerator can take when the two subsets are identical ($r = k$), ensuring that the index is bounded in $[-1, 1]$. A value of 1 indicates that the two subsets are identical, a value of 0 indicates an overlap no greater than what would be expected by chance, and negative values indicate an overlap smaller than randomness. Notably, when k is close to m , both numerator and denominator tend to zero, making the index numerically unstable and reflecting the fact that selecting almost all available features provides no meaningful discriminative information about stability. The overall stability of a given subset size k is then obtained by averaging the pairwise similarity in Equation 2.1 across all pairs of feature subsets produced by the repeated inner cross-validation splits. This correction for randomness is what distinguishes the Kuncheva index from simpler overlap measures and is particularly important in a setting such as this one, where the number of candidate features is large relative to the number of patients, so that a non-trivial overlap between two randomly chosen subsets is significant.

2.5.4 Predictability and Performance Metrics

Alongside stability, the *predictability* of a feature subset assesses how well a model trained on that subset performs at the prediction task itself. For the **classification** task, following Pereira et al. [35], predictability is defined as a weighted combination of the Area Under the ROC Curve (AUC) and the minimum between sensitivity and specificity, computed on the held-out validation portion of each inner fold:

$$\text{Predictability}_{\text{class}} = \alpha \cdot \text{AUC} + \gamma \cdot \min(\text{Sensitivity}, \text{Specificity}) \quad (2.2)$$

where α and γ are weighting parameters, with $\alpha + \gamma = 1$, that control the relative contribution of overall discrimination (AUC) versus balanced classification performance (the minimum between sensitivity and specificity) to the predictability score. α and γ values chosen in all experiments reported in this thesis are described in the Materials and Methods chapter (Section 3.3.2). In the binary, one-vs-all setting, AUC, sensitivity, and specificity are computed directly; in the multiclass setting, AUC is computed as the macro-averaged one-vs-rest AUC across the four time windows, while sensitivity and specificity are likewise averaged across classes before the minimum is taken. Using the minimum between sensitivity and specificity, rather than their average or AUC alone, ensures that the optimal subset size is not chosen on the basis of strong overall discrimination achieved at the expense of a severe imbalance between the ability to correctly identify converters and non-converters: a feature subset that yields high sensitivity but poor specificity, or vice versa, is penalised by Equation 2.2 even when its AUC remains high, an imbalance to which AUC alone can be comparatively insensitive.

Model fit for the classification task is additionally assessed, for interpretive purposes, via the Matthews Correlation Coefficient (MCC) [25], defined for the binary case as

$$\text{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad (2.3)$$

where TP , TN , FP , and FN denote, respectively, the number of true positives, true negatives, false positives, and false negatives. The MCC is bounded in $[-1, 1]$, with 1 indicating a perfect prediction, 0 indicating a prediction no better than random, and -1 indicating total disagreement between prediction and observation; unlike AUC, sensitivity, or specificity taken in isolation, it accounts for all four entries of the confusion matrix simultaneously, making it a comparatively robust summary statistic under the marked class imbalance that characterises several of the time windows considered in this study. In the multiclass setting, the MCC is computed using its generalisation to an arbitrary number of classes proposed by

Gorodkin [15] and implemented in the scikit-learn library [33], which extends the binary formulation in Equation 2.3 to the full $K \times K$ confusion matrix. Unlike α , γ , and the predictability score itself, the MCC does not enter the RPT- β optimisation but is used alongside AUC, sensitivity, and specificity as a primary metric for the evaluation of final model performance, given its robustness to class imbalance.

For the **regression** task, no equivalent formulation exists in the original framework of Pereira et al. [35], since that work addresses only classification. A dedicated predictability metric, adapted to the continuous time-to-conversion target, was therefore developed as part of this thesis. Its formulation, and the choice of error metric on which it is based, are described in detail in the Materials and Methods chapter (Section 3.3.2).

2.5.5 The RPT- β Score

Stability and predictability are finally combined into a single scalar criterion, the *Robustness-Performance Trade-off* score, following the formulation of Pereira et al. [35]:

$$\text{RPT}_\beta(k) = (1 + \beta^2) \cdot \frac{\text{Stability}(k) \cdot \text{Predictability}(k)}{\beta^2 \cdot \text{Stability}(k) + \text{Predictability}(k)} \quad (2.4)$$

This expression is the harmonic mean of stability and predictability for a feature subset of size k , weighted by the parameter β , which governs the relative importance assigned to the two components. When $\beta = 1$, equal weight is given to stability and predictability, and the expression reduces to their unweighted harmonic mean; values of $\beta < 1$ favour stability, while values of $\beta > 1$ favour predictability. The optimal feature subset size k^* is then identified as the value of k that maximises the RPT- β score:

$$k^* = \arg \max_k \text{RPT}_\beta(k) \quad (2.5)$$

and the top- k^* features from the aggregated ensemble ranking are selected as the input to the final model for that fold. The choice of β adopted in this thesis, together with the practical procedure used to scan candidate values of k , is discussed in the Materials and Methods chapter (Section 3.3.2).

2.6 Related Work

Recent literature on predicting conversion from MCI to Alzheimer’s disease converges on a central observation: patients who share the same clinical profile and the same categorical biomarkers may conceal substantially different pathophysiological

processes, and this underlying heterogeneity limits both the biological understanding of the disease and the reliability of prognostic models built by treating cohorts as homogeneous groups. Three complementary lines of research have approached this problem from different angles: the identification of pathophysiological subtypes through CSF proteomics, the study of how tau status modulates the biological processes associated with cognitive decline, and the development of machine learning frameworks capable of handling the clinical and dimensional heterogeneity of neuropsychological data. The present thesis sits at the intersection of these three lines of work.

CSF proteomics and biological subtypes. The work of Moutinho et al. [29] represents the most direct point of origin for the LC cohort used in this thesis, and demonstrates how CSF proteomics can reveal distinct biological subtypes already at the MCI stage due to AD. The study compared MCI patients with biomarkers indicative of a high likelihood of AD (MCI_{AD}) against MCI patients without any biomarkers of amyloid deposition or neuronal injury (MCI_{Other}), using the Portuguese Cognitive Complaints Cohort (CCC) as the primary cohort and the EMIF-AD cohort as independent validation.

Through discriminant analysis (PLS-DA) followed by unsupervised clustering (nNMF), the authors identified two subtypes that were consistent across both cohorts, despite differences in sample preparation protocols and in the quantification platforms used (DDA in EMIF-AD, DIA in CCC): a first cluster characterised by blood-brain barrier dysfunction and inflammatory/immune processes (coagulation, complement activation), and a second cluster characterised by neuronal hyperplasticity. The two clusters did not differ in age, gender, years of education, or MMSE score, but showed significantly different levels of total tau, p-tau, and $A\beta_{42}$ in the CSF, suggesting that the proteomic classification captures a pathophysiological dimension distinct from purely clinical criteria.

The study by Moutinho et al. [29] is relevant to the present thesis because it methodologically confirms that a relatively small number of patients (45 MCI_{AD} in the primary cohort) can be sufficient to reveal reproducible proteomic patterns across independent cohorts, an assumption that underpins the validation approach adopted in this work, where LC is a comparably small dataset; but the study is also useful because it illustrates the complexity arising from the heterogeneity of the disease, and the relevance of proteomic features within this broad landscape, potentially also for temporal prediction. However, Moutinho et al. [29] does not pursue the question of whether this same signal can be exploited for predictive purposes: the goal of subtype identification is inherently qualitative, characterising the biological landscape of the disease without directly addressing the temporal

dimension of conversion. The present thesis takes this signal as its starting point and asks a complementary question: whether the proteomic profiles of MCI A+ patients at baseline are informative not only of the biological subtype they belong to, but of the time at which they are likely to convert to Alzheimer’s dementia, moving from a qualitative characterisation of the proteomic landscape towards an explicitly predictive and clinically applicable framework.

Cox Proportional Hazards Modelling of CSF Proteomics and Cognitive Decline. A second line of work has investigated the relationship between baseline CSF protein levels and the rate of cognitive decline, progressively introducing tau status as a crucial stratification variable, and employing Cox proportional hazards (Cox PH) modelling as the primary statistical framework. The Cox PH model is a semi-parametric regression method for survival analysis that estimates the association between one or more covariates and the hazard of a time-to-event outcome, without making assumptions about the baseline hazard function [52]. In the context of AD research, the event of interest is progression to dementia, the time variable is the interval between baseline assessment and clinical conversion, and each protein is tested as an independent covariate, yielding a hazard ratio (HR) that quantifies the direction and magnitude of its association with the risk of conversion per unit change in normalised protein abundance. This framework is particularly well suited to cohorts with variable follow-up duration and censored observations, both of which are characteristic of longitudinal MCI studies, and allows the identification of proteins whose baseline CSF levels are informative of conversion risk independently of the binary converter/non-converter classification.

In the first study in this line of research [51], the authors measured 671 CSF proteins using the Olink platform (Proximity Extension Assay) in 49 individuals with prodromal AD (amyloid-positive MCI) from the Amsterdam Dementia Cohort, testing the association between baseline protein levels and time to clinical conversion using Cox PH models. A total of 128 proteins were found to be associated with the risk of progression, with enrichment for immune system processes, signal transduction, neuronal death, and neurodevelopmental biology. This study, despite its limited sample size, established a methodological principle that the present thesis shares: the use of the quantitative protein signal in relation to time to conversion, rather than the binary stable/converter classification alone, as the primary variable of interest in the analysis of proteomic biomarkers.

The more recent work in the same line [52] substantially extends this approach in three aspects that make it particularly relevant to the present thesis. First, the authors moved from a targeted proteomic platform (Olink, approximately 1000 proteins) to an untargeted approach based on Tandem Mass Tag mass spectrometry (TMT-MS),

measuring a total of 2302 proteins in the primary cohort (Amsterdam Dementia Cohort, $n = 80$ amyloid-positive MCI) and replicating the results in an independent ADNI cohort ($n = 245$) that employed the SOMAscan platform, a methodological choice analogous to the strategy adopted in this thesis, where the LC and ADNI cohorts likewise use different proteomic platforms (Short-GeLC LC-SWATH-MS and SOMAscan 7K, respectively) and are compared as independent evaluations. Second, and more importantly for the framing of this thesis, the authors stratified the entire analysis by tau status ($A + T+$ versus $A + T-$), demonstrating that the biological processes associated with cognitive decline differ substantially between the two subgroups: in $A + T+$ individuals, higher levels of proteins related to synaptic plasticity and lower levels of immune system proteins predicted a steeper decline on the Mini-Mental State Examination (MMSE) and faster progression to dementia; in $A + T-$ individuals, by contrast, higher levels of proteins indicative of blood-brain barrier dysfunction and lower levels of synaptic plasticity proteins predicted decline, with an opposite direction of effect for a subset of proteins shared between the two groups. Third, independent replication in ADNI confirmed between 58% ($A + T-$) and 67% ($A + T+$) of the biological pathways identified in the discovery cohort, providing solid evidence that these processes are not artefacts of platform or cohort.

The relevance of this work to the present thesis is direct: the ADNI cohort used here is entirely $A + T-$, and the findings of Vromen et al. [52] suggest caution when comparing results obtained on ADNI with those obtained on the LC cohort, whose composition is instead skewed towards $T+$ patients. Furthermore, the Cox PH framework adopted by Vromen et al. [51, 52] is applied in the present thesis as a complementary analytical tool, used to assess the univariate survival associations of the proteomic candidates identified by the machine learning pipeline, and to provide an independent biological validation of the selected features.

Although the two studies by Vromen et al. [51] and Vromen et al. [52] represent a fundamental methodological reference for the use of CSF proteomics in relation to time to conversion, both share a common limitation: the Cox PH framework is employed exclusively as a tool for univariate statistical association, identifying which individual proteins are correlated with the hazard of conversion without testing their actual predictive capacity on unseen patients, and without exploring potential non-linear patterns or interactions between features that a single semi-parametric model cannot capture. The present thesis addresses these limitations by embedding the Cox PH within a broader machine learning pipeline: the FSE-StabPred ensemble integrates multiple ranking methods to capture associations from complementary perspectives, and the selected features are subsequently evaluated for predictive capacity through Random Forest and Support Vector Machine models within a nested cross-validation framework. The Cox PH model is retained as a complementary tool

for the biological validation of the identified candidates, rather than as the primary predictive instrument.

Machine learning frameworks for MCI conversion. The third line of work, developed by Pereira, Madeira, and colleagues [34], addresses the problem from a methodological perspective oriented towards machine learning, and is the one from which this thesis most directly inherits its analytical framework. The foundational study in this line [34] introduced the Time Windows approach for predicting conversion from MCI to dementia using neuropsychological data from the Portuguese Cognitive Complaints Cohort (CCC). The starting observation is that the approach prevailing in the literature for constructing learning examples, termed by the authors the First Last approach (which combines the baseline assessment with the last available evaluation regardless of the time elapsed between the two), disregards the intrinsic heterogeneity of an MCI patient cohort, whose members lie at different stages of the same continuous neurodegenerative process. The authors therefore proposed stratifying patients into temporally homogeneous groups, constructing separate learning examples for time windows of 2, 3, 4, and 5 years, and showing that this approach systematically outperforms the First Last approach in terms of AUC, sensitivity, and specificity, both in the cross-validation set and in an independent validation set. The most clinically relevant finding is that the 5-year window achieved the best performance (AUC 0.88 in cross-validation, 0.76 in independent validation), suggesting that longer follow-up periods allow more stable decline patterns to be learned.

The subsequent work in the same line [35] extends this framework by introducing the element that the present thesis adopts most directly: feature selection through an ensemble combining stability and predictability, termed by the authors *FSE-StabPred*. The motivating problem is that the neuropsychological test batteries used in clinical practice comprise dozens of measures (79 in ADNI, 40 in CCC after filtering for missing values), a number that is large relative to the typical sample size of these studies, making feature selection not merely useful but necessary in order to avoid overfitting and to improve the clinical interpretability of the models. The authors combine seven base feature selection methods into an ensemble, then aggregate the resulting rankings by averaging and optimise the size of the final feature subset by maximising the *Robustness-Performance Trade-off* (RPT- β), balancing the stability of the selected features against classification performance (further details of the framework are provided in Section 2.5.2). The approach, validated on ADNI and CCC across four time windows (2, 3, and 4 years), identified subsets of approximately 30 and 20 features respectively, achieving stability above 0.89 and 0.95 and AUC above 0.87 and 0.82, consistently outperforming the use of the entire feature set and

proving competitive with, and at times superior to, each individual selection method.

The *FSE-StabPred* framework constitutes the direct methodological foundation of the machine learning pipeline developed in this thesis, with a relevant extension represented by the formulation of predictability also for regression problems, made necessary by the fact that the goal here is not only binary stable/converter classification but also, and primarily, the direct prediction of time to conversion.

Despite the robustness and methodological rigour demonstrated by Pereira et al. [34, 35], a key limitation of this line of work is that the FSE-StabPred framework has been applied exclusively to neuropsychological data, and has never been tested on high-dimensional proteomic feature spaces, where the number of candidate features is orders of magnitude larger relative to the sample size and the biological interpretation of the selected subset carries additional clinical relevance. The present thesis addresses this gap by applying and extending the FSE-StabPred framework to CSF proteomic data, assessing whether the stability and predictability gains demonstrated on neuropsychological batteries generalise to a fundamentally different and more challenging feature space. Furthermore, although Pereira et al. [34, 35] focus exclusively on binary classification tasks, the present thesis extends the pipeline to encompass multiclass classification across four time windows simultaneously, and introduces a regression formulation targeting the continuous time-to-conversion variable directly, requiring the development of a dedicated predictability metric within the RPT- β framework.

2.7 Chapter Summary and Research Motivation

The three lines of work reviewed in Section 2.6 delineate the present thesis is situated and collectively define both its starting point and its open questions. From Moutinho et al. [29] comes evidence that CSF proteomic profiles in MCI A+ patients carry sufficient biological information to reveal reproducible pathophysiological subtypes across independent cohorts, confirming that proteomic features are informative of the underlying disease heterogeneity even at the MCI stage.

From Vromen et al. [51, 52] comes the choice to employ CSF proteomic biomarkers as the primary informative substrate and the Cox proportional hazards framework as a complementary analytical tool for relating baseline protein levels to time to conversion.

From Pereira et al. [34, 35] comes the machine learning methodological framework of feature selection through an ensemble combining stability and predictability, and the attention to time to conversion as an informative dimension that a static binary classification cannot fully capture.

Each of these lines of work leaves open questions that the present thesis addresses

directly. Moutinho et al. [29] demonstrated that the proteomic signal is biologically meaningful, but did not pursue its exploitation for temporally explicit prediction of conversion. Vromen et al. [51, 52] established statistical associations between baseline CSF proteins and conversion risk, but stopped short of testing the actual predictive capacity of these proteins on unseen patients, and relied on a single semi-parametric formulation that does not capture non-linear patterns or interactions between features. Pereira et al. [34, 35] validated a robust machine learning framework for stable feature selection and time-window-based prediction, but applied it exclusively to neuropsychological data and to binary classification tasks, leaving open the question of whether it generalises to high-dimensional proteomic feature spaces and to regression formulations targeting the continuous time-to-conversion variable directly.

These gaps collectively motivate the contributions of the present thesis, detailed in the following Chapter 3 (*Materials and Methods*): the extension of the FSE-StabPred framework to CSF proteomics and to regression tasks; the development of predictive models evaluated across multiple modelling approaches and two independent cohorts; the identification of strong candidate biomarkers through convergent evidence from ensemble feature selection, Gini importance, and Cox proportional hazards analysis; and the cross-cohort consistency analysis on the shared proteomic panel, providing a methodological foundation for future multi-cohort proteomic studies.

3. Materials and Methods

A pipeline of the project structure, described in detail in this Chapter, is represented in Figure 3.1.

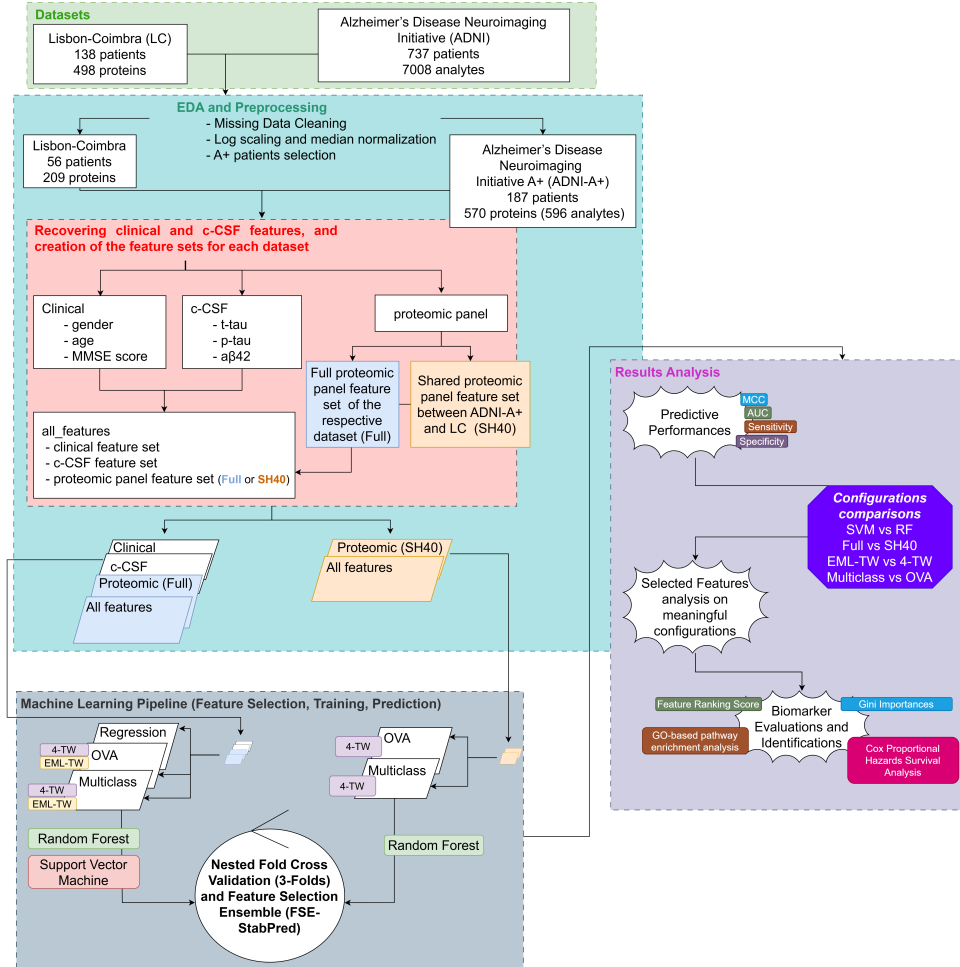


Figure 3.1: Overview of the thesis pipeline. After exploratory data analysis and preprocessing (Sections 3.1.1 and 3.1.2), the LC cohort is reduced to 56 patients and 209 proteins, and the ADNI-A+ cohort to 187 patients and 570 proteins. Four feature sets are constructed for each dataset: **clinical** (gender, age, MMSE), **c-CSF** (t-tau, p-tau, A β 42), **proteomic**, and **all_features**; the latter two exist in a Full and a Shared Proteomic Panel (SH40) variant. All modelling approaches, regression, OVA classification (4-TW and EML-TW formulations), and multiclass classification, are evaluated on the Full panel using both Random Forest and SVM; the SH40 variant is evaluated using Random Forest only, restricted to the 4-TW OVA and multiclass formulations. All configurations are trained within a nested cross-validation framework with FSE-StabPred ensemble feature selection (Sections 2.5.1 and 2.5.2). Results are analysed in terms of predictive performances, and configuration comparisons; selected feature of the meaningful configurations are analysed, and the most robust proteomic candidates are identified through the integration of Gini importance, Cox proportional hazards survival analysis, and FSE-StabPred feature ranking, and further characterised via GO-based pathway enrichment (Section 4.3).

3.1 Study Cohorts and Preprocessing

This section describes the two clinical datasets used in the present study and the preprocessing steps applied to each. The primary dataset is the Lisbon-Coimbra (LC) cohort (Section 3.1.1). The secondary dataset is obtained from the A+ patients of the Alzheimer’s Disease Neuroimaging Initiative (ADNI) cohort (Section 3.1.2). An additional proteomic feature set was generated, comprising the 40 proteins shared between the two proteomic panels, and is described further in detail in Section 3.1.3.

3.1.1 Lisbon-Coimbra (LC) Cohort

The primary dataset used in this study is a sub-cohort of the Portuguese Cognitive Complaints Cohort (CCC) and was derived from two Portuguese clinical centres: the *Memoclínica* private memory clinic in Lisbon, and the *Centro Hospitalar e Universitário de Coimbra – Polo Hospitais da Universidade de Coimbra (CHUC)*, in Coimbra. The CCC study was conducted in accordance with the Declaration of Helsinki and was approved by the local ethics committee (Hospital de Santa Maria e Faculdade de Medicina da Universidade de Lisboa). All participants provided written informed consent [29]. Within this sub-cohort, referred to as the Lisbon-Coimbra cohort (LC), a total of 138 patients were initially available, of whom 78 were recruited from the Lisbon centre and 60 from the Coimbra centre. All participants had received a clinical diagnosis of Mild Cognitive Impairment (MCI); however, for a subset of patients the etiological status was not determined with certainty, making their classification as MCI due to Alzheimer’s disease uncertain. The availability of clinical and proteomic data varied across patients, with some discrepancies between the different data files available; the specific inclusion criteria applied for the patient selection, including the confirmation of amyloid-positive (A+) status are described in Moutinho et al. [29] while the resolution of inconsistencies across data sources, are described below.

CSF samples were collected via lumbar puncture following standard clinical procedures. Proteomic profiling was performed using the Short-GeLC LC-SWATH-MS methodology; the general principles underlying this technique are described in Section 2.3, while the specific protocol applied to this cohort follows Moutinho et al. [29]. A total of 498 proteins were initially quantified across the cohort. Clinical variables available for each patient included the date of CSF sampling, CSF concentrations of A β 42, total tau (t-tau), and phosphorylated tau (p-tau), the Mini-Mental State Examination (MMSE) score at baseline, conversion status to Alzheimer’s dementia, date of conversion, and duration of follow-up.

Preprocessing The preprocessing of the LC cohort involved three distinct data sources: the clinical file, the proteomic metadata file, and the proteomic expression matrix obtained from the DIA-NN output. In a first consistency check phase, the three datasets were compared to verify the coherence of patient identifiers across sources. The clinical file contained 71 patients, the proteomic metadata file 134 patients, and the proteomic matrix 138 columns, of which 6 corresponded to technical replicates of three patients (LIS021, LIS023, and LIS034, each with one replicate). Since both clinical and proteomic information are required for every patient included in the analysis, only patients present in all three data sources were retained at this stage; patients lacking either clinical or proteomic data, were excluded. Of the 71 remaining patients some presented an etiological status classified as “Unknown”, meaning it was not possible to determine with certainty whether the MCI was due to Alzheimer’s disease; these patients were also excluded from the analysis.

The three patients with technical replicates (LIS021, LIS023, and LIS034) were inspected by computing the Pearson correlation between each pair of replicates as a quality check. The second replicate of each patient was subsequently removed following the recommendation of the specialised laboratory responsible for CSF proteomic data acquisition, which advised retaining only the first measurement in cases of technical duplication.

Protein filtering was performed in two steps. First, contaminant proteins belonging to the cRAP (Common Repository of Adventitious Proteins) database were removed. Second, only proteins characterised by at least 2 proteotypic sequences were retained. Proteotypic sequences are peptides uniquely attributable to a single protein and not shared with other proteins in the database; requiring at least 2 of such sequences per protein ensures greater reliability in protein identification and quantification within DIA-NN-based analyses, and reduces the risk of false identifications. This filtering reduced the number of proteins from 498 to 349.

Subsequently, patients with more than 20% missing values across the total protein set were excluded. In particular patient 100145 and LIS079, that although present in the proteomic matrix, was found to have entirely missing values across the whole protein expression matrix. The 20% threshold was chosen as a compromise between the need to preserve proteomic data quality and the need to retain a sufficient number of patients in an already limited cohort: more permissive thresholds would have increased the risk of introducing noise in downstream analyses, while more stringent thresholds would have led to the loss of additional patients. Proteins not detected in at least one patient were then removed, bringing the final number of proteins to 209. At this point, it was verified that every patient had a recorded follow-up. It was found that some patients did not. For this reason these patients were removed, bringing the dataset to 56 patients.

Raw proteomic intensities exhibited a strongly right-skewed distribution. A $\log_2$ transformation was therefore applied to reduce skewness and improve comparability across samples, followed by median normalisation to align sample distributions and remove residual technical variation between the two sub-centres (Lisbon and Coimbra). Median normalisation was performed by subtracting from each sample its own median and adding the global median computed across all samples.

The normalised proteomic matrix was then merged with the clinical dataset via an inner join on patient identifier.

During deeper clinical data inspection, a special case was identified: patient LIS004 had the tTau value recorded as the string “>1200” rather than as a numeric value. This was converted to the numeric value 1200 to allow downstream analyses. In addition was found that follow-up times in the Coimbra cohort were recorded as whole years, while Lisbon values included decimal precision. For this reason a follow-up time alignment procedure was applied between the two sub-cohorts. To ensure comparability, Lisbon values were floor-rounded to the nearest integer, with the exception of values between 0 and 1, which were rounded to 0.5. The final LC dataset comprised **56 patients** and **209 proteins**. A representative pipeline of the filtering procedure is schematized in Figure 3.2

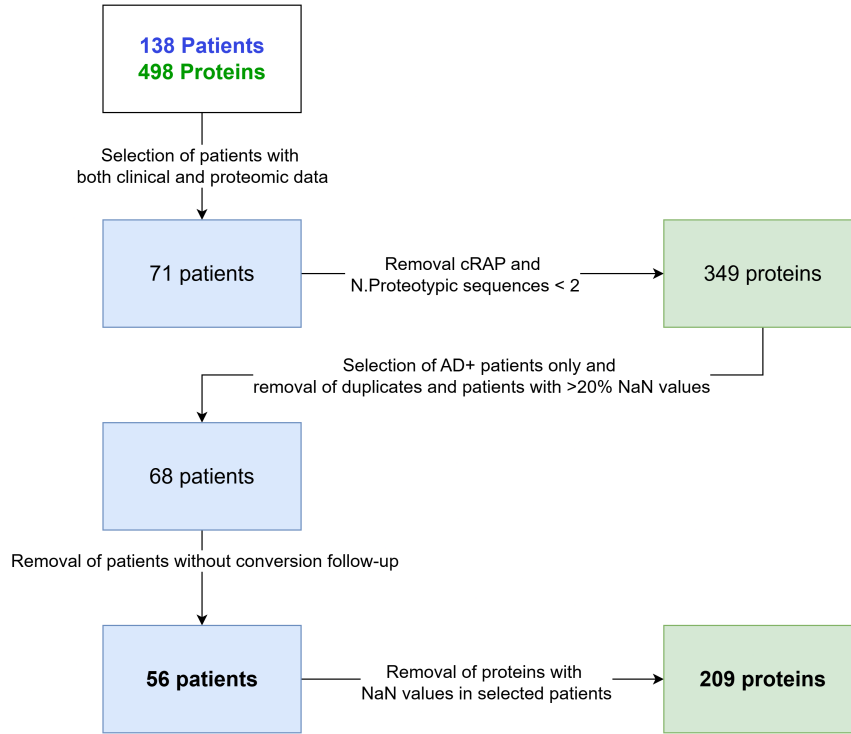


Figure 3.2: Filtering procedure applied to the LC dataset. Blue boxes represent filtering steps applied to the patient set; green boxes represent the corresponding state of the proteomic panel. The 20% NaN threshold for patient removal was chosen as a compromise between preserving data quality and retaining a sufficient number of patients in an already limited cohort; at the protein level, a stricter criterion was applied: any protein presenting at least one missing value across the retained patients was removed, to avoid introducing bias through imputation in an already sparse and delicate dataset.

3.1.2 Alzheimer’s Disease Neuroimaging Initiative (ADNI) A+ Cohort

The other dataset used in this study was obtained from the *Alzheimer’s Disease Neuroimaging Initiative (ADNI) database*¹. The ADNI was launched in 2003 as a public-private partnership, led by Principal Investigator Michael W. Weiner, MD. The primary goal of ADNI has been to test whether serial magnetic resonance imaging (MRI), positron emission tomography (PET), other biological markers, and clinical and neuropsychological assessment can be combined to measure the progression of mild cognitive impairment (MCI) and early Alzheimer’s disease (AD).

The ADNI study was conducted according to Good Clinical Practice guidelines, the Declaration of Helsinki, and US 21 CFR Part 50 (Protection of Human Subjects) and Part 56 (Institutional Review Boards), and in compliance with HIPAA regulations. Written informed consent was obtained from all participants and/or their authorised representatives. Data from the ADNI database were accessed in accordance with the

¹<https://adni.loni.usc.edu>

ADNI Data Use Agreement ², and were anonymised in both cohorts [29, 54].

The CSF proteomic data used in this study were acquired using the SomaScan 7K (v4.1) platform developed by SomaLogic; the general principles underlying this technique are described in Section 2.3. A subsequent quality control protocol was done at the *NeuroGenomics and Informatics Center, Washington University School of Medicine*, as described in detail in Wang et al. [54], and yielded a final matrix comprising 7,008 analytes measured across a subset of 737 ADNI samples used in the present study.

It is important to note that the dataset used in the present study does not correspond to the full ADNI cohort, but rather to a specific sub-cohort selected on the basis of amyloid positivity and MCI-to-AD conversion status, referred to throughout this thesis as ADNI-A+. Clinical variables available for each ADNI-A+ participant included longitudinal clinical diagnosis at each visit, Mini-Mental State Examination (MMSE) score, demographic information including gender and age, and CSF concentrations of A β 42, total tau (t-tau), and phosphorylated tau (p-tau), measured by immunoassay at the *ADNI Biomarker Core, University of Pennsylvania*. The selection criteria applied to identify MCI-to-AD converters are described in Wang et al. [54], while preprocessing operations performed on the ADNI-A+ sub-cohort, are described below.

Preprocessing The preprocessing of the ADNI cohort followed the same criteria applied to the LC cohort, with adaptations to account for the longitudinal structure of the ADNI database. The starting point was the post-QC SomaScan 7K proteomic matrix (comprising, as already mentioned, 737 ADNI samples and 7,008 analytes).

Since the ADNI dataset is longitudinal, with multiple CSF measurements per patient collected at different time points, only baseline visits were retained, ensuring that each patient contributed a single proteomic profile corresponding to their first CSF sample collected at study entry. Patients were then classified based on their longitudinal diagnosis trajectory: patients already diagnosed with AD at baseline and patients who never converted to AD during follow-up were excluded, retaining only MCI-to-AD converters. The conversion date was defined as the date of the first clinical visit at which the diagnosis changed to AD (DIAGNOSIS = 3). The time-to-conversion was computed as the difference in days between the baseline CSF collection date and the conversion date, converted to years, and rounded using the same floor-rounding procedure applied to the LC cohort. This procedure yielded 187 MCI-to-AD converters from the 737 baseline samples.

Missing values were handled using the same criteria as the LC cohort. No patient

²https://adni.loni.usc.edu/wp-content/themes/adni_2023/documents/ADNI_Data_Use_Agreement.pdf

exceeded the 20% missing value threshold and all 187 patients were retained. Proteins not detected in at least one patient were subsequently removed, reducing the number of analytes from 7,008 to 596, corresponding to 570 unique proteins based on UniProt identifiers.

Clinical variables were retrieved from multiple ADNI data files and merged with the proteomic dataset. Baseline MMSE scores were obtained from the screening visit of the MMSE file; all 187 patients had a valid baseline MMSE score available, with a negligible temporal difference between the MMSE visit date and the proteomic CSF collection date (mean difference: 0.13 years). Age was derived from the difference between the baseline CSF collection date and the patient’s date of birth, retrieved from the demographics file; gender was retrieved from the same file. CSF concentrations of A β 42, total tau (t-tau), and phosphorylated tau (p-tau) were retrieved from the UPENN Biomarker Core file, using baseline visits. Duplicate biomarker entries for the same patient were resolved by selecting only the first replicate, to maintain consistency with the LC preprocessing procedure. Two patients presented missing t-tau values. Since p-tau and t-tau are known to be strongly correlated CSF biomarkers reflecting the same underlying tau pathology (a relationship confirmed in the present dataset by a Spearman correlation coefficient of $r = 0.78$, $p < 0.001$), missing t-tau values were imputed using K-Nearest Neighbours (KNN) imputation with $k = 5$ and distance-based weighting, using p-tau as the reference feature. This approach was preferred over patient exclusion to preserve the already limited sample size. The temporal difference between the CSF biomarker collection date and the proteomic collection date was verified and found to be zero for all patients, confirming full consistency between the two datasets.

Although the SomaScan 7K data had already undergone technical normalisation by SomaLogic prior to release, visual inspection of the protein intensity distributions across samples revealed residual inter-sample variability. To ensure comparability with the LC cohort pipeline and to reduce this variability, the same normalisation procedure was applied: a $\log_2$ transformation followed by median normalisation, whereby each sample’s distribution was shifted to align with the global median computed across all samples. The final ADNI-A+ sub-cohort comprised **187 patients** and **570 proteins**. A representative pipeline of the filtering procedure is schematized in Figure 3.3

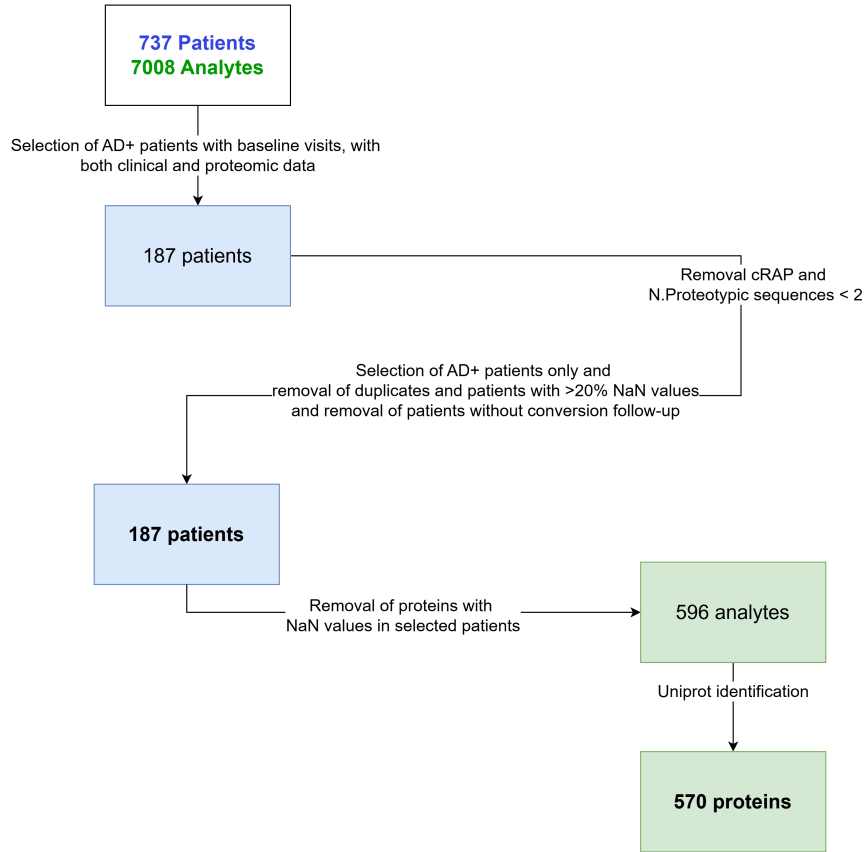


Figure 3.3: Filtering procedure applied to the ADNI-A+ dataset. Blue boxes represent filtering steps applied to the patient set; green boxes represent the corresponding state of the proteomic panel. Compared to the LC cohort, two additional preprocessing steps were required: the selection of baseline visits only, as ADNI is a longitudinal dataset comprising multiple visits per patient; and the mapping of SOMAscan analytes to UniProt protein identifiers, as multiple analytes may correspond to the same protein target, reducing the panel from 596 analytes to 570 unique proteins. The NaN-based filtering criteria follow the same rationale described for the LC cohort (Figure 3.2).

3.1.3 Shared Proteomic Panel between LC and ADNI-A+ Cohorts (SH40)

As discussed in the previous section (Sections 3.1.1 and 3.1.2), the LC and ADNI-A+ cohorts were profiled on substantially different proteomic platforms (Short-GeLC LC-SWATH-MS and SomaScan 7K, respectively), and consequently capture largely non-overlapping sets of proteins: 209 proteins for the LC panel and 570 for the ADNI-A+ panel after preprocessing. Therefore, a comparative analysis was conducted on a shared feature space, restricting the proteomic panel to the 40 proteins common to both cohorts (referred to as SH40 throughout this thesis), to investigate whether the two cohorts showed consistent patterns when evaluated on the same set of features.

The two preprocessed proteomic matrices were matched on UniProt protein identifiers, yielding only 40 proteins present in both the LC and the ADNI panel.

Given this, both cohorts were filtered to retain exclusively these 40 shared proteins, alongside their respective non-proteomic clinical variables (e.g. amyloid and tau status, age, gender, MMSE score, and time-to-conversion). In addition, the two cohorts were kept as separate datasets throughout, rather than being pooled into a single combined sample. The resulting LC overlap dataset comprised the same 56 patients as the full LC cohort, restricted to 40 proteomic features; the ADNI-A+ overlap dataset likewise comprised the same 187 patients as the full ADNI-A+ cohort, restricted to the same 40 proteins. In the ADNI panel, 9 of these 40 proteins corresponded to two distinct SOMAscan analytes each, rather than a single one; both analytes were retained for these proteins, bringing the total number of proteomic features in the ADNI-A+ overlap dataset to 49. This minor redundancy was not resolved manually, and was instead left to be handled internally by the ensemble feature selection procedure (Section 3.3.2), consistent with the redundancy-penalising behaviour already built into methods such as mRMR.

The original motivation for constructing this overlapping dataset was to assess whether the LC and ADNI-A+ pipelines, run independently on their respective full proteomic panels, tended to select the same proteins as most relevant. Since the two panels share only 40 proteins out of 209 (LC) and 570 (ADNI-A+), a meaningful direct comparison of this kind is only possible within that shared subset: the question was therefore narrowed to whether, when both pipelines are restricted to the same 40-protein feature space, they converge on selecting the same proteins among them, which would indicate a degree of cross-cohort consistency in the proposed feature selection framework. The same machine learning pipeline used for the full LC and ADNI-A+ panels was accordingly applied, separately, to the LC and ADNI overlap datasets.

This analysis is best understood as a methodological consistency check rather than as a validation of predictive performance, for a reason inherent to the construction of the overlapping dataset itself: restricting the feature space to only 40 shared proteins necessarily discards the large majority of each cohort’s proteomic profile, so that proteins that may be genuinely most informative for either cohort, but that happen not to be among the 40 shared ones, are excluded from consideration from the outset. Consistent selection of a subset of these 40 proteins across both cohorts would therefore support the robustness of the feature selection framework, but leaving open the possibility that more strongly predictive proteins exist outside the shared feature space in either cohort.

3.2 Exploratory Data Analysis

This section presents the exploratory analyses performed on the two cohorts to better understand the data characteristics. Section 3.2.1 assesses the homogeneity of the two clinical centres composing the LC cohort through a comparison of baseline characteristics. Section 3.2.2 describes the discretisation of the continuous time-to-conversion variable into four time windows following Pereira et al. [35], and the resulting class distributions for both cohorts, including the merged three-class formulation. Finally, Section 3.2.3 reports the amyloid and tau stratification of both datasets, which is relevant to the biological interpretation of the results.

3.2.1 Inter-cohort Comparison (LC Dataset)

To assess whether the patients recruited from the two clinical centres (Lisbon and Coimbra) could be treated as a single homogeneous dataset, a systematic comparison of baseline characteristics was performed, including clinical variables (age, gender, and MMSE score) and conventional CSF biomarkers (total tau, phosphorylated tau, and amyloid- β_{42}). Continuous variables were compared using the Mann-Whitney U test, chosen for its robustness against non-normal distributions and outliers; gender distribution was compared using the Chi-squared test. Results are summarised in Table 3.1.

Table 3.1: Inter-cohort comparison of baseline characteristics between Lisbon and Coimbra sub-cohorts. Continuous variables (age, follow-up time, MMSE score, and conventional CSF biomarkers) were compared using the Mann-Whitney U test; gender was compared using the Chi-squared test. Significance threshold: $p < 0.05$. No statistically significant differences were found between the two centres.

Feature	Coimbra (Mean $\pm$ SD)	Lisbon (Mean $\pm$ SD)	p-value	Significant
Time-to-conversion (years)	2.26 $\pm$ 1.72	2.99 $\pm$ 1.73	0.1140	No
Age (years)	64.32 $\pm$ 8.14	68.54 $\pm$ 5.48	0.0687	No
CSF tTau (pg/ml)	621.85 $\pm$ 264.81	771.08 $\pm$ 424.16	0.1743	No
CSF pTau (pg/ml)	86.93 $\pm$ 41.79	86.54 $\pm$ 33.48	0.5333	No
CSF A β_{42} (pg/ml)	460.02 $\pm$ 92.41	507.30 $\pm$ 105.27	0.1716	No
MMSE	25.11 $\pm$ 1.63	25.05 $\pm$ 2.26	0.7637	No
Gender (F/M)	F: 63.2%, M: 36.8%	F: 64.9%, M: 35.1%	1.0000	No

Overall, the two sub-cohorts showed a remarkably similar distribution across most baseline parameters, with no statistically significant differences detected for any of the features examined. The Coimbra cohort showed a trend towards earlier conversion compared to Lisbon (2.26 $\pm$ 1.72 vs. 2.99 $\pm$ 1.73 years; $p = 0.1140$),

although this difference did not reach statistical significance. The age distribution was also comparable between the two groups (64.32 ± 8.14 vs. 68.54 ± 5.48 years; $p = 0.0687$), with the higher standard deviation in Coimbra reflecting the inclusion of a younger subset of patients largely absent in the Lisbon cohort. CSF pTau levels were highly consistent across both sites ($p = 0.5333$), while slight numerical divergences were observed for tTau and A β 42, though neither reached statistical significance ($p = 0.1743$ and $p = 0.1716$, respectively). MMSE scores (25.11 ± 1.63 vs. 25.05 ± 2.26 ; $p = 0.7637$) and gender distribution (approximately 64% female and 35% male in both cohorts; $p = 1.000$) were virtually identical between centres. These results support the validity of combining the two sub-cohorts into a single analytical dataset.

A centre-stratified analysis using KDE plots further revealed that the two sub-cohorts differ in their longitudinal coverage (Figure 3.4): the Coimbra cohort is predominantly composed of patients with short time-to-conversion (0–2 years), providing more detailed characterisation of early-stage progression, while the Lisbon cohort better represents patients with longer follow-up times (up to 6–7 years). This complementary distribution, while not introducing a statistically significant bias, should be taken into account when interpreting the results of downstream analyses.

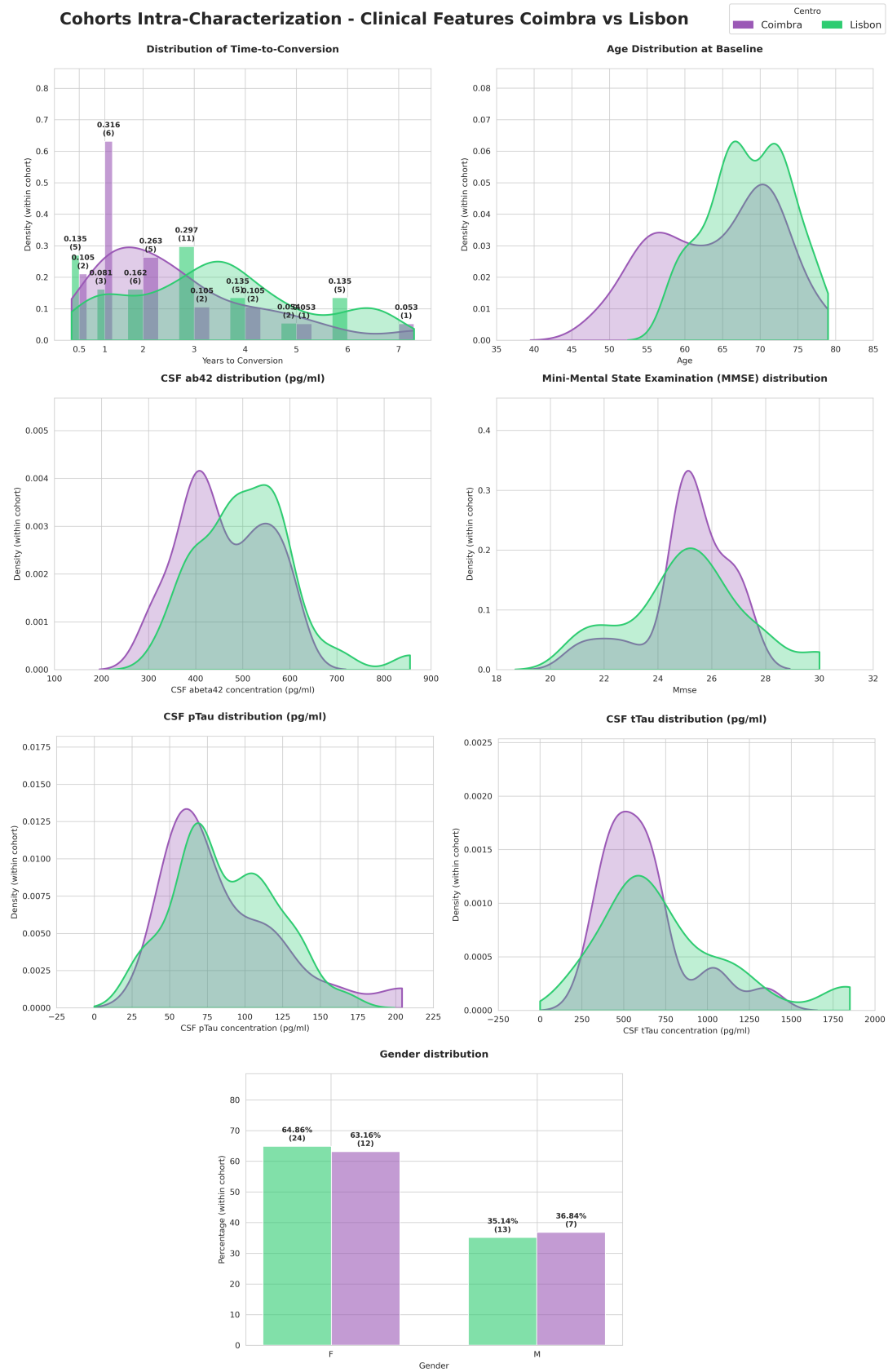


Figure 3.4: Distribution of clinical features across the Lisbon and Coimbra sub-cohorts. Time-to-conversion histogram shows that Coimbra patients are predominantly concentrated in the 0–2 year window, while Lisbon patients are more representative of longer follow-up times. Age, CSF biomarkers (tTau, pTau, A β 42) and MMSE score distributions are shown as KDE curves normalised within each cohort. Gender distribution is shown as percentage bar chart.

3.2.2 Class Distribution and Time Window Definition

Following the approach proposed by Pereira et al. [35], patients were classified into discrete time windows based on their time-to-conversion value. Specifically, the time-to-conversion variable was discretised into four mutually exclusive windows using the following boundaries: patients converting within 2 years ($0 < t \leq 2.00$ years) were assigned to the **2-year window**; those converting between 2.01 and 3.00 years to the **3-year window**; those converting between 3.01 and 4.00 years to the **4-year window**; and those converting after 4.01 years to the **5+ year window**. This discretisation transforms the time-to-conversion prediction problem into a multiclass classification task, where each class represents a clinically meaningful prognostic horizon.

LC Cohort. The resulting class distribution across the 56 patients of the LC cohort is shown in Figure 3.5. The dataset is markedly unbalanced, with the 2-year window representing nearly half of all patients ($n = 27$, 48.2%), followed by the 3-year window ($n = 13$, 23.2%), the 5+ year window ($n = 9$, 16.1%), and the 4-year window ($n = 7$, 12.5%). This class imbalance represents a methodological challenge for the machine learning pipeline and was explicitly accounted for in the model design, as described in Section 3.3.

LC Target Distribution: Patients per Time Window (%) (Total n=56)

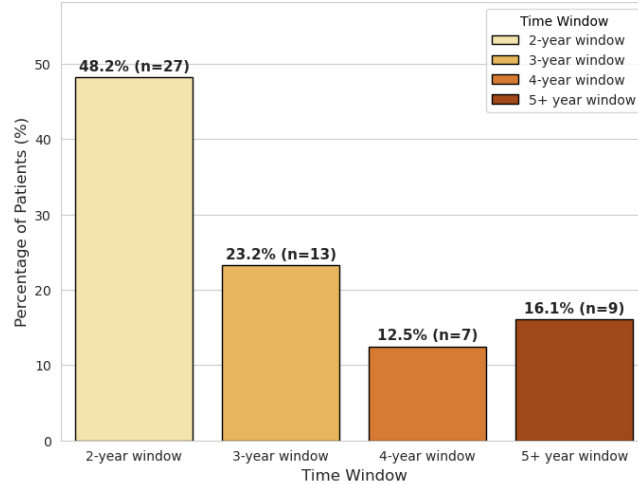


Figure 3.5: Target distribution of patients per time window in the LC cohort (total $n = 56$).

Given the pronounced imbalance of the 4-year window, the smallest of the four classes with only 7 patients (12.5%), an additional, merged class definition was also considered, in which the 3-year and 4-year windows, the two intermediate minority classes, were combined into a single class. For clarity and narrative consistency, the resulting three classes are referred to throughout this thesis as **early** (corresponding

to the original 2-year window), **mid** (corresponding to the merged 3-year and 4-year windows), and **late** (corresponding to the original 5+ year window) converters. This was explored as a way of testing whether a modest improvement in class balance would translate into a measurable gain in predictive performance, relative to the original four-class formulation. The resulting class distribution across the same 56 patients is shown in Figure 3.6.

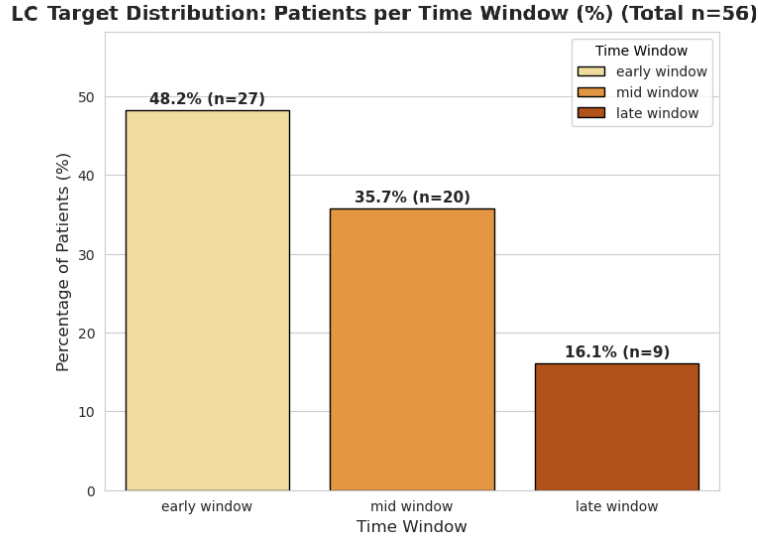


Figure 3.6: Target distribution of patients per time window in the LC cohort after merging the 3-year and 4-year windows into a single class (total $n = 56$).

ADNI-A+ Cohort. The 187 patients in the final ADNI-A+ dataset were classified into the same four time windows applied to the LC cohort. The resulting class distribution is shown in Figure 3.7. Similarly to the LC cohort, the dataset is markedly imbalanced, with the 2-year window representing the majority of patients ($n = 115$, 61.5%), followed by the 5+ year window ($n = 45$, 24.1%), the 3-year window ($n = 16$, 8.6%), and the 4-year window ($n = 11$, 5.9%). Notably, the imbalance in ADNI-A+ is even more pronounced than in the LC cohort, with the 2-year window alone accounting for more than 60% of all patients, and the 3-year and 4-year windows being particularly underrepresented.

ADNI-A+ Target Distribution: Patients per Time Window (%) (Total n=187)

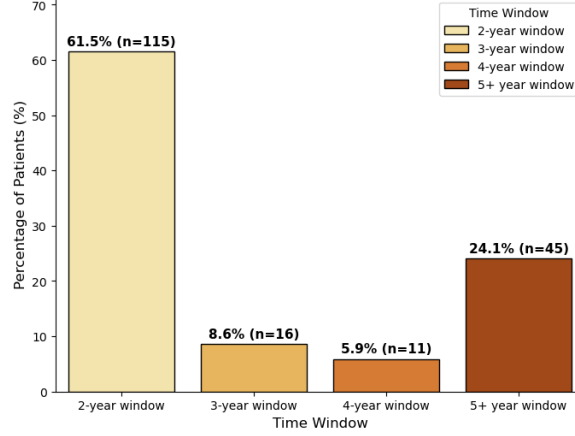


Figure 3.7: Target distribution of patients per time window in the ADNI-A+ cohort ($n = 187$).

As for the LC cohort, an additional, merged class definition was considered for the ADNI-A+ cohort, in which the 3-year and 4-year windows, the two smallest classes with 16 and 11 patients respectively, were combined into a single class. Following the same convention adopted for the LC cohort, the resulting three classes are referred to as **early**, **mid**, and **late** converters. This was explored as a way of testing whether a modest improvement in class balance would translate into a measurable gain in predictive performance, relative to the original four-class formulation, and to allow a direct comparison with the equivalent analysis performed on the LC cohort. The resulting class distribution across the same 187 patients is shown in Figure 3.8. Even after merging, the early class remains by far the largest, reflecting the more pronounced overall unbalancement already noted for the four-class formulation of the ADNI-A+ cohort.

ADNI-A+ Target Distribution: Patients per Time Window (%) (Total n=187)

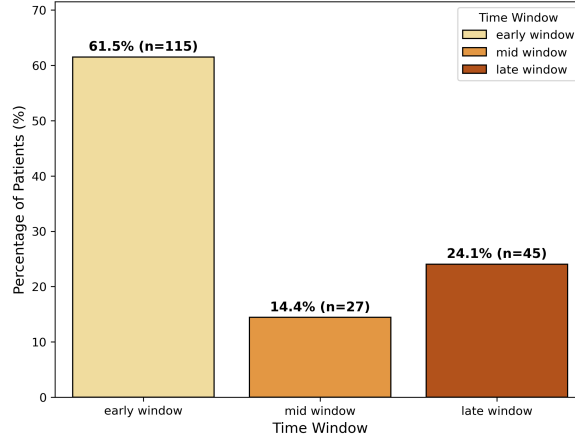


Figure 3.8: Target distribution of patients per time window in the ADNI-A+ cohort after merging the 3-year and 4-year windows into a single class ($n = 187$).

3.2.3 Amyloid and Tau Stratification

LC Cohort. Amyloid and tau positivity were assessed using the CSF cut-offs established in Vromen et al. [52]: $A\beta_{42} < 813$ pg/ml for amyloid positivity and $t\text{Tau} > 375$ pg/ml for tau positivity (Figure 3.9). As expected by design, the vast majority of patients (55 out of 56, 98.2%) were confirmed $A\beta+$. The single patient below the threshold (LIS013, $A\beta_{42} = 855$ pg/ml) presented a value only marginally above the cut-off and was therefore considered a borderline case, likely a false negative, and retained in the analysis as $A\beta+$. Regarding tau status, 49 patients (87.5%) were classified as $\text{Tau}+$ and 7 patients (12.5%) as $\text{Tau}-$ (LIS008, LIS051, LIS057, LIS090, LIS104 from the Lisbon centre, and 100111, 105080 from Coimbra). The limited number of $\text{Tau}-$ patients precluded a stratified analysis by tau status, as performed in Vromen et al. [52]; all patients were therefore retained in the combined analysis regardless of tau status, resulting in a cohort that is effectively $A+$ for all practical purposes.

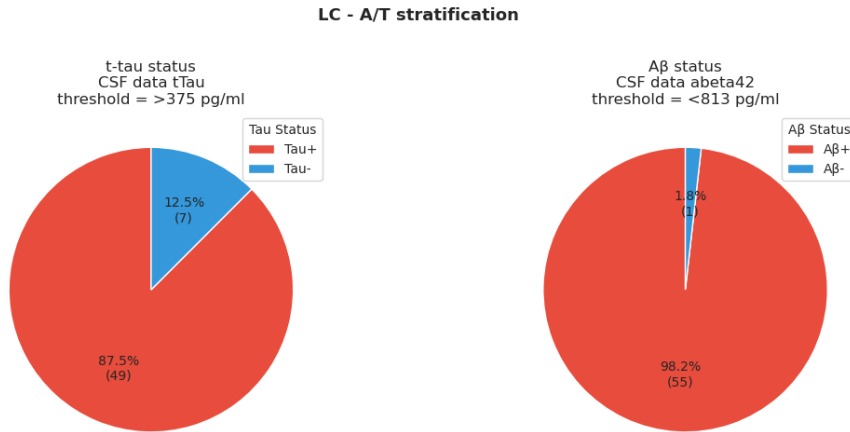


Figure 3.9: CSF amyloid and tau status distribution in the LC cohort, assessed using the cut-offs of Vromen et al. [52].

ADNI-A+ Cohort. Amyloid and tau positivity were assessed using the same CSF cut-offs applied to the LC cohort, as established in Vromen et al. [52]: $A\beta_{42} < 813$ pg/ml for amyloid positivity and $t\text{Tau} > 375$ pg/ml for tau positivity (Figure 3.10). All 187 patients were confirmed $A\beta+$ (100%), consistent with the inclusion criteria applied during preprocessing. Surprisingly, all 187 patients were also classified as $\text{Tau}-$ (100%), in sharp contrast with the LC cohort where 87.5% of patients were $\text{Tau}+$. This marked difference in tau status between the two cohorts represents a key biological distinction that should be taken into account when interpreting and comparing the results of the machine learning pipeline across the two datasets. As discussed by Vromen et al. [52], $A+T-$ and $A+T+$ individuals are characterised by distinct CSF proteomic signatures and different underlying pathophysiological

mechanisms, which may limit the direct comparability of proteomic biomarkers identified in the two cohorts.

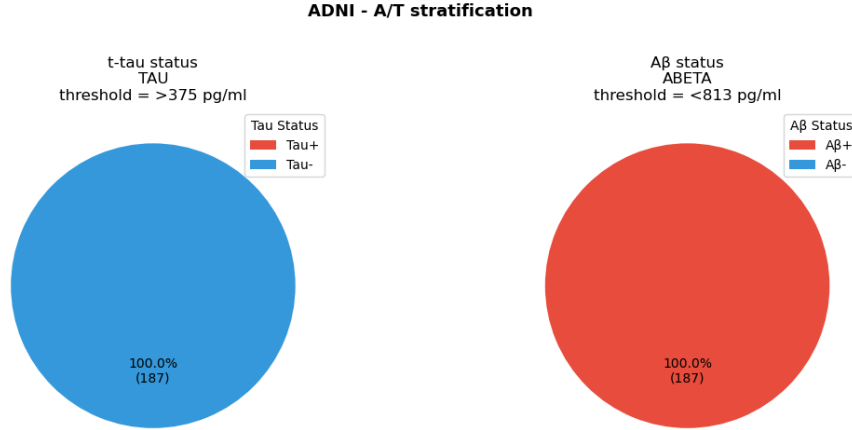


Figure 3.10: CSF amyloid and tau status distribution in the ADNI-A+ cohort, assessed using the cut-offs of Vromen et al. [52].

3.3 Machine Learning Pipeline

The machine learning pipeline described in this section was applied across multiple experimental configurations. Four feature sets were evaluated independently: clinical variables alone (`clinical`), conventional CSF biomarkers alone (`c-CSF`), proteomic data alone (`proteomic`), and their combination (`all_features`) (Table 3.2). Each feature set was tested under three modelling approaches: a one-vs-all (OVA) binary classification strategy, in which each time window is treated as the positive class against all remaining windows; a single multiclass classification model, in which all time windows are predicted simultaneously; and a regression approach, in which the model predicts the continuous time-to-conversion value directly. For the classification approaches (multiclass and OVA), both the original four-class time windows formulation (4-TW) and the merged three-class early/mid/late (EML-TW) time windows formulation introduced in Section 3.2.2 were evaluated. The OVA and multiclass strategies were compared to assess whether decomposing the problem into independent binary tasks, each optimised to discriminate a specific time window against all others, yields more discriminative models than a joint multiclass formulation. The latter is expected to favour the selection of biomarkers with a consistent expression gradient across the full temporal axis of conversion, whereas the OVA strategy may capture window-specific proteomic signatures that would otherwise be obscured in a global model. The regression approach was included as an alternative to classification, given that time-to-conversion is intrinsically a continuous variable.

Two predictive models were evaluated across all modelling approaches and feature

sets: a Random Forest (RF) and a Support Vector Machine (SVM) (Sections 2.4.2 and 2.4.1). For feature selection, two complementary strategies were compared. The primary strategy is the FSE-StabPred ensemble (Section 2.5.2), which jointly optimises stability and predictability to identify a robust subset of features.

The complete pipeline, illustrated in Figure 3.11, follows a sequential workflow: for each experimental configuration, the FSE-StabPred ensemble feature selection is first applied within the inner loop of the nested cross-validation (Section 3.3.1) to identify the optimal feature subset (Section 3.3.2); the selected features are then used to tune the final model via grid search on the development set (Section 3.3.3); the tuned model is trained on the full development set and evaluated on the held-out benchmark set; and performance metrics are aggregated across the three outer folds. The selected features are subsequently carried forward to the biomarker analysis described in Section 3.4. The following subsections describe each of these steps in detail.

Table 3.2: Composition of the four feature sets used across all experimental configurations. Age is expressed in years; gender is encoded as a binary variable (0 = Female, 1 = Male). c-CSF biomarkers are expressed as concentration values; proteomic features represent relative protein abundance levels.

Feature Set	Features
clinical	Age, gender, MMSE score.
c-CSF	Total tau (t-tau), phosphorylated tau (p-tau), amyloid- β 42 (A β 42) concentration signals.
proteomic	Protein abundance profiles: 209 proteins in the LC cohort (Short-GeLC LC-SWATH-MS) and 570 proteins in the ADNI-A+ cohort (SomaScan 7K), after preprocessing and missing value filtering. See the entire proteomic profile in Table 8.1 in Appendix chapter.
all_features	All clinical, conventional CSF biomarker, and proteomic features combined.

3.3.1 Nested Cross-Validation Design

The limited size of the LC dataset ($n = 56$) and the critical unbalancement imposes severe constraints on the validation strategy. With such a restricted sample, standard approaches such as a single training/test split risk to produce unstable performances that are strongly dependent on the particular partition chosen. To address this, a *nested cross-validation* (Section 2.5.1) was adopted. With this strategy was possible to achieve a statistically more robust estimate of model generalisation

by strictly separating the phases of feature selection, hyperparameter tuning, and final evaluation.

The *outer loop* partitions the dataset into three folds using a Stratified K-Fold: at each iteration, approximately 75% of the data forms the *development set*, while the remaining 25% constitutes the *benchmark set*, kept completely isolated throughout the inner loop. This cycle is repeated three times by rotating the benchmark set, so that every patient is evaluated exactly once. The *inner loop* operates exclusively on the development set of each outer fold and is responsible for two tasks: feature ranking (via the Ensemble Ranking method (Section 2.5.2)) and selection (optimizing the RPT- β score described in Section 2.5.5). For this purpose, a *Repeated Stratified K-Fold* with 5 folds and 10 repetitions (50 splits in total) is employed, so as to obtain more reliable estimates of stability and predictability despite the limited sample size. Stratification is applied on the multiclass target variable (**time-window**) to preserve the class distribution across every split.

3.3.2 FSE-StabPred: implementation and adaptations

Feature ranking is performed within each inner fold of the repeated stratified cross-validation, using the FSE-StabPred framework described in Sections 2.5.2, 2.5.3, 2.5.4, 2.5.5, with three main adaptations compared to the original implementation of Pereira et al. [35]: the replacement of the Chi-squared ranking method with the ANOVA F-test to accommodate standardised continuous features; the extension of the ensemble to a regression setting, including a dedicated set of regression-adapted ranking methods; and the formulation of a dedicated predictability metric for the continuous time-to-conversion target within the RPT- β framework. These adaptations constitute one of the main methodological contributions of the present thesis (Section 1.2).

Data preparation for the ensemble method. Prior to ranking, features are standardised within each inner training fold by applying a z-score transformation via `StandardScaler` [33], which centres each feature to zero mean and scales it to unit variance. Although protein abundance values had already undergone $\log_2$ transformation and median normalisation during preprocessing (Sections 3.1.1 and 3.1.2), these steps reduce distributional skewness and correct for systematic inter-patient bias but do not bring all features to a common scale. Z-score standardisation is therefore necessary to ensure that scale-sensitive ranking methods such as LR-L1, SVM-RFE, and ReliefF assign importance based on the informative content of each feature rather than on its numerical magnitude. To prevent data leakage, the `StandardScaler` parameters are estimated exclusively on the training partition of each inner fold and then applied to the corresponding validation partition. Inside the Ensemble Feature

Selection, ReliefF was applied with $k = 10$ neighbours in all experiments, and, when the clinical feature set is used, the **gender** variable is declared as categorical to ensure correct handling by the mRMR algorithm.

Adaptations of the FSE-StabPred framework As already anticipated three adaptations from the original FSE-StabPred framework were introduced:

1. **The ANOVA F-test was used in place of the Chi-squared test** originally employed by Pereira et al. [35], since the Chi-squared test requires non-negative input values and is therefore incompatible with standardised features, which may take negative values after z-scoring; this substitution was not required by Pereira et al. [35], whose neuropsychological features were not z-scored prior to ranking.
2. **The original framework was extended to a regression setting**, since the present thesis targets the continuous time-to-conversion variable in addition to discrete time windows, while Pereira et al. [35] addresses only classification. An analogous ensemble of five ranking methods, adapted to this continuous target, was developed by mirroring the six classification methods (Section 2.5.2):
 - (a) **Mutual Information for regression** operates analogously to its classification counterpart, measuring the statistical dependence between each feature and the continuous target, including non-linear associations.
 - (b) **F-Regression** computes the linear correlation between each feature and the continuous target and converts it into an F-score, efficiently identifying features with a monotonic linear relationship with time to conversion.
 - (c) **Lasso regression** applies an L1 penalty to the regression coefficients, shrinking those of uninformative or redundant features to exactly zero, a property that is particularly valuable in a high-dimensional, low-sample-size setting such as the LC cohort.
 - (d) **Recursive Feature Elimination with a linear Support Vector Regressor** (SVR-RFE) extends the wrapper approach to the regression setting, iteratively eliminating the feature contributing least to the predictive accuracy of the support vector regression model.
 - (e) **mRMR for regression** applies the same minimum-redundancy-maximum-relevance criterion used in classification to a continuous target, selecting features that collectively maximise predictive relevance while minimising redundancy among them.
3. Since Pereira et al. [35] likewise addresses only classification, a **dedicated predictability metric was developed for the regression task**. Predictability

is defined as a monotonic transformation of the Root Mean Squared Error (RMSE) of the model on the held-out validation portion of each inner fold, computed over the n patients in that validation portion as:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (3.1)$$

where y_i is the observed time to conversion for patient i and $\hat{y}_i$ is the corresponding model prediction. Predictability for the regression task is then obtained as:

$$\text{Predictability}_{\text{reg}} = \frac{1}{1 + \text{RMSE}} \quad (3.2)$$

This transformation maps $\text{RMSE} \in [0, +\infty]$ onto a predictability score in $[0, 1]$, making it directly comparable in scale to the classification predictability (Section 2.5.4), and therefore usable within the same RPT- β framework. The choice of this particular transformation follows directly from the fact that RMSE, as a square root of a sum of squares, is by construction non-negative ($\text{RMSE} \geq 0$): the monotonically decreasing function $1/(1 + \text{RMSE})$ maps this unbounded, non-negative error measure onto a strictly bounded interval, $[0, 1]$, with a perfect prediction ($\text{RMSE} = 0$) yielding a predictability of exactly 1 and predictability approaching 0 as RMSE grows arbitrarily large, without ever requiring the score itself to be clipped or otherwise constrained. RMSE was preferred over the Mean Absolute Error (MAE) as the underlying error measure because it penalises larger prediction errors more heavily, so that a model producing occasional but severe mispredictions of the time to conversion is assigned a correspondingly lower predictability score.

Rank aggregation and per- k estimation of stability and predictability. At this point, features were then ranked independently by the ensemble methods at each of the 50 inner splits. Rank aggregation is performed by converting the raw ranks produced by each method into scores, assigning the value m to the top-ranked feature and 1 to the lowest-ranked, where m denotes the total number of features in the active feature set; the ensemble score for each feature is then computed as the arithmetic mean of its scores across all methods. This procedure is repeated across all 50 inner splits, yielding 50 ensemble rankings per outer fold. Each feature ranking, is used within each respective inner repeated stratified fold, to test each candidate subset size k from $k = 1$ to $k = m$ of features, and to compute an estimate of stability and predictability as follow: for each of the 10 repetitions, stability is computed by

applying the Kuncheva index pairwise to the 5 feature subsets of size k produced by the 5 folds of that repetition, yielding one stability estimate per repetition; the 10 per-repetition estimates are then averaged to obtain the final stability value for that k . Predictability is computed analogously: for each of the 50 inner splits, a model is trained on the top- k features of the ensemble ranking for that split and evaluated on the corresponding test fold, yielding one predictability value per split; these are averaged first within each repetition (across the 5 folds) and then across the 10 repetitions to obtain the final predictability estimate for that k . Regarding classification tasks, $\alpha = \gamma = 0.5$ was used for the predictability score evaluation (Equation 2.2), giving equal weight to AUC and to the minimum between sensitivity and specificity.

Inner-loop predictive models. The model used for this inner-loop predictability evaluation is intentionally kept simple and fixed, to avoid introducing additional computing-intensive hyperparameter optimisation within the scan over k . For the tasks the hyperparameter used are the ones that are considered standard in scikit-learn documentation. In particular, for classification tasks (both multiclass and one-vs-all), a Random Forest classifier with 100 estimators and `class_weight='balanced'` is used; for the regression task, a Random Forest regressor with 100 estimators is used. For the SVM-based experiments, the corresponding inner-loop models are an SVC with `kernel='rbf'`, $C = 1.0$, `gamma='scale'`, `class_weight='balanced'`, and an SVR with `kernel='rbf'`, $C = 1.0$, `gamma='scale'`, $\varepsilon = 0.1$. In all cases, the predictability metric is computed as described in the FSE-StabPred paragraph above: the classification predictability formula (Equation 2.2) for classification tasks, and the regression predictability formula (Equation 3.2) for the regression task.

RPT- β score evaluation. These two quantities are combined via Equation 2.4 into a single RPT- β score for that value of k , and the optimal subset size k^* is identified by an exhaustive scan across all candidate values, selecting the value that maximises this score (Equation 2.5)(Section 2.5.5).

In all experiments reported in this thesis, $\beta = 1$ was adopted, giving equal weight to stability and predictability. This choice was made in light of the limited sample size of the LC cohort ($n = 56$): placing disproportionate weight on predictability alone ($\beta > 1$) would risk selecting feature subsets that perform well on a particular partition of the data by chance but are not robust to resampling, whereas the equal weighting at $\beta = 1$ explicitly penalises also instability.

Recovery of the final feature subset. Once k^* has been identified, the final feature subset for that outer fold is recovered as follows. The 50 ensemble rankings

produced by the inner loop are aggregated by averaging their scores feature by feature, yielding a single consensus leaderboard for that outer fold. The top- k^* features from this consensus leaderboard constitute the feature subset used to train the final model on the development set of that fold and to evaluate it on the corresponding benchmark set. Since this procedure is applied independently to each of the three outer folds, it produces three potentially distinct feature subsets and three independent performance estimates, as reported in Chapter 4.

3.3.3 Final performance evaluation

Once the optimal feature subset has been identified by the inner loop, hyperparameter tuning of the final model is performed on the development set via an exhaustive grid search, implemented using the `GridSearchCV` function from the scikit-learn library [33]. Tuning is carried out within a stratified 5-fold cross-validation applied to the development set, keeping the benchmark set strictly isolated.

Grid Search Space For the Random Forest classifier, the grid search explores the following hyperparameter space: number of estimators (`n_estimators` $\in \{100, 200, 300, 500\}$), maximum tree depth (`max_depth` $\in \{\text{None}, 10, 20\}$), minimum number of samples required to split an internal node (`min_samples_split` $\in \{2, 5, 10\}$), and class weighting strategy (`class_weight` $\in \{\text{balanced}, \text{balanced_subsample}\}$). For the Random Forest regressor, the grid is analogous but omits `class_weight`, replaces it with minimum number of samples required at a leaf node (`min_samples_leaf` $\in \{1, 2, 4\}$), and uses a finer grid for maximum tree depth (`max_depth` $\in \{\text{None}, 5, 10, 20\}$).

For the Support Vector Machine, both the classification (SVC) and regression (SVR) variants share most of the hyperparameter grid: kernel $\in \{\text{rbf}, \text{linear}\}$, regularisation parameter $C \in \{0.01, 0.1, 1, 10, 100\}$, and kernel coefficient $\gamma \in \{\text{scale}, \text{auto}, 0.001, 0.01, 0.1\}$. For the SVC, the class weighting strategy is fixed to `balanced` to account for class imbalance across time windows. For the SVR, the insensitive loss tube parameter $\varepsilon \in \{0.01, 0.1, 0.5, 1\}$ is additionally explored, controlling the width of the margin within which no penalty is incurred for prediction errors.

Hyperparameter selection scoring method. The optimal hyperparameter combination is selected according to a modelling approach specific scoring criterion.

For classification approaches, whether multiclass or one-vs-all, the selection criterion is the AUC under a one-vs-rest scheme (`roc_auc_ovr`), computed across the inner validation folds: in the multiclass setting this corresponds to the macro-

averaged AUC across the four time windows, while in the one-vs-all setting it reduces to the standard binary AUC for the target window against all others.

For the regression approach, the selection criterion is the negative root mean squared error (`neg_root_mean_squared_error`), so that the configuration minimising prediction error on the inner folds is retained. All preprocessing steps, including standard scaling, are encapsulated within a scikit-learn `Pipeline` to prevent any form of data leakage between training and validation splits.

Final Evaluation on benchmark set. The best hyperparameter configuration identified during tuning is then used to train the final model on the entire development set. This model is subsequently evaluated on the benchmark set, which has never participated in either the ranking or the tuning phase, ensuring a performance estimate free of data leakage. Performance metrics are finally aggregated across the three outer folds, reporting mean and standard deviation as measures of predictive capacity and model stability for each modelling approach, respectively.

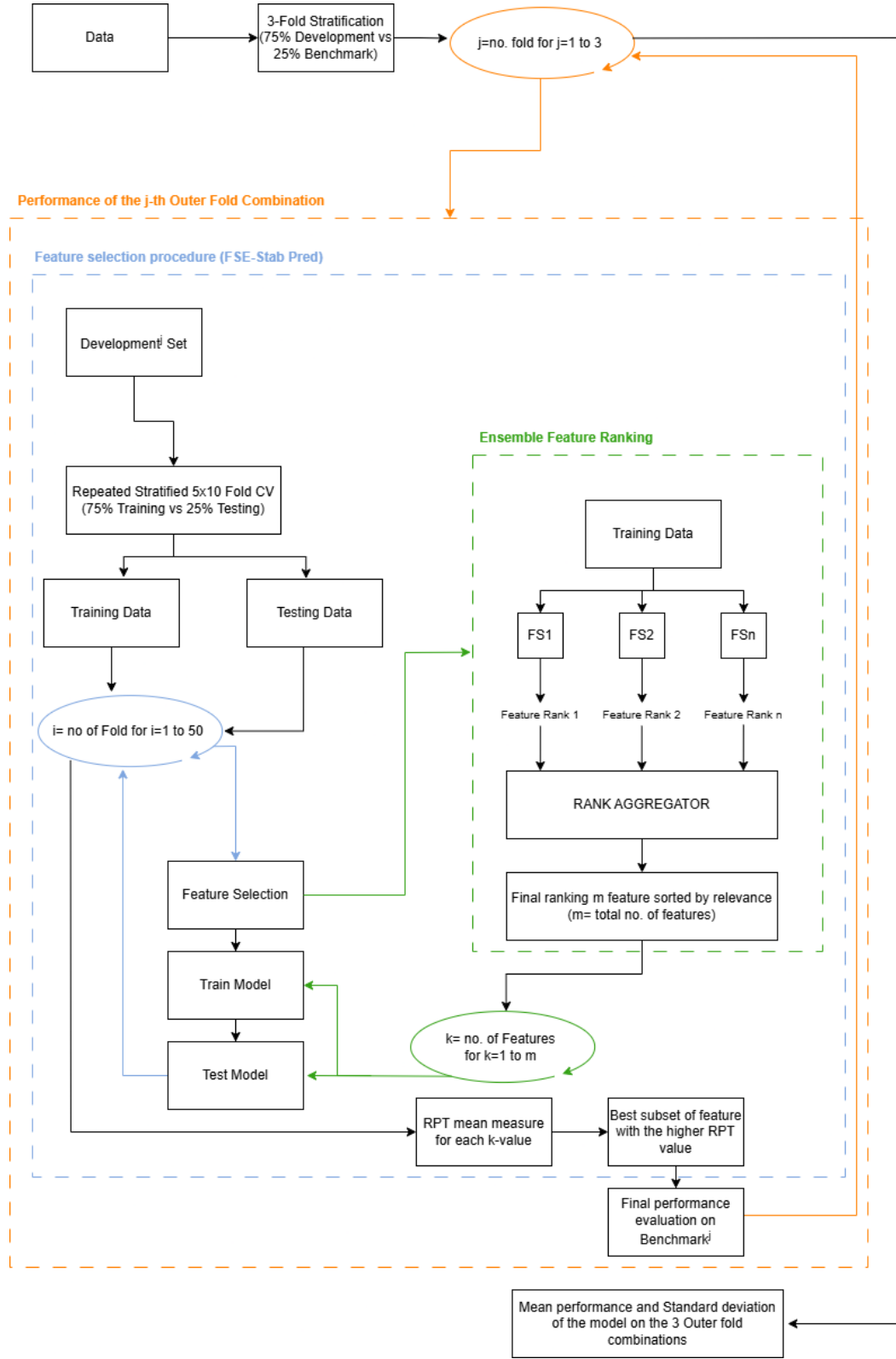


Figure 3.11: Schematic overview of the nested cross-validation pipeline. The outer loop performs a 3-fold stratified split, assigning two folds to the development set and one to the benchmark set at each iteration. Within the development set, a repeated stratified 5×10 inner CV drives the feature ranking ensemble (FSE-StabPred) and RPT- β -based feature subset optimisation. The best feature subset identified in the inner loop is used to train the final model on the full development set, which is subsequently evaluated on the benchmark set. Mean performance and standard deviation are computed across the three outer folds.

3.4 Proteomics Biomarker Analysis

The machine learning pipeline described in Section 3.3 produces, for each outer fold, a ranked list of features and a selected subset of size k^* . While predictive performance metrics quantify the overall discriminative power of these subsets, they do not by themselves identify which individual proteins are the most robust and biologically meaningful candidates for downstream investigation. Three complementary analyses were therefore performed to interpret the features selected by the pipeline and to consolidate the evidence for candidate biomarkers: **Cox proportional hazards survival analysis**, **Gini importance analysis** and **pathway enrichment analysis**.

Cox Proportional Hazards Survival Analysis. A Cox proportional hazards survival analysis was performed on the datasets, following the approach of Vromen et al. [52] (Section 2.6). For each protein in the respective panel, a separate Cox model was fitted with time-to-conversion as the duration variable and conversion to Alzheimer’s dementia as the event, adjusted for age and gender as covariates. Prior to fitting, each protein’s expression values were z-scored (zero mean, unit variance) so that hazard ratios are expressed per one standard deviation increase in protein level and are directly comparable across proteins. The full results and the significant features subset for each cohort are used for the downstream pathway enrichment analysis (Section 3.4).

Gini Importance Analysis. Gini importance scores were extracted from the Random Forest models trained during the nested cross-validation pipeline, to provide a complementary feature ranking independent of the FSE-StabPred ensemble. The motivation is that if a protein is ranked as highly relevant by both the ensemble procedure and by the model-intrinsic Gini criterion, the evidence for its importance is stronger than if it emerged from only one of the two approaches.

The analysis was applied separately to the **proteomic** and **both** feature sets. In both cases, Gini importance scores were extracted directly from the Random Forest models saved at the end of each outer fold of the nested cross-validation: accordingly to the feature set names criteria already described in Table 3.2 for the **proteomic** feature set, the respective saved model extracted the gini importance on the complete panel of proteomic feature only; for the **both** feature set, instead, importances were extracted across all feature types simultaneously. In both cases the gini importance score evaluation was done for the entire respective feature set, and not only on the features selected by FSE-StabPred. In this way also features not detected by FSE-Stab pred could also receive an importance estimate and can

be compared on equal terms with the selected ones, to assess whether features excluded by FSE-StabPred carry additional discriminative information beyond what is captured by the selected subset.

In both cases, the procedure was applied independently to each of the three outer folds. Gini importance scores were then aggregated across the three outer folds by computing mean and standard deviation per feature.

The analysis was applied to both the LC and ADNI-A+ cohorts. For ADNI-A+, SOMAscan analyte names were mapped to UniProt identifiers via the SOMAscan annotation file, to ensure consistent labelling with the LC cohort. In a final step, proteins appearing in the top 20 by mean Gini importance in both cohorts were identified for each feature set, constituting the cross-cohort Gini overlap.

Pathway Enrichment Analysis. Pathway enrichment analysis was performed to place the individual protein findings into a broader biological context and to identify molecular processes overrepresented among the candidate biomarkers. The analysis was applied to the set of *strong candidates*, proteins satisfying two independent criteria simultaneously: appearing in the top 20 of the RPT-based ensemble ranking in at least 2 out of 3 outer folds (for the `proteomic` and `both` feature sets), and appearing in the top 20 by mean Gini importance across folds (again for `proteomic` and `both` feature sets). This double filter ensures that only proteins consistently identified as highly relevant by two methodologically independent approaches are submitted to enrichment, reducing the risk of including spuriously ranked proteins. For each strong candidate, the Cox proportional hazards analysis (Section 3.4) was used to assign a directional label: proteins with $\log(\text{HR}) > 0$ were classified as *risk proteins* (higher CSF levels associated with faster progression), and proteins with $\log(\text{HR}) < 0$ as *protective proteins* (higher CSF levels associated with slower progression). For proteins not reaching statistical significance in the Cox model ($p \geq 0.05$), the HR direction was retained as an exploratory annotation only. Enrichment analysis was performed using the `gProfiler` Python library [22], querying Gene Ontology Biological Processes (GO:BP) annotation databases [47]. Statistical significance was assessed using the `g:SCS` multiple testing correction method implemented in `gProfiler`, with a significance threshold of $\text{FDR} < 0.05$. UniProt identifiers were mapped to gene symbols via the UniProt REST API prior to submission, as required by `gProfiler`. Risk and protective proteins were submitted as two separate queries: one joint query for risk proteins and one joint query for protective proteins. To focus the interpretation on pathways directly relevant to Alzheimer’s disease pathophysiology, the enriched terms were filtered using a curated set of AD-relevant GO ancestor terms, retrieved via the QuickGO REST API [5]. For each enriched GO:BP term, the full set of ancestor terms in the GO hierarchy was fetched, and terms mapping to one

of the predefined AD-relevant categories were retained. The categories used cover the major biological processes implicated in AD progression, and extensively discussed in 2.1 including: Neural Development and Regeneration, Neuroinflammation, Synaptic Signaling, Oxidative Stress and Mitochondrial Function, Apoptosis, Autophagy and the Ubiquitin-Proteasome System, IGF and Insulin Signaling, Lipid and Cholesterol Metabolism, ECM Remodelling and Cell Adhesion, Amyloid and Tau Pathology, and Complement and Immune Activation.

4. Results

This chapter presents the results of the machine learning pipeline described in Chapter 3, organised into three main sections. Section 4.1 evaluates the predictive performance of the developed models across all experimental configurations, identifying the modelling approaches and feature sets that yield meaningful discriminative capacity for each cohort. Section 4.2 then reports the feature subsets selected by the FSE-StabPred ensemble, focusing on the configurations that produced relevant predictive results; complete feature selection tables for all configurations are provided in the Appendix. Finally, Section 4.3 presents the biomarker evaluation of the most robust proteomic candidates, integrating convergent evidence from ensemble feature selection, Gini importance analysis, and Cox proportional hazards survival analysis, complemented by a pathway enrichment analysis. The biological interpretation of these candidates is addressed in Chapter 5.

4.1 Predictive Performances

This section reports the predictive performance of the machine learning pipeline across all experimental configurations described in Section 3.3 and addresses the following questions:

- (a) Did the machine learning pipeline achieve meaningful predictive performance for time-to-conversion prediction?
- (b) Under which cohort, time window, and feature set configuration were the best results obtained?
- (c) Did proteomic features provide a genuine discriminative contribution beyond standard clinical and biochemical measures?

The primary metric used to assess classification performance is the Matthews Correlation Coefficient (MCC), which is robust to class imbalance and provides a single balanced measure of discriminative ability; AUC is reported as a complementary metric. For regression, Root Mean Squared Error (RMSE) is used as the primary metric, as it directly corresponds to the predictability formulation adopted in the RPT- β framework.

This section is organised into three subsections, one per cohort. For the LC and ADNI-A+ cohorts, each subsection is structured around the following analytical paragraphs: **One-vs-All Classification**, reporting performance across the four binary OVA time windows (2-, 3-, 4-, and 5+ year); **Multiclass Classification**, assessing the four-class time window formulation and including a per-class comparison against the OVA; **Regression**, reporting RMSE for the continuous time-to-conversion task; **RF vs SVM**, comparing the two predictive models across all configurations; and **Four-class time windows vs early/mid/late time windows**, comparing the 4-TW formulation against the merged three-class EML-TW scheme.

For the Shared Proteomic Panel (SH40) subsection, a more concise analysis is presented, structured around the **One-vs-All Classification** and **Multiclass Classification** paragraphs only, reporting the performance difference relative to the full proteomic panel separately for each cohort.

4.1.1 LC Cohort

This subsection evaluates the predictive performance of the pipeline on the LC dataset, comparing the various task formulations and the two predictive models (RF and SVM). The analysis also includes a comparison of the informative power of the different feature sets, to assess which combination of features is most predictive of time to conversion, and an assessment of whether merging the intermediate time windows into a single class yields any gain in predictive performance. Complete MCC results are reported in Table 4.1; RMSE results for the regression task are reported in Table 4.3.

One-vs-All Classification. Table 4.1 reports the classification performance for RF and SVM across all feature sets and time windows. The 2-year window was the only one to yield consistently positive and meaningful MCC values for both models. In this window, the `proteomic` and `all_features` feature sets were the only ones to achieve meaningful discriminative performance, with RF reaching MCC of 0.439 ± 0.062 and 0.448 ± 0.072 respectively, compared with only 0.030 ± 0.215 using `clinical` features alone. A similar advantage of proteomic information was observed with SVM. All other time windows showed only negative or near-zero MCC across all feature sets and both models, sustained also by very low sensitivity values. Meaningful discriminative performance was therefore observed only for the 2-year conversion horizon within this cohort.

Multiclass Classification. Table 4.1 reports the multiclass classification performance for RF and SVM across all feature sets. Both models produced near-zero or negative MCC values across most feature sets, indicating performance close to or

below chance. A marginal improvement was observed only under the `all_features` feature set ($\text{MCC}_{\text{RF}}: 0.132 \pm 0.157$, $\text{MCC}_{\text{SVM}}: 0.042 \pm 0.047$), which nonetheless remained considerably below the performance achieved by the 2-year OVA configuration.

To allow a fairer comparison with the OVA formulation, a per-class analysis was performed (Table 4.2). The RF model reveals a consistent pattern across feature sets: the 2-year and 3-year windows show a modest sensitivity increase relative to the corresponding OVA models, accompanied by a drop in specificity. For example, under the `all_features` feature set, the 2-year window shows $\text{Sens}_{\text{OVA}} 0.667 \pm 0.192 \rightarrow \text{Sens}_{\text{Multi}} 0.778 \pm 0.222$, while specificity drops from $\text{Spec}_{\text{OVA}} 0.763 \pm 0.148$ to $\text{Spec}_{\text{Multi}} 0.385 \pm 0.178$. No such effect is observed for the 4-year and 5+ year windows. A possible explanation is that the multiclass formulation redistributes the aggregated negative class of the OVA across the remaining time windows, reducing the dominance of the negative class and making the 2-year window relatively more prevalent, which may favour its correct identification at the cost of increased false positives. For the 3-year window, the same mechanism may allow occasional correct classifications that would otherwise be suppressed by the large aggregated negative class in the OVA setting. This interpretation should be treated with caution, however, as the overall and per-class performance metrics remain consistently low, suggesting that the sensitivity gain reflects a systematic bias introduced by the multiclass structure rather than a genuine capture of proteomic patterns across time windows.

Overall, the multiclass strategy did not yield any meaningful improvement over the OVA formulation for this cohort. The sensitivity gains observed in the per-class analysis are not accompanied by a corresponding improvement in MCC, and are more consistent with a redistribution artefact than with robust discriminative capacity. Furthermore, the multiclass formulation requires the model to identify a consistent proteomic signal across all four time windows simultaneously, a considerably stronger requirement than the window-specific discrimination targeted by the OVA strategy, and one that is unlikely to be met given the severe class imbalance and limited sample size of the LC cohort. For these reasons, the OVA formulation is retained as the primary framework for subsequent analyses.

Regression. Table 4.3 reports the regression performance for both models across all feature sets. RMSE values were consistently high relative to the range of the target variable (approximately 1–6 years), with the best configuration reaching 1.667 ± 0.183 (RF, `proteomic`), and r^2 values (reported in Figure 4.1) near zero across all configurations, indicating that neither model learned a meaningful mapping between proteomic profiles and the continuous time-to-conversion variable. For this reason, subsequent analyses focused exclusively on the classification formulation.

RF vs SVM. Figure 4.2 summarise the comparison between the two models across all configurations. As shown in Table 4.1, the 2-year OVA window, with proteomic and combined feature sets, was the only configuration yielding robust positive MCC in both models, with RF showing a slight advantage in terms of MCC and lower variance, particularly under the `all_features` feature set ($\text{MCC}_{\text{RF}}: 0.448 \pm 0.072$ vs $\text{MCC}_{\text{SVM}}: 0.368 \pm 0.191$). For all other windows and the multiclass task, neither model produced meaningful predictions.

A notable pattern emerges when comparing AUC and MCC jointly: SVM achieved higher AUC than RF in several configurations where MCC remained near zero or negative. Inspection of the corresponding sensitivity and specificity values reveals that these AUC gains are driven by a tendency of SVM to predict a larger number of positive cases, occasionally recovering some true positives from configurations where RF predicts exclusively negative, but at the cost of a substantially higher false positive rate. This behaviour is particularly evident for the 4-year window with `clinical` features, where SVM achieves a higher AUC ($\Delta_{\text{RF-SVM}} = -0.480$) at the cost of substantially lower specificity (RF: 0.78 ± 0.09 vs SVM: 0.51 ± 0.42), while sensitivity, although slightly higher for SVM, remains poor for both models. Since AUC measures the ability of the model to rank positive cases above negative ones across all possible thresholds, a more permissive prediction strategy that assigns higher scores to positive cases, even at the cost of many false positives, can inflate AUC without reflecting robust discriminative capacity. This pattern suggests that SVM’s AUC gains do not reflect an effective capture of discriminative proteomic signal, but rather a more permissive prediction strategy. RF was therefore selected as the primary model for subsequent biomarker analyses: when the proteomic signal is insufficient to support a positive classification, RF tends to predict the negative class, providing a more meaningful estimate of true discriminative capacity.

Four-class time windows vs early/mid/late time windows. Figure 4.4 summarise the comparison between the EML-TW and 4-TW formulations for RF and SVM respectively, showing MCC and AUC differences for the mid window and the multiclass task. Neither model showed a consistent improvement under the merged formulation. For RF, MCC was lower under EML-TW for the mid window across all feature sets, with the largest drop observed for `proteomic` ($\Delta = -0.219$), while AUC showed a mixed picture with marginal gains for `proteomic` and `all_features` that did not translate into better MCC. For SVM, the picture was broadly similar, with EML-TW outperforming 4-TW only for the `clinical` feature set in the mid window ($\Delta = +0.165$), while all other feature sets favoured the four-class formulation.

The expected gain from better class balance in the mid window was not realised in terms of MCC, suggesting that the fusion of the 3-year and 4-year windows did

not provide either model with a more learnable target. For this reason, subsequent analyses were conducted using the 4-TW formulation, with focus on the 2-year OVA window as the only configuration yielding meaningful predictive performance within this cohort.

Table 4.1: Classification performance (mean $\pm$ SD across 3 outer folds) for RF and SVM on the LC cohort. Bold underlined values indicate the best-performing feature set per task, model and metric.

Model	Task	Feature set	MCC	AUC	Sensitivity	Specificity
RF	Multiclass	clinical	-0.131 ± 0.052	0.368 ± 0.050	0.148 ± 0.042	0.721 ± 0.012
		proteomic	0.035 ± 0.156	0.495 ± 0.045	0.250 ± 0.062	0.760 ± 0.039
		c-CSF	0.062 ± 0.140	0.531 ± 0.070	0.287 ± 0.094	0.766 ± 0.029
		all_features	0.132 ± 0.157	0.502 ± 0.048	0.308 ± 0.034	0.779 ± 0.042
	2-year	clinical	0.030 ± 0.215	0.466 ± 0.102	0.481 ± 0.128	0.548 ± 0.134
		proteomic	0.439 ± 0.062	0.728 ± 0.128	0.630 ± 0.128	0.796 ± 0.094
		c-CSF	-0.109 ± 0.139	0.509 ± 0.063	0.444 ± 0.111	0.448 ± 0.150
		all_features	0.448 ± 0.072	0.732 ± 0.067	0.667 ± 0.192	0.763 ± 0.148
	3-year	clinical	-0.199 ± 0.170	0.342 ± 0.137	0.283 ± 0.301	0.487 ± 0.114
		proteomic	0.000 ± 0.000	0.467 ± 0.108	0.000 ± 0.000	1.000 ± 0.000
		c-CSF	-0.022 ± 0.179	0.534 ± 0.181	0.067 ± 0.115	0.930 ± 0.003
		all_features	-0.059 ± 0.102	0.467 ± 0.075	0.000 ± 0.000	0.956 ± 0.077
	4-year	clinical	-0.182 ± 0.036	0.208 ± 0.073	0.000 ± 0.000	0.776 ± 0.094
		proteomic	0.000 ± 0.000	0.277 ± 0.062	0.000 ± 0.000	1.000 ± 0.000
		c-CSF	0.016 ± 0.265	0.511 ± 0.163	0.111 ± 0.192	0.879 ± 0.057
		all_features	0.000 ± 0.000	0.224 ± 0.094	0.000 ± 0.000	1.000 ± 0.000
	5+ year	clinical	-0.039 ± 0.095	0.471 ± 0.223	0.222 ± 0.192	0.749 ± 0.185
		proteomic	0.000 ± 0.000	0.386 ± 0.017	0.000 ± 0.000	1.000 ± 0.000
		c-CSF	-0.136 ± 0.030	0.361 ± 0.121	0.000 ± 0.000	0.893 ± 0.039
		all_features	0.147 ± 0.347	0.356 ± 0.108	0.111 ± 0.192	0.979 ± 0.036
SVM	Multiclass	clinical	-0.038 ± 0.037	0.510 ± 0.059	0.214 ± 0.034	0.741 ± 0.010
		proteomic	0.031 ± 0.054	0.529 ± 0.046	0.244 ± 0.032	0.763 ± 0.018
		c-CSF	-0.040 ± 0.055	0.530 ± 0.017	0.222 ± 0.024	0.741 ± 0.018
		all_features	0.042 ± 0.047	0.527 ± 0.081	0.253 ± 0.018	0.766 ± 0.017
	2-year	clinical	-0.019 ± 0.032	0.354 ± 0.068	0.370 ± 0.339	0.611 ± 0.347
		proteomic	0.437 ± 0.160	0.723 ± 0.149	0.593 ± 0.170	0.830 ± 0.051
		c-CSF	0.075 ± 0.129	0.547 ± 0.060	0.185 ± 0.321	0.889 ± 0.192
		all_features	0.368 ± 0.191	0.667 ± 0.089	0.667 ± 0.222	0.693 ± 0.089
	3-year	clinical	-0.215 ± 0.148	0.447 ± 0.170	0.217 ± 0.202	0.535 ± 0.036
		proteomic	-0.031 ± 0.202	0.524 ± 0.165	0.317 ± 0.161	0.651 ± 0.073
		c-CSF	-0.131 ± 0.115	0.517 ± 0.083	0.133 ± 0.231	0.741 ± 0.339
		all_features	0.073 ± 0.047	0.478 ± 0.096	0.400 ± 0.173	0.678 ± 0.165
	4-year	clinical	-0.212 ± 0.259	0.688 ± 0.263	0.167 ± 0.289	0.515 ± 0.421
		proteomic	-0.059 ± 0.102	0.456 ± 0.291	0.000 ± 0.000	0.922 ± 0.136
		c-CSF	0.037 ± 0.263	0.274 ± 0.022	0.333 ± 0.577	0.735 ± 0.251
		all_features	0.000 ± 0.000	0.434 ± 0.317	0.000 ± 0.000	1.000 ± 0.000
	5+ year	clinical	-0.106 ± 0.226	0.622 ± 0.160	0.222 ± 0.385	0.665 ± 0.234
		proteomic	-0.166 ± 0.030	0.480 ± 0.129	0.111 ± 0.192	0.717 ± 0.274
		c-CSF	-0.109 ± 0.112	0.499 ± 0.161	0.000 ± 0.000	0.896 ± 0.130
		all_features	-0.109 ± 0.210	0.464 ± 0.152	0.111 ± 0.192	0.767 ± 0.069

Table 4.2: Per-class performance comparison between the OVA and multiclass formulations on the LC cohort. **(a)** Random Forest model. **(b)** Support Vector Machine model. For the OVA columns, values are taken directly from the outer fold evaluation of the corresponding binary classifier. For the multiclass columns, per-class metrics are derived by binarising the multiclass predictions in a one-vs-rest fashion for each time window: sensitivity and specificity are computed from the resulting binary confusion matrix, while AUC is computed using the predicted class probability for that window against the remaining classes. Values are mean $\pm$ standard deviation across 3 outer folds. **Bold** values indicate the higher mean between OVA and multiclass for that window, feature set, and metric.

Feature set	Window	Sensitivity			Specificity			AUC			MCC		
		OVA	Multi		OVA	Multi		OVA	Multi		OVA	Multi	
clinical	2-year	0.481 $\pm$ 0.128	0.370 $\pm$ 0.231	0.548 $\pm$ 0.134	0.581 $\pm$ 0.277	0.466 $\pm$ 0.102	0.454 $\pm$ 0.083	0.030 $\pm$ 0.215	-0.046 $\pm$ 0.065				
	3-year	0.283 $\pm$ 0.301	0.000 $\pm$ 0.000	0.487 $\pm$ 0.114	0.675 $\pm$ 0.036	0.342 $\pm$ 0.137	0.236 $\pm$ 0.038	-0.199 $\pm$ 0.170	-0.317 $\pm$ 0.036				
	4-year	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.776 $\pm$ 0.094	0.124 $\pm$ 0.253	0.208 $\pm$ 0.073	0.239 $\pm$ 0.253	-0.182 $\pm$ 0.036	-0.112 $\pm$ 0.225				
	5+ year	0.222 $\pm$ 0.192	0.222 $\pm$ 0.385	0.749 $\pm$ 0.185	0.768 $\pm$ 0.127	0.471 $\pm$ 0.223	0.544 $\pm$ 0.120	-0.039 $\pm$ 0.095	-0.044 $\pm$ 0.225				
c-CSF	2-year	0.444 $\pm$ 0.111	0.667 $\pm$ 0.193	0.448 $\pm$ 0.150	0.415 $\pm$ 0.202	0.509 $\pm$ 0.063	0.562 $\pm$ 0.083	-0.109 $\pm$ 0.139	0.085 $\pm$ 0.136				
	3-year	0.067 $\pm$ 0.115	0.150 $\pm$ 0.132	0.930 $\pm$ 0.003	0.930 $\pm$ 0.072	0.534 $\pm$ 0.181	0.554 $\pm$ 0.058	-0.022 $\pm$ 0.179	0.131 $\pm$ 0.258				
	4-year	0.111 $\pm$ 0.192	0.333 $\pm$ 0.289	0.879 $\pm$ 0.057	0.782 $\pm$ 0.276	0.511 $\pm$ 0.163	0.678 $\pm$ 0.244	0.016 $\pm$ 0.265	0.099 $\pm$ 0.188				
	5+ year	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.893 $\pm$ 0.039	0.937 $\pm$ 0.109	0.361 $\pm$ 0.121	0.331 $\pm$ 0.065	-0.136 $\pm$ 0.030	-0.063 $\pm$ 0.109				
proteomic	2-year	0.630 $\pm$ 0.128	0.741 $\pm$ 0.231	0.796 $\pm$ 0.094	0.352 $\pm$ 0.184	0.728 $\pm$ 0.128	0.705 $\pm$ 0.178	0.439 $\pm$ 0.062	0.124 $\pm$ 0.348				
	3-year	0.000 $\pm$ 0.000	0.150 $\pm$ 0.132	1.000 $\pm$ 0.000	0.862 $\pm$ 0.120	0.467 $\pm$ 0.108	0.539 $\pm$ 0.052	0.000 $\pm$ 0.000	0.072 $\pm$ 0.347				
	4-year	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	0.277 $\pm$ 0.062	0.317 $\pm$ 0.108	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000				
	5+ year	0.000 $\pm$ 0.000	0.111 $\pm$ 0.192	1.000 $\pm$ 0.000	0.825 $\pm$ 0.205	0.386 $\pm$ 0.017	0.420 $\pm$ 0.089	0.000 $\pm$ 0.000	-0.067 $\pm$ 0.076				
all_features	2-year	0.667 $\pm$ 0.192	0.778 $\pm$ 0.222	0.763 $\pm$ 0.148	0.385 $\pm$ 0.178	0.732 $\pm$ 0.067	0.676 $\pm$ 0.127	0.448 $\pm$ 0.072	0.190 $\pm$ 0.399				
	3-year	0.000 $\pm$ 0.000	0.233 $\pm$ 0.029	0.956 $\pm$ 0.077	0.862 $\pm$ 0.120	0.467 $\pm$ 0.075	0.537 $\pm$ 0.138	-0.059 $\pm$ 0.102	0.163 $\pm$ 0.254				
	4-year	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	0.224 $\pm$ 0.094	0.322 $\pm$ 0.179	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000				
	5+ year	0.111 $\pm$ 0.192	0.222 $\pm$ 0.192	0.979 $\pm$ 0.036	0.868 $\pm$ 0.177	0.356 $\pm$ 0.108	0.473 $\pm$ 0.092	0.147 $\pm$ 0.347	0.107 $\pm$ 0.186				
(a) RF model													
Feature set	Window	Sensitivity			Specificity			AUC			MCC		
		OVA	Multi		OVA	Multi		OVA	Multi		OVA	Multi	
clinical	2-year	0.370 $\pm$ 0.339	0.111 $\pm$ 0.192	0.611 $\pm$ 0.347	0.852 $\pm$ 0.256	0.354 $\pm$ 0.068	0.491 $\pm$ 0.095	-0.019 $\pm$ 0.032	-0.038 $\pm$ 0.066				
	3-year	0.217 $\pm$ 0.202	0.133 $\pm$ 0.231	0.535 $\pm$ 0.036	0.905 $\pm$ 0.165	0.447 $\pm$ 0.170	0.524 $\pm$ 0.206	-0.215 $\pm$ 0.148	0.036 $\pm$ 0.062				
	4-year	0.167 $\pm$ 0.289	0.167 $\pm$ 0.289	0.515 $\pm$ 0.421	0.775 $\pm$ 0.254	0.688 $\pm$ 0.263	0.417 $\pm$ 0.085	-0.212 $\pm$ 0.259	-0.050 $\pm$ 0.086				
	5+ year	0.222 $\pm$ 0.385	0.444 $\pm$ 0.509	0.665 $\pm$ 0.234	0.435 $\pm$ 0.434	0.622 $\pm$ 0.160	0.608 $\pm$ 0.070	-0.106 $\pm$ 0.226	-0.108 $\pm$ 0.094				
c-CSF	2-year	0.185 $\pm$ 0.321	0.222 $\pm$ 0.192	0.889 $\pm$ 0.192	0.748 $\pm$ 0.282	0.547 $\pm$ 0.060	0.615 $\pm$ 0.014	0.075 $\pm$ 0.129	-0.024 $\pm$ 0.189				
	3-year	0.133 $\pm$ 0.231	0.167 $\pm$ 0.289	0.741 $\pm$ 0.339	0.810 $\pm$ 0.271	0.517 $\pm$ 0.083	0.554 $\pm$ 0.034	-0.131 $\pm$ 0.115	-0.047 $\pm$ 0.081				
	4-year	0.333 $\pm$ 0.577	0.167 $\pm$ 0.289	0.735 $\pm$ 0.251	0.822 $\pm$ 0.256	0.274 $\pm$ 0.022	0.300 $\pm$ 0.129	0.037 $\pm$ 0.263	-0.023 $\pm$ 0.056				
	5+ year	0.000 $\pm$ 0.000	0.333 $\pm$ 0.577	0.896 $\pm$ 0.130	0.583 $\pm$ 0.520	0.499 $\pm$ 0.161	0.649 $\pm$ 0.171	-0.109 $\pm$ 0.112	-0.075 $\pm$ 0.129				
proteomic	2-year	0.593 $\pm$ 0.170	0.296 $\pm$ 0.280	0.830 $\pm$ 0.051	0.859 $\pm$ 0.122	0.723 $\pm$ 0.149	0.488 $\pm$ 0.230	0.437 $\pm$ 0.160	0.164 $\pm$ 0.187				
	3-year	0.317 $\pm$ 0.161	0.233 $\pm$ 0.252	0.651 $\pm$ 0.073	0.786 $\pm$ 0.189	0.524 $\pm$ 0.165	0.556 $\pm$ 0.099	-0.031 $\pm$ 0.202	0.013 $\pm$ 0.169				
	4-year	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.922 $\pm$ 0.136	0.980 $\pm$ 0.034	0.456 $\pm$ 0.291	0.622 $\pm$ 0.138	-0.059 $\pm$ 0.102	-0.027 $\pm$ 0.047				
	5+ year	0.111 $\pm$ 0.192	0.444 $\pm$ 0.509	0.717 $\pm$ 0.274	0.428 $\pm$ 0.386	0.480 $\pm$ 0.129	0.453 $\pm$ 0.173	-0.166 $\pm$ 0.030	-0.096 $\pm$ 0.229				
all_features	2-year	0.667 $\pm$ 0.222	0.333 $\pm$ 0.294	0.693 $\pm$ 0.089	0.822 $\pm$ 0.168	0.667 $\pm$ 0.089	0.474 $\pm$ 0.249	0.368 $\pm$ 0.191	0.161 $\pm$ 0.188				
	3-year	0.400 $\pm$ 0.173	0.233 $\pm$ 0.252	0.678 $\pm$ 0.165	0.810 $\pm$ 0.180	0.478 $\pm$ 0.096	0.583 $\pm$ 0.176	0.073 $\pm$ 0.047	0.039 $\pm$ 0.210				
	4-year	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	1.000 $\pm$ 0.000	0.980 $\pm$ 0.034	0.434 $\pm$ 0.317	0.581 $\pm$ 0.134	0.000 $\pm$ 0.000	-0.027 $\pm$ 0.047				
	5+ year	0.111 $\pm$ 0.192	0.444 $\pm$ 0.509	0.767 $\pm$ 0.069	0.450 $\pm$ 0.397	0.464 $\pm$ 0.152	0.468 $\pm$ 0.136	-0.109 $\pm$ 0.210	-0.082 $\pm$ 0.205				
(b) SVM model													

Table 4.3: RMSE (mean $\pm$ SD across 3 outer folds) for the regression task on the LC cohort. Lower values are better. Bold values indicate the best-performing feature set per model.

Model	Clinical (RMSE)	Proteomic (RMSE)	c-CSF (RMSE)	all_features (RMSE)
RF (regression)	1.913 $\pm$ 0.324	1.667 $\pm$ 0.183	1.875 $\pm$ 0.033	1.717 $\pm$ 0.085
SVM (regression)	1.884 $\pm$ 0.329	1.754 $\pm$ 0.284	1.849 $\pm$ 0.042	1.682 $\pm$ 0.231

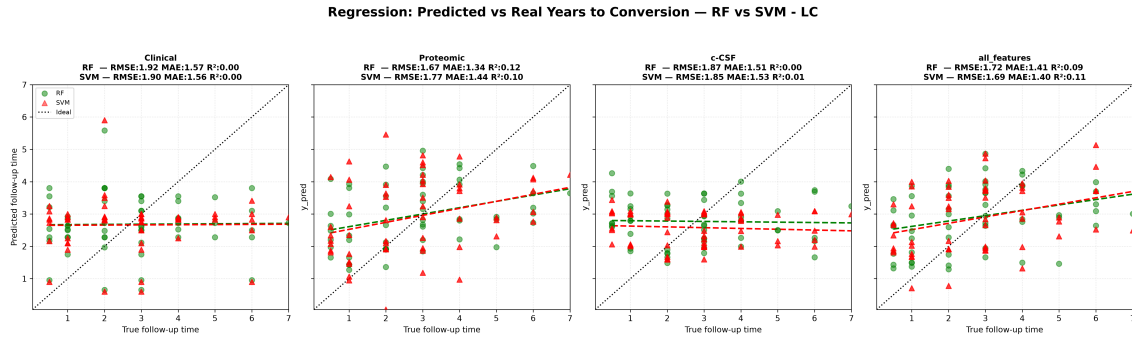


Figure 4.1: Predicted vs. true time-to-conversion (years) for the regression task on the LC cohort, comparing RF (green circles) and SVM (red triangles) across all four feature sets. RMSE, MAE and r^2 are reported per model and feature set. The predicted values are nearly flat across all feature sets except proteomic and all_features, where a slight positive slope is visible; however, r^2 values remain close to zero in all configurations.

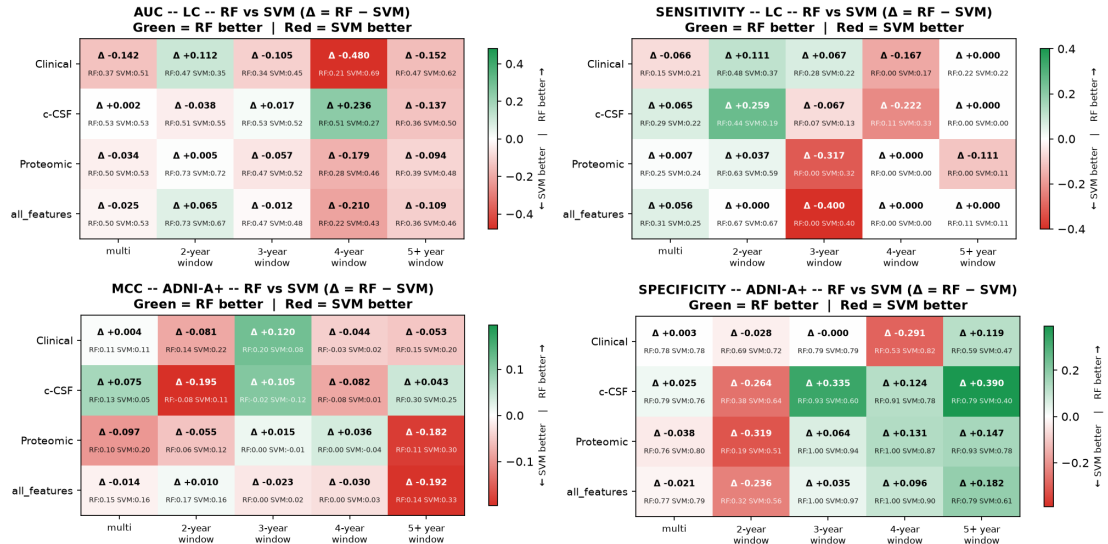


Figure 4.2: RF vs SVM comparison on the LC cohort across all task formulations and feature sets, showing MCC, AUC, Sensitivity and Specificity ($\Delta = \text{RF} - \text{SVM}$). Green cells indicate configurations where RF outperforms SVM; red cells indicate where SVM outperforms RF.

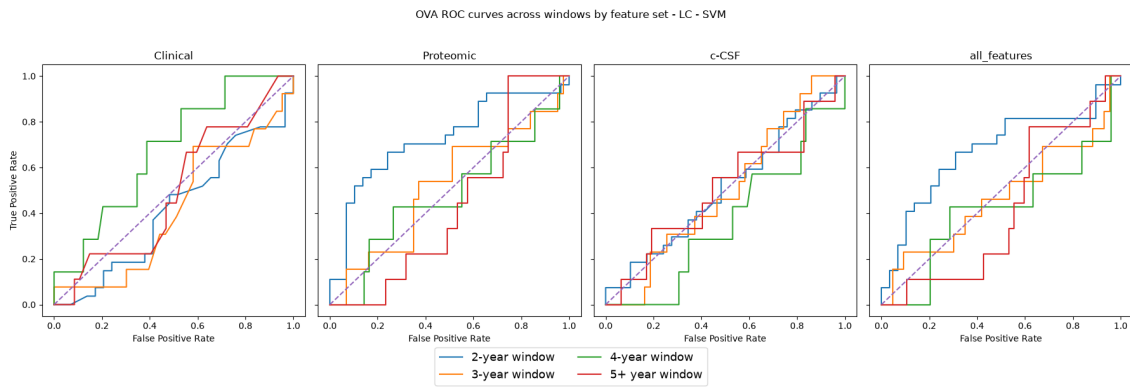
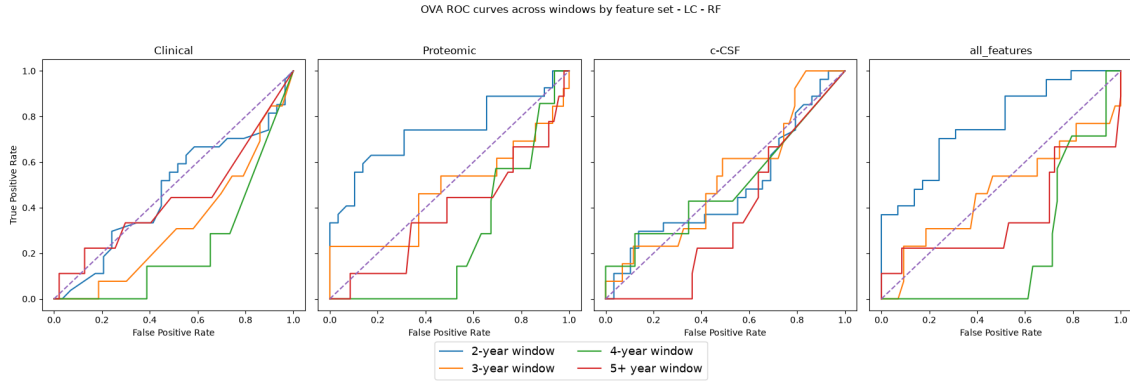
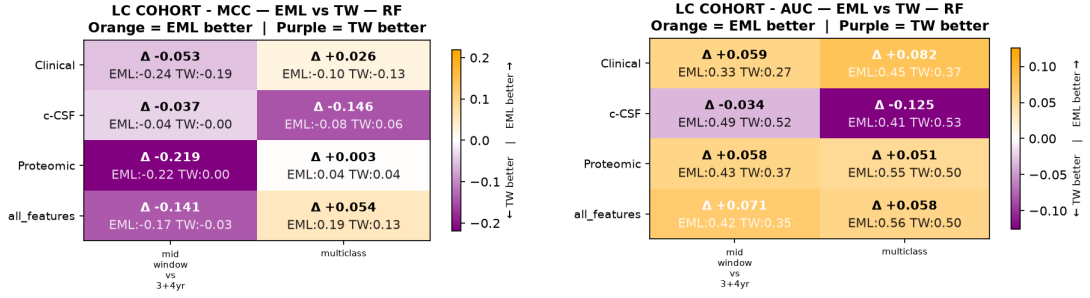
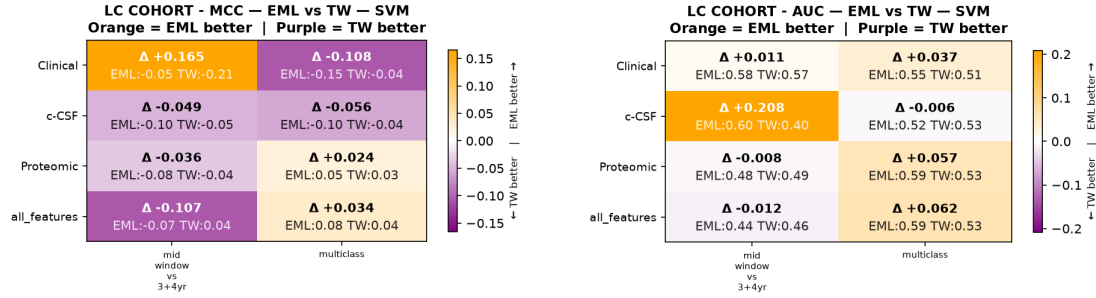


Figure 4.3: OVA ROC curves for the LC cohort across all four feature sets (clinical, proteomic, c-CSF, all_features) and all four time windows (2-year in blue, 3-year in yellow, 4-year in green, 5+ year in red). **(a)** Random Forest model: the 2-year window shows the clearest separation from the diagonal in the proteomic and all_features feature sets, reflecting the strongest gain in performance after the addition of the proteomic feature set. **(b)** SVM model.



(a) RF model.



(b) SVM model.

Figure 4.4: MCC (left) and AUC (right) comparison between the EML-TW (early/mid/late) and 4-TW (four-class time window) formulations on the LC cohort. (a) Random Forest model. (b) SVM model. Orange cells indicate configurations where EML-TW outperforms 4-TW ($\Delta > 0$); purple cells indicate where 4-TW outperforms EML-TW ($\Delta < 0$). For the mid window comparison, the EML-TW mid class performance was compared against the mean of the 3-year and 4-year OVA performances under the 4-TW formulation. Only the mid window and the multiclass task are shown, as the early and late windows are identical under both formulations by construction. The complete performance panel is reported in Appendix (Table 8.7).

4.1.2 ADNI-A+ Cohort

This subsection repeats the analysis performed on the LC cohort on the larger ADNI-A+ dataset, with the aim of assessing whether the patterns observed in LC generalise to an independent cohort and whether a larger sample size translates into improved predictive performance. Complete MCC results are reported in Table 4.4; RMSE results for the regression task are reported in Table 4.5.

One-vs-All Classification. Table 4.4 reports the OVA classification performance for the ADNI-A+ cohort. Unlike the LC cohort, no single time window yielded consistently strong performance across feature sets and models. The 2-year and intermediate windows (3-year, 4-year) remained at low MCC values across all configurations, consistent with the severe class imbalance of these windows: the 3-year and 4-year groups contain only 16 and 11 patients respectively out of 187, a more

extreme imbalance than in LC despite the larger overall sample size.

The most notable finding was the 5+ year window, which showed a meaningful improvement relative to all other windows: RF achieved MCC 0.298 ± 0.064 with `c-CSF` features, and SVM 0.328 ± 0.018 with `all_features`, a pattern not observed in LC where the 5+ year window consistently produced near-zero MCC. Unlike LC, where proteomic features drove the best performance, in ADNI-A+ the most informative contribution came from `c-CSF` features. The `proteomic` feature set alone showed a reasonable AUC under SVM (0.706 ± 0.017), with high specificity (0.782 ± 0.046) but modest sensitivity (0.533 ± 0.133), suggesting that proteomic features contribute primarily to the correct rejection of negative samples. By contrast, `c-CSF` features drove a peak in sensitivity (0.867 ± 0.115) at the cost of lower specificity (0.400 ± 0.245). The `all_features` feature set achieved the most balanced result under SVM (MCC 0.328 ± 0.018 , sensitivity 0.756 ± 0.154 , specificity 0.612 ± 0.180), benefiting from the complementary contributions of the two modalities. Notably, this complementary effect was more pronounced for SVM than for RF, where the addition of `c-CSF` features did not yield a comparable gain in sensitivity. For this reason, both the feature selection analysis and the biomarker evaluation presented in Sections 4.2 and 4.3 will focus on the `all_features` feature set for the 5+ year window using SVM as the primary configuration for the ADNI-A+ cohort.

Multiclass Classification. Table 4.4 reports the multiclass classification performance for the ADNI-A+ cohort. Unlike LC, both models produced low but consistently positive MCC values across all feature sets, with RF achieving its best performance under `all_features` (MCC 0.147 ± 0.087) and SVM under `proteomic` (MCC 0.198 ± 0.040). This marginal improvement over the LC multiclass results is consistent with the larger sample size of ADNI-A+ ($n = 187$), which provides slightly greater statistical power; however, performance remains too low to support meaningful pattern analysis.

The per-class analysis (Table 4.6) does not reveal the same pattern observed for LC. Notably, the small improvement in overall MCC of the multiclass relative to LC is reflected in the per-class results: unlike LC, where the 5+ year window showed near-zero sensitivity in almost all configurations, in ADNI-A+ the multiclass model maintains a poor but slightly higher sensitivity for this window, ranging approximately between 0.2 and 0.5, particularly under `all_features` for RF and `proteomic` for SVM, the configurations showing the highest overall multiclass MCC, likely benefiting from the discriminative signal carried by a bigger 5+ year window sample size in this cohort. For `clinical` and `c-CSF` feature sets, however, the multi-class formulation leads to an overall reduction in sensitivity across most time windows relative to the OVA. For `proteomic` and `all_features` feature sets the picture is

more mixed, without a consistent directional advantage for either formulation. The OVA formulation is therefore retained as the primary framework, as it yields more focused and clinically relevant results for the most informative time window.

Regression. The regression performance for the ADNI-A+ cohort are reported in Table 4.5. RMSE values were consistently high relative to the range of the target variable (approximately 1–8 years), with the best configuration reaching 2.967 ± 0.446 (SVM, `all_features`), indicating that both models perform close to a naive mean predictor. Consistent with the LC findings, the regression approach was considered underpowered also on the larger ADNI-A+ dataset and was not pursued further.

RF vs SVM. The relative comparison between the two models on the ADNI-A+ cohort is summarised in Figure 4.6. In general, the two models perform differently depending on the metric considered. Overall, SVM shows a slight advantage over RF in terms of MCC, while RF shows a slight advantage in terms of AUC. As observed in LC, RF tends to achieve higher specificity than SVM across most configurations, while SVM achieves higher sensitivity, confirming RF’s more conservative prediction behaviour.

However, unlike LC, in ADNI-A+ the sensitivity gains of SVM are notably more meaningful, particularly for the 5+ year window. Under `c-CSF` features, SVM achieves sensitivity 0.867 ± 0.115 with specificity 0.400 ± 0.245 , and under `all_features` sensitivity 0.756 ± 0.154 with specificity 0.612 ± 0.180 , compared to substantially lower sensitivity for RF. In contrast to LC, where SVM’s sensitivity gains were driven by a higher false positive rate rather than genuine pattern recognition, in ADNI-A+ the more balanced sensitivity-specificity profile of SVM for the 5+ year window suggests that the model is capturing a robust discriminative signal for this conversion horizon.

Four-class time windows vs early/mid/late time windows. The comparison between the EML-TW and 4-TW formulations for RF and SVM on the ADNI-A+ cohort is summarised in Figure 4.8, showing MCC and AUC differences for the mid window and the multiclass task.

As observed for LC, neither formulation yields a consistent advantage over the other. For RF, the mid window MCC was predominantly lower under EML-TW across most feature sets, with the largest drop for `clinical` ($\Delta_{\text{MCC}} = -0.133$), and differences in the multiclass task were negligible across all feature sets. For SVM, no relevant differences were observed in the mid window, with only a marginal gain under EML-TW for the multiclass task using `c-CSF` features ($\Delta_{\text{MCC}} = +0.129$). Given the absence of any consistent improvement under the merged formulation,

the 4-TW scheme was retained as the primary framework for subsequent analyses, consistent with the decision made for the LC cohort.

Table 4.4: Classification performance (mean $\pm$ SD across 3 outer folds) for RF and SVM on the ADNI-A+ cohort. Bold underlined values indicate the best-performing feature set per task, model and metric.

Model	Task	Feature set	MCC	AUC	Sensitivity	Specificity
RF	Multiclass	clinical	0.111 $\pm$ 0.072	0.620 $\pm$ 0.029	0.306 $\pm$ 0.098	0.782 $\pm$ 0.021
		proteomic	0.101 $\pm$ 0.123	0.577 $\pm$ 0.063	0.264 $\pm$ 0.017	0.759 $\pm$ 0.011
		c-CSF	0.125 $\pm$ 0.127	0.504 $\pm$ 0.031	0.273 $\pm$ 0.061	0.785 $\pm$ 0.037
		all_features	0.147 $\pm$ 0.087	0.612 $\pm$ 0.079	0.281 $\pm$ 0.020	0.772 $\pm$ 0.018
	2-year	clinical	0.142 $\pm$ 0.043	0.650 $\pm$ 0.030	0.444 $\pm$ 0.100	0.694 $\pm$ 0.120
		proteomic	0.060 $\pm$ 0.162	0.614 $\pm$ 0.110	0.852 $\pm$ 0.081	0.194 $\pm$ 0.096
		c-CSF	-0.081 $\pm$ 0.174	0.476 $\pm$ 0.087	0.541 $\pm$ 0.176	0.375 $\pm$ 0.000
		all_features	0.170 $\pm$ 0.143	0.607 $\pm$ 0.059	0.825 $\pm$ 0.062	0.319 $\pm$ 0.064
	3-year	clinical	0.199 $\pm$ 0.204	0.531 $\pm$ 0.067	0.367 $\pm$ 0.153	0.788 $\pm$ 0.206
		proteomic	0.000 $\pm$ 0.000	0.652 $\pm$ 0.141	0.000 $\pm$ 0.000	1.000 $\pm$ 0.000
		c-CSF	-0.020 $\pm$ 0.077	0.414 $\pm$ 0.050	0.067 $\pm$ 0.115	0.930 $\pm$ 0.052
		all_features	0.000 $\pm$ 0.000	0.604 $\pm$ 0.194	0.000 $\pm$ 0.000	1.000 $\pm$ 0.000
	4-year	clinical	-0.026 $\pm$ 0.083	0.546 $\pm$ 0.055	0.417 $\pm$ 0.382	0.534 $\pm$ 0.225
		proteomic	0.000 $\pm$ 0.000	0.387 $\pm$ 0.137	0.000 $\pm$ 0.000	1.000 $\pm$ 0.000
		c-CSF	-0.076 $\pm$ 0.015	0.403 $\pm$ 0.181	0.000 $\pm$ 0.000	0.909 $\pm$ 0.027
		all_features	0.000 $\pm$ 0.000	0.427 $\pm$ 0.135	0.000 $\pm$ 0.000	1.000 $\pm$ 0.000
	5+ year	clinical	0.148 $\pm$ 0.031	0.686 $\pm$ 0.026	0.578 $\pm$ 0.102	0.591 $\pm$ 0.131
		proteomic	0.113 $\pm$ 0.170	0.631 $\pm$ 0.082	0.133 $\pm$ 0.067	0.929 $\pm$ 0.054
		c-CSF	0.298 $\pm$ 0.064	0.683 $\pm$ 0.038	0.511 $\pm$ 0.154	0.790 $\pm$ 0.124
		all_features	0.136 $\pm$ 0.097	0.695 $\pm$ 0.064	0.311 $\pm$ 0.252	0.794 $\pm$ 0.205
SVM	Multiclass	clinical	0.107 $\pm$ 0.097	0.620 $\pm$ 0.054	0.260 $\pm$ 0.062	0.779 $\pm$ 0.024
		proteomic	0.198 $\pm$ 0.040	0.557 $\pm$ 0.062	0.343 $\pm$ 0.047	0.797 $\pm$ 0.011
		c-CSF	0.051 $\pm$ 0.047	0.540 $\pm$ 0.083	0.228 $\pm$ 0.026	0.760 $\pm$ 0.018
		all_features	0.160 $\pm$ 0.079	0.573 $\pm$ 0.066	0.326 $\pm$ 0.088	0.794 $\pm$ 0.018
	2-year	clinical	0.224 $\pm$ 0.135	0.646 $\pm$ 0.079	0.503 $\pm$ 0.151	0.722 $\pm$ 0.087
		proteomic	0.115 $\pm$ 0.080	0.556 $\pm$ 0.050	0.601 $\pm$ 0.142	0.514 $\pm$ 0.168
		c-CSF	0.115 $\pm$ 0.026	0.604 $\pm$ 0.014	0.478 $\pm$ 0.059	0.639 $\pm$ 0.048
		all_features	0.160 $\pm$ 0.161	0.587 $\pm$ 0.061	0.609 $\pm$ 0.021	0.556 $\pm$ 0.168
	3-year	clinical	0.079 $\pm$ 0.170	0.350 $\pm$ 0.150	0.344 $\pm$ 0.435	0.789 $\pm$ 0.192
		proteomic	-0.015 $\pm$ 0.070	0.565 $\pm$ 0.126	0.067 $\pm$ 0.115	0.936 $\pm$ 0.052
		c-CSF	-0.125 $\pm$ 0.062	0.536 $\pm$ 0.195	0.189 $\pm$ 0.019	0.595 $\pm$ 0.097
		all_features	0.023 $\pm$ 0.122	0.508 $\pm$ 0.054	0.067 $\pm$ 0.115	0.965 $\pm$ 0.017
	4-year	clinical	0.018 $\pm$ 0.055	0.514 $\pm$ 0.048	0.194 $\pm$ 0.173	0.825 $\pm$ 0.198
		proteomic	-0.036 $\pm$ 0.045	0.373 $\pm$ 0.158	0.083 $\pm$ 0.144	0.869 $\pm$ 0.146
		c-CSF	0.006 $\pm$ 0.117	0.564 $\pm$ 0.236	0.250 $\pm$ 0.433	0.784 $\pm$ 0.251
		all_features	0.030 $\pm$ 0.083	0.367 $\pm$ 0.151	0.167 $\pm$ 0.289	0.904 $\pm$ 0.152
	5+ year	clinical	0.201 $\pm$ 0.099	0.547 $\pm$ 0.205	0.756 $\pm$ 0.154	0.471 $\pm$ 0.069
		proteomic	0.296 $\pm$ 0.076	0.706 $\pm$ 0.017	0.533 $\pm$ 0.133	0.782 $\pm$ 0.046
		c-CSF	0.254 $\pm$ 0.082	0.599 $\pm$ 0.213	0.867 $\pm$ 0.115	0.400 $\pm$ 0.245
		all_features	0.328 $\pm$ 0.018	0.585 $\pm$ 0.230	0.756 $\pm$ 0.154	0.612 $\pm$ 0.180

Table 4.5: RMSE (mean $\pm$ SD across 3 outer folds) for the regression task on the ADNI-A+ cohort. Lower values are better. Bold underlined values indicate the best-performing feature set per model.

Model	Clinical (RMSE)	Proteomic (RMSE)	c-CSF (RMSE)	all_features (RMSE)
RF	2.994 ± 0.126	3.119 ± 0.254	3.335 ± 0.164	<u>2.992 ± 0.239</u>
SVM	3.096 ± 0.273	3.144 ± 0.230	3.151 ± 0.228	<u>2.967 ± 0.446</u>

Table 4.6: Per-class performance comparison between the OVA and multiclass formulations on the ADNI-A+ cohort. (a) Random Forest model. (b) Support Vector Machine model. For the OVA columns, values are taken directly from the outer fold evaluation of the corresponding binary classifier. For the multiclass columns, per-class metrics are derived by binarising the multiclass predictions in a one-vs-rest fashion for each time window: sensitivity and specificity are computed from the resulting binary confusion matrix, while AUC is computed using the predicted class probability for that window against the remaining classes. Values are mean $\pm$ standard deviation across 3 outer folds. **Bold** values indicate the higher mean between OVA and multiclass for that window, feature set, and metric.

Feature set	Window	Sensitivity			Specificity			AUC			MCC		
		OVA	Multi	Multi	OVA	Multi	Multi	OVA	Multi	Multi	OVA	Multi	Multi
clinical	2-year	0.444 $\pm$ 0.100	0.235 $\pm$ 0.048	0.235 $\pm$ 0.048	0.694 $\pm$ 0.120	0.889 $\pm$ 0.048	0.889 $\pm$ 0.048	0.650 $\pm$ 0.030	0.602 $\pm$ 0.019	0.602 $\pm$ 0.019	0.142 $\pm$ 0.043	0.153 $\pm$ 0.119	0.153 $\pm$ 0.119
	3-year	0.367 $\pm$ 0.153	0.311 $\pm$ 0.102	0.311 $\pm$ 0.102	0.788 $\pm$ 0.206	0.814 $\pm$ 0.069	0.814 $\pm$ 0.069	0.531 $\pm$ 0.067	0.595 $\pm$ 0.128	0.595 $\pm$ 0.128	0.199 $\pm$ 0.204	0.097 $\pm$ 0.108	0.097 $\pm$ 0.108
	4-year	0.417 $\pm$ 0.382	0.167 $\pm$ 0.289	0.167 $\pm$ 0.289	0.534 $\pm$ 0.225	0.750 $\pm$ 0.048	0.750 $\pm$ 0.048	0.546 $\pm$ 0.055	0.580 $\pm$ 0.079	0.580 $\pm$ 0.079	-0.026 $\pm$ 0.083	-0.041 $\pm$ 0.174	-0.041 $\pm$ 0.174
	5+ year	0.578 $\pm$ 0.102	0.511 $\pm$ 0.077	0.511 $\pm$ 0.077	0.591 $\pm$ 0.131	0.675 $\pm$ 0.088	0.675 $\pm$ 0.088	0.686 $\pm$ 0.026	0.703 $\pm$ 0.075	0.703 $\pm$ 0.075	0.148 $\pm$ 0.031	0.168 $\pm$ 0.016	0.168 $\pm$ 0.016
c-CSF	2-year	0.541 $\pm$ 0.176	0.558 $\pm$ 0.113	0.558 $\pm$ 0.113	0.375 $\pm$ 0.000	0.583 $\pm$ 0.084	0.583 $\pm$ 0.084	0.476 $\pm$ 0.087	0.543 $\pm$ 0.069	0.543 $\pm$ 0.069	-0.081 $\pm$ 0.174	0.138 $\pm$ 0.173	0.138 $\pm$ 0.173
	3-year	0.067 $\pm$ 0.115	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.930 $\pm$ 0.052	0.860 $\pm$ 0.063	0.860 $\pm$ 0.063	0.414 $\pm$ 0.050	0.385 $\pm$ 0.128	0.385 $\pm$ 0.128	-0.020 $\pm$ 0.077	-0.116 $\pm$ 0.033	-0.116 $\pm$ 0.033
	4-year	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.909 $\pm$ 0.027	0.943 $\pm$ 0.020	0.943 $\pm$ 0.020	0.403 $\pm$ 0.181	0.436 $\pm$ 0.108	0.436 $\pm$ 0.108	-0.076 $\pm$ 0.015	-0.058 $\pm$ 0.010	-0.058 $\pm$ 0.010
	5+ year	0.511 $\pm$ 0.154	0.533 $\pm$ 0.134	0.533 $\pm$ 0.134	0.790 $\pm$ 0.124	0.754 $\pm$ 0.021	0.754 $\pm$ 0.021	0.683 $\pm$ 0.038	0.650 $\pm$ 0.044	0.650 $\pm$ 0.044	0.298 $\pm$ 0.064	0.263 $\pm$ 0.136	0.263 $\pm$ 0.136
proteomic	2-year	0.852 $\pm$ 0.081	0.991 $\pm$ 0.015	0.991 $\pm$ 0.015	0.194 $\pm$ 0.096	0.042 $\pm$ 0.042	0.042 $\pm$ 0.042	0.614 $\pm$ 0.110	0.615 $\pm$ 0.021	0.615 $\pm$ 0.021	0.060 $\pm$ 0.162	0.091 $\pm$ 0.123	0.091 $\pm$ 0.123
	3-year	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	0.652 $\pm$ 0.141	0.552 $\pm$ 0.117	0.552 $\pm$ 0.117	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000
	4-year	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	0.387 $\pm$ 0.137	0.465 $\pm$ 0.214	0.465 $\pm$ 0.214	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000
	5+ year	0.133 $\pm$ 0.067	0.067 $\pm$ 0.067	0.067 $\pm$ 0.067	0.929 $\pm$ 0.054	0.993 $\pm$ 0.012	0.993 $\pm$ 0.012	0.631 $\pm$ 0.082	0.675 $\pm$ 0.064	0.675 $\pm$ 0.064	0.113 $\pm$ 0.170	0.145 $\pm$ 0.165	0.145 $\pm$ 0.165
all_features	2-year	0.825 $\pm$ 0.062	0.746 $\pm$ 0.418	0.746 $\pm$ 0.418	0.319 $\pm$ 0.064	0.347 $\pm$ 0.494	0.347 $\pm$ 0.494	0.607 $\pm$ 0.059	0.603 $\pm$ 0.059	0.603 $\pm$ 0.059	0.170 $\pm$ 0.143	0.165 $\pm$ 0.107	0.165 $\pm$ 0.107
	3-year	0.000 $\pm$ 0.000	0.111 $\pm$ 0.192	0.111 $\pm$ 0.192	1.000 $\pm$ 0.000	0.964 $\pm$ 0.062	0.964 $\pm$ 0.062	0.604 $\pm$ 0.194	0.604 $\pm$ 0.143	0.604 $\pm$ 0.143	0.000 $\pm$ 0.000	0.066 $\pm$ 0.115	0.066 $\pm$ 0.115
	4-year	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	1.000 $\pm$ 0.000	0.927 $\pm$ 0.127	0.927 $\pm$ 0.127	0.427 $\pm$ 0.135	0.588 $\pm$ 0.152	0.588 $\pm$ 0.152	0.000 $\pm$ 0.000	-0.039 $\pm$ 0.067	-0.039 $\pm$ 0.067
	5+ year	0.311 $\pm$ 0.252	0.267 $\pm$ 0.291	0.267 $\pm$ 0.291	0.794 $\pm$ 0.205	0.851 $\pm$ 0.240	0.851 $\pm$ 0.240	0.695 $\pm$ 0.064	0.654 $\pm$ 0.010	0.654 $\pm$ 0.010	0.136 $\pm$ 0.097	0.195 $\pm$ 0.114	0.195 $\pm$ 0.114

(a) RF model

Feature set	Window	Sensitivity			Specificity			AUC			MCC		
		OVA	Multi	Multi	OVA	Multi	Multi	OVA	Multi	Multi	OVA	Multi	Multi
clinical	2-year	0.503 $\pm$ 0.151	0.351 $\pm$ 0.304	0.351 $\pm$ 0.304	0.722 $\pm$ 0.087	0.778 $\pm$ 0.197	0.778 $\pm$ 0.197	0.646 $\pm$ 0.079	0.660 $\pm$ 0.058	0.660 $\pm$ 0.058	0.224 $\pm$ 0.135	0.126 $\pm$ 0.117	0.126 $\pm$ 0.117
	3-year	0.344 $\pm$ 0.435	0.067 $\pm$ 0.115	0.067 $\pm$ 0.115	0.789 $\pm$ 0.192	0.879 $\pm$ 0.209	0.879 $\pm$ 0.209	0.350 $\pm$ 0.150	0.420 $\pm$ 0.075	0.420 $\pm$ 0.075	0.079 $\pm$ 0.170	-0.031 $\pm$ 0.053	-0.031 $\pm$ 0.053
	4-year	0.194 $\pm$ 0.173	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.825 $\pm$ 0.198	0.876 $\pm$ 0.113	0.876 $\pm$ 0.113	0.514 $\pm$ 0.048	0.691 $\pm$ 0.242	0.691 $\pm$ 0.242	0.018 $\pm$ 0.055	-0.074 $\pm$ 0.064	-0.074 $\pm$ 0.064
	5+ year	0.756 $\pm$ 0.154	0.622 $\pm$ 0.204	0.622 $\pm$ 0.204	0.471 $\pm$ 0.069	0.585 $\pm$ 0.103	0.585 $\pm$ 0.103	0.547 $\pm$ 0.205	0.708 $\pm$ 0.024	0.708 $\pm$ 0.024	0.201 $\pm$ 0.099	0.180 $\pm$ 0.087	0.180 $\pm$ 0.087
c-CSF	2-year	0.478 $\pm$ 0.059	0.235 $\pm$ 0.116	0.235 $\pm$ 0.116	0.639 $\pm$ 0.048	0.750 $\pm$ 0.000	0.750 $\pm$ 0.000	0.604 $\pm$ 0.014	0.645 $\pm$ 0.077	0.645 $\pm$ 0.077	0.115 $\pm$ 0.026	-0.025 $\pm$ 0.129	-0.025 $\pm$ 0.129
	3-year	0.189 $\pm$ 0.019	0.056 $\pm$ 0.096	0.056 $\pm$ 0.096	0.595 $\pm$ 0.097	0.876 $\pm$ 0.108	0.876 $\pm$ 0.108	0.536 $\pm$ 0.195	0.397 $\pm$ 0.089	0.397 $\pm$ 0.089	-0.125 $\pm$ 0.062	-0.051 $\pm$ 0.070	-0.051 $\pm$ 0.070
	4-year	0.250 $\pm$ 0.433	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.784 $\pm$ 0.251	0.745 $\pm$ 0.171	0.745 $\pm$ 0.171	0.564 $\pm$ 0.236	0.461 $\pm$ 0.194	0.461 $\pm$ 0.194	0.006 $\pm$ 0.117	-0.137 $\pm$ 0.068	-0.137 $\pm$ 0.068
	5+ year	0.867 $\pm$ 0.115	0.622 $\pm$ 0.077	0.622 $\pm$ 0.077	0.400 $\pm$ 0.245	0.669 $\pm$ 0.053	0.669 $\pm$ 0.053	0.599 $\pm$ 0.213	0.656 $\pm$ 0.106	0.656 $\pm$ 0.106	0.254 $\pm$ 0.082	0.255 $\pm$ 0.029	0.255 $\pm$ 0.029
proteomic	2-year	0.601 $\pm$ 0.142	0.740 $\pm$ 0.109	0.740 $\pm$ 0.109	0.514 $\pm$ 0.168	0.444 $\pm$ 0.105	0.444 $\pm$ 0.105	0.556 $\pm$ 0.050	0.583 $\pm$ 0.037	0.583 $\pm$ 0.037	0.115 $\pm$ 0.080	0.195 $\pm$ 0.059	0.195 $\pm$ 0.059
	3-year	0.067 $\pm$ 0.115	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.936 $\pm$ 0.052	0.942 $\pm$ 0.039	0.942 $\pm$ 0.039	0.565 $\pm$ 0.126	0.558 $\pm$ 0.142	0.558 $\pm$ 0.142	-0.015 $\pm$ 0.070	-0.070 $\pm$ 0.022	-0.070 $\pm$ 0.022
	4-year	0.083 $\pm$ 0.144	0.167 $\pm$ 0.289	0.167 $\pm$ 0.289	0.869 $\pm$ 0.146	0.977 $\pm$ 0.026	0.977 $\pm$ 0.026	0.373 $\pm$ 0.158	0.415 $\pm$ 0.221	0.415 $\pm$ 0.221	-0.036 $\pm$ 0.045	0.124 $\pm$ 0.244	0.124 $\pm$ 0.244
	5+ year	0.533 $\pm$ 0.133	0.467 $\pm$ 0.067	0.467 $\pm$ 0.067	0.782 $\pm$ 0.046	0.824 $\pm$ 0.030	0.824 $\pm$ 0.030	0.706 $\pm$ 0.017	0.672 $\pm$ 0.025	0.672 $\pm$ 0.025	0.296 $\pm$ 0.076	0.289 $\pm$ 0.058	0.289 $\pm$ 0.058
all_features	2-year	0.609 $\pm$ 0.021	0.600 $\pm$ 0.064	0.600 $\pm$ 0.064	0.556 $\pm$ 0.168	0.583 $\pm$ 0.072	0.583 $\pm$ 0.072	0.587 $\pm$ 0.061	0.578 $\pm$ 0.043	0.578 $\pm$ 0.043	0.160 $\pm$ 0.161	0.179 $\pm$ 0.070	0.179 $\pm$ 0.070
	3-year	0.067 $\pm$ 0.115	0.133 $\pm$ 0.115	0.133 $\pm$ 0.115	0.965 $\pm$ 0.017	0.930 $\pm$ 0.070	0.930 $\pm$ 0.070	0.508 $\pm$ 0.054	0.547 $\pm$ 0.182	0.547 $\pm$ 0.182	0.023 $\pm$ 0.122	0.059 $\pm$ 0.066	0.059 $\pm$ 0.066
	4-year	0.167 $\pm$ 0.289	0.083 $\pm$ 0.144	0.083 $\pm$ 0.144	0.904 $\pm$ 0.152	0.921 $\pm$ 0.122	0.921 $\pm$ 0.122	0.367 $\pm$ 0.151	0.467 $\pm$ 0.221	0.467 $\pm$ 0.221	0.030 $\pm$ 0.083	0.069 $\pm$ 0.228	0.069 $\pm$ 0.228
	5+ year	0.756 $\pm$ 0.154	0.489 $\pm$ 0.077	0.489 $\pm$ 0.077	0.612 $\pm$ 0.180	0.740 $\pm$ 0.060	0.740 $\pm$ 0.060	0.585 $\pm$ 0.230	0.702 $\pm$ 0.019	0.702 $\pm$ 0.019	0.328 $\pm$ 0.018	0.213 $\pm$ 0.117	0.213 $\pm$ 0.117

(b) SVM model

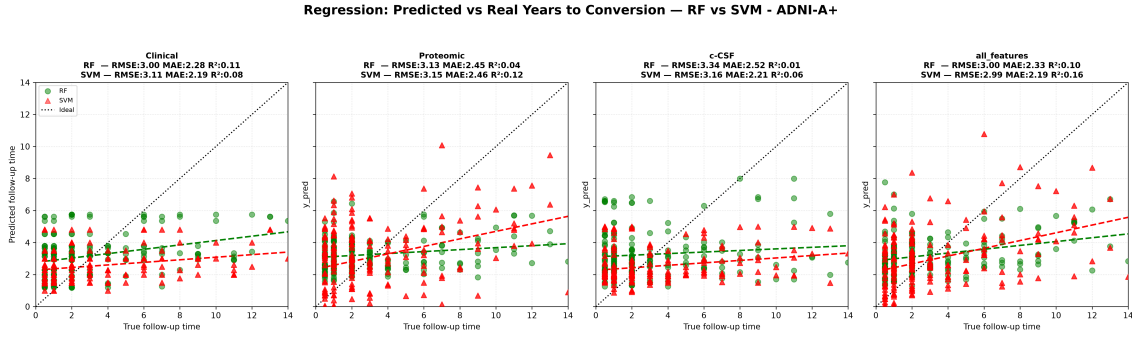


Figure 4.5: Predicted vs. true time-to-conversion (years) for the regression task on the ADNI-A+ cohort, comparing RF (green circles) and SVM (red triangles) across all four feature sets. RMSE, MAE and r^2 are reported per model and feature set. Predicted values show no meaningful association with true time-to-conversion in any configuration, confirming the poor regression performance observed on this cohort.

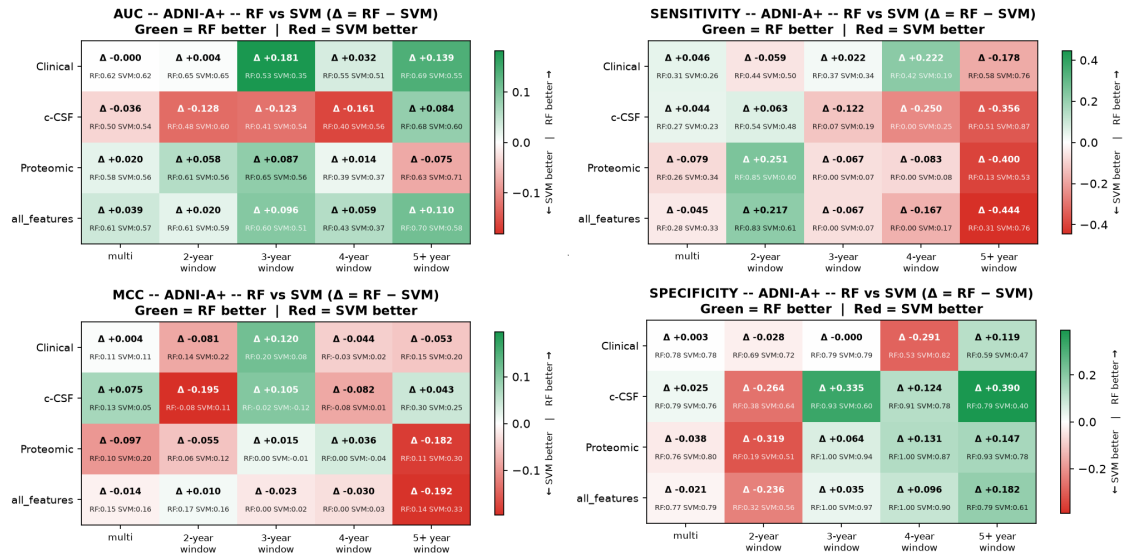


Figure 4.6: RF vs SVM comparison on the ADNI-A+ cohort across all task formulations and feature sets, showing MCC, AUC, Sensitivity and Specificity ($\Delta = \text{RF} - \text{SVM}$). Green cells indicate configurations where RF outperforms SVM; red cells indicate where SVM outperforms RF.

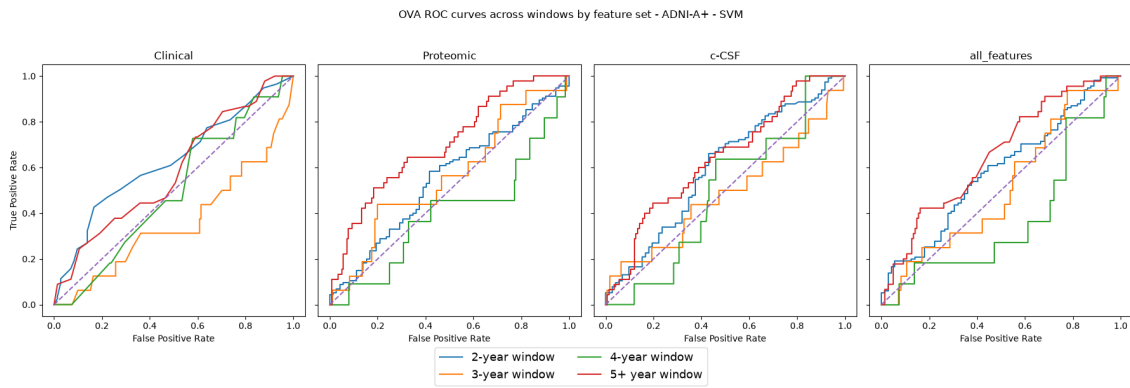
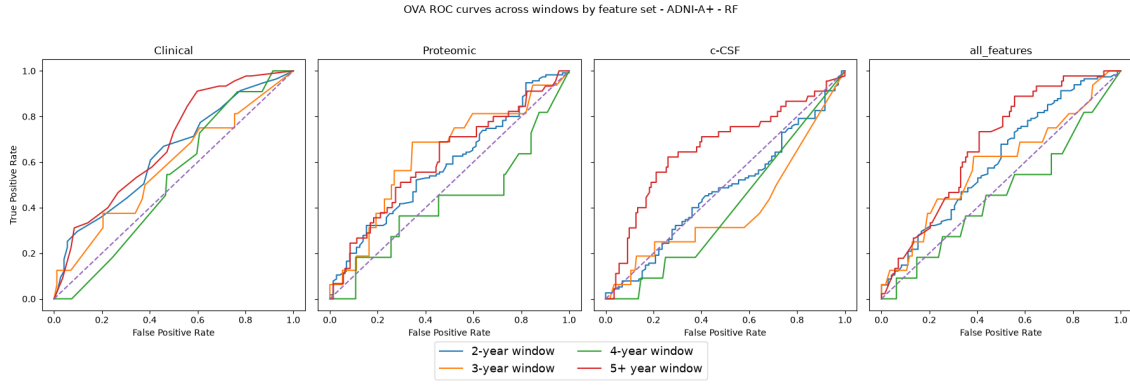
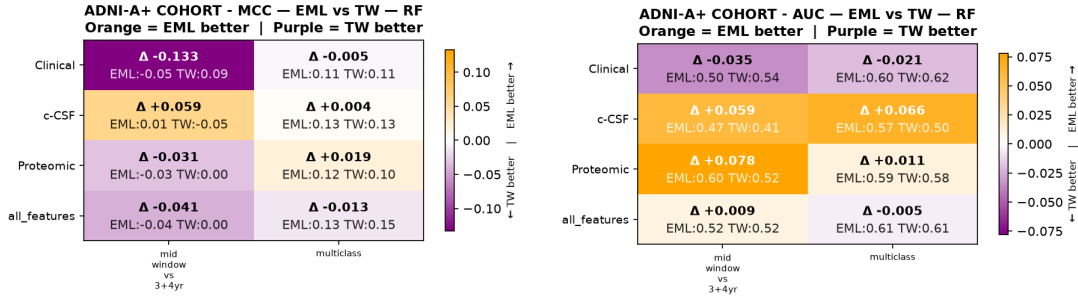
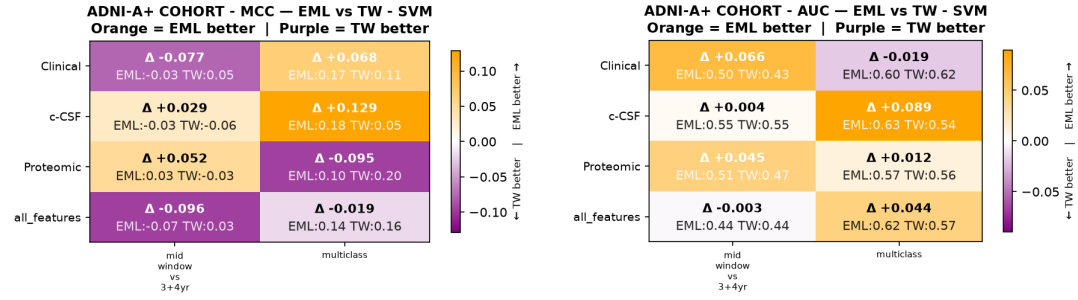


Figure 4.7: OVA ROC curves for the ADNI-A+ cohort across all four feature sets (clinical, proteomic, c-CSF, all_features) and all four time windows (2-year in blue, 3-year in yellow, 4-year in green, 5+ year in red). **(a)** Random Forest model: the 5+ year window (red) shows the clearest separation from the diagonal, particularly for the c-CSF feature set. **(b)** SVM model: the 5+ year window shows a consistent separation across all feature sets, with the all_features feature set yielding the highest AUC.



(a) RF model.



(b) SVM model.

Figure 4.8: MCC (left) and AUC (right) comparison between the EML-TW (early/mid/late) and 4-TW (four-class time window) formulations on the ADNI-A+ cohort. **(a)** Random Forest model. **(b)** SVM model. Orange cells indicate configurations where EML-TW outperforms 4-TW ($\Delta > 0$); purple cells indicate where 4-TW outperforms EML-TW ($\Delta < 0$). Only the mid window and the multiclass task are shown, as the early and late windows are identical under both formulations by construction. The complete performance panel is reported in Appendix (Table 8.8).

4.1.3 Shared Proteomic Panel between LC and ADNI-A+ Cohorts (SH40)

This subsection reports the results obtained by restricting the proteomic feature space to the 40 proteins shared between the LC and ADNI-A+ panels (Section 3.1.3), evaluated separately on each cohort. The analysis was conducted using the Random Forest model only, on the `proteomic` and `all_features` feature sets, across all modelling approaches. The primary aim is to quantify the loss of information incurred by restricting the proteomic panel to the 40 shared proteins, and to assess whether the two cohorts select consistent features within this shared space (Section 4.2.3). The comparison between full panel and SH40 performances is reported in Table 4.7.

One-vs-All Classification. Table 4.7 reports the complete numerical comparison between the full proteomic panel and the SH40 panel across all OVA configurations, cohorts, and metrics. Figure 4.9 provide a visual summary of the differences in MCC,

AUC, sensitivity, and specificity between the two panels for the LC and ADNI-A+ cohorts respectively.

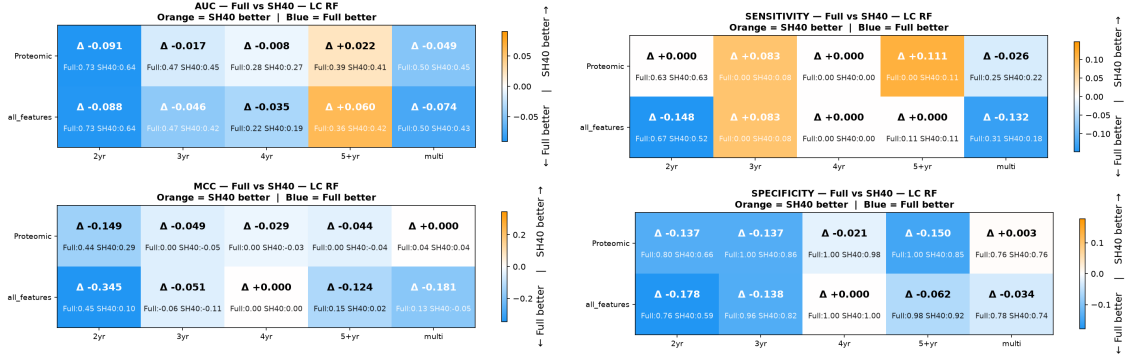
For the LC cohort, the most pronounced performance drop is observed for the 2-year window, the only configuration yielding meaningful MCC on the full panel. Under `all_features`, MCC drops from 0.448 ± 0.072 to 0.103 ± 0.188 (approximately -77%), while under `proteomic` the loss is less severe (MCC from 0.439 ± 0.062 to 0.290 ± 0.147 , -34%). These results confirm that the most informative proteins for the LC 2-year window are largely outside the shared panel. For all remaining OVA windows, where the full panel already yielded near-zero or negative MCC, restricting to SH40 does not substantially change the outcome; isolated marginal gains in AUC are accompanied by high variance and do not reflect genuine discriminative capacity.

For the ADNI-A+ cohort, the picture is more complex. For the 5+ year window under `all_features`, the only configuration yielding relevant performance on the full panel, MCC marginally improves from 0.136 ± 0.097 to 0.164 ± 0.056 , but this is accompanied by a shift in the sensitivity-specificity trade-off rather than a genuine improvement, consistent with the increased relative weight of `c-CSF` features when the proteomic panel is restricted. For the `proteomic` feature set, performance drops substantially (MCC from 0.113 ± 0.172 to 0.044 ± 0.094), driven by a drastic fall in specificity (from 0.929 ± 0.054 to 0.753 ± 0.056), confirming that the most informative proteins for the correct rejection of negative samples in ADNI-A+ are lost when restricting to the shared panel. For all remaining windows, no meaningful change in performance is observed relative to the full panel.

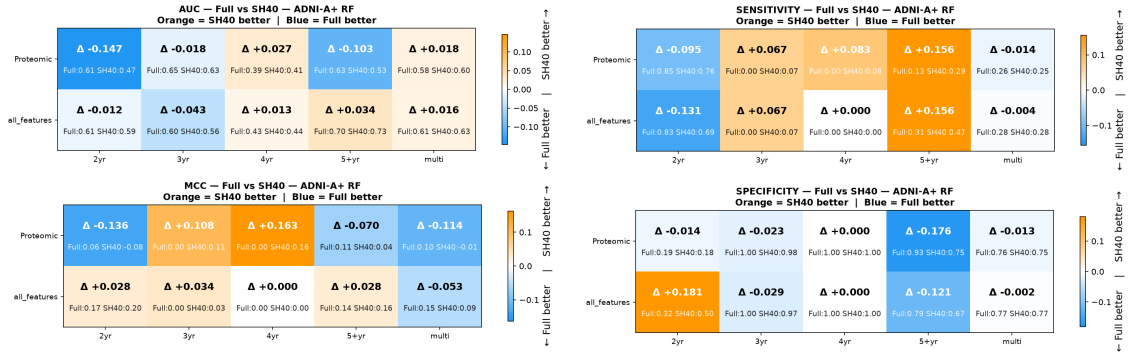
The OVA ROC curves for the SH40 panel (Figure 4.10) provide a visual complement to these results. For LC, the 2-year window curve remains visibly separated from the diagonal under `proteomic` and `all_features`, confirming that a residual discriminative signal is preserved within the shared panel, albeit weaker than in the full panel. For ADNI-A+, no window shows a clear separation from the diagonal in either feature set, consistent with the scarcity of meaningful proteomic informativeness across all OVA configurations on this cohort. This divergence will be discussed further in Chapter 5.

Multiclass Classification. For both cohorts, the multiclass formulation already yielded near-zero or negative MCC on the full panel, and restricting to the SH40 panel does not substantially change this outcome. For LC, the most notable change is a drop in MCC under `all_features`, while `proteomic` remains broadly comparable. For ADNI-A+, MCC drops moderately under both feature sets, with only marginal and inconsistent AUC changes. Overall, the restriction to the shared panel does not recover any discriminative capacity in the multiclass setting for either cohort, confirming that the limited signal present in the full panel is further reduced within

the SH40 feature space.



(a) LC cohort.

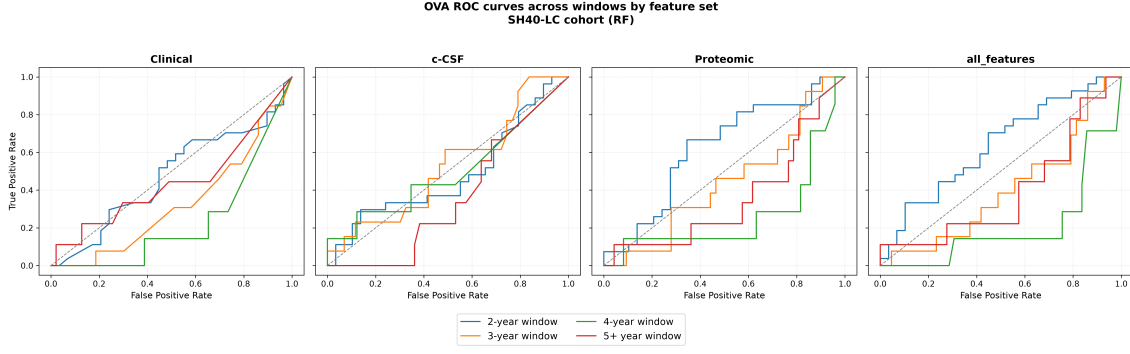


(b) ADNI-A+ cohort.

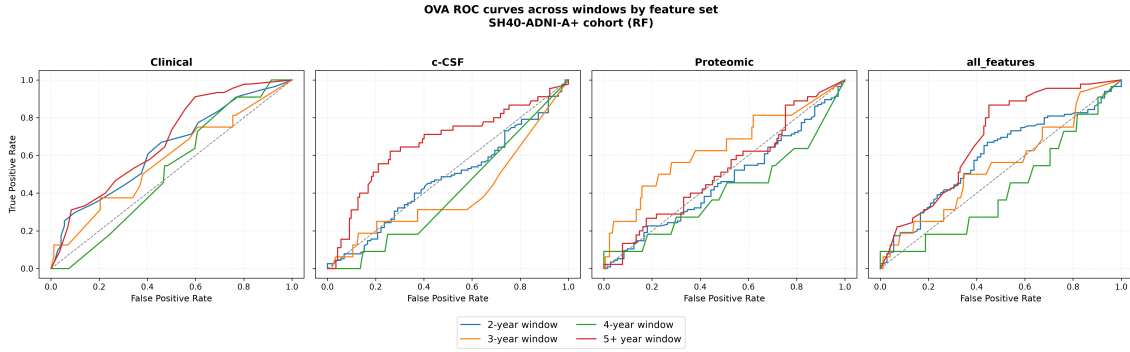
Figure 4.9: Full panel vs SH40 performance comparison (MCC, AUC, Sensitivity, Specificity) across all task formulations and feature sets, using Random Forest. (a) LC cohort. (b) ADNI-A+ cohort. Orange cells indicate configurations where SH40 outperforms the full panel ($\Delta > 0$); blue cells indicate where the full panel is better ($\Delta < 0$).

Table 4.7: Performance comparison between the full proteomic panel and the overlapping 40-protein panel (SH40) across all task formulations and both cohorts, using Random Forest. Values are mean $\pm$ SD across 3 outer folds.

Cohort	Task	Feature set	MCC		AUC		Sensitivity		Specificity	
			Full	SH40	Full	SH40	Full	SH40	Full	SH40
LC	2-year	all_features	0.448 $\pm$ 0.072	0.103 $\pm$ 0.188	0.732 $\pm$ 0.067	0.643 $\pm$ 0.041	0.667 $\pm$ 0.192	0.519 $\pm$ 0.280	0.763 $\pm$ 0.148	0.585 $\pm$ 0.103
		proteomic	0.439 $\pm$ 0.062	0.290 $\pm$ 0.147	0.728 $\pm$ 0.128	0.637 $\pm$ 0.045	0.630 $\pm$ 0.128	0.630 $\pm$ 0.064	0.796 $\pm$ 0.094	0.659 $\pm$ 0.103
	3-year	all_features	-0.059 $\pm$ 0.102	-0.110 $\pm$ 0.216	0.467 $\pm$ 0.075	0.421 $\pm$ 0.115	0.000 $\pm$ 0.000	0.083 $\pm$ 0.144	0.956 $\pm$ 0.077	0.817 $\pm$ 0.135
		proteomic	0.000 $\pm$ 0.000	-0.049 $\pm$ 0.198	0.467 $\pm$ 0.108	0.450 $\pm$ 0.056	0.000 $\pm$ 0.000	0.083 $\pm$ 0.144	1.000 $\pm$ 0.000	0.863 $\pm$ 0.133
	4-year	all_features	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.224 $\pm$ 0.094	0.190 $\pm$ 0.103	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000
		proteomic	0.000 $\pm$ 0.000	-0.029 $\pm$ 0.050	0.277 $\pm$ 0.062	0.269 $\pm$ 0.185	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	1.000 $\pm$ 0.000	0.979 $\pm$ 0.036
	5+ year	all_features	0.147 $\pm$ 0.347	0.023 $\pm$ 0.040	0.356 $\pm$ 0.108	0.415 $\pm$ 0.129	0.111 $\pm$ 0.192	0.111 $\pm$ 0.192	0.979 $\pm$ 0.036	0.917 $\pm$ 0.144
		proteomic	0.000 $\pm$ 0.000	-0.044 $\pm$ 0.140	0.386 $\pm$ 0.017	0.408 $\pm$ 0.124	0.000 $\pm$ 0.000	0.111 $\pm$ 0.192	1.000 $\pm$ 0.000	0.850 $\pm$ 0.132
	multiclass	all_features	0.132 $\pm$ 0.157	-0.050 $\pm$ 0.107	0.502 $\pm$ 0.048	0.428 $\pm$ 0.010	0.308 $\pm$ 0.034	0.176 $\pm$ 0.058	0.779 $\pm$ 0.042	0.744 $\pm$ 0.030
		proteomic	0.035 $\pm$ 0.156	0.035 $\pm$ 0.081	0.495 $\pm$ 0.045	0.446 $\pm$ 0.051	0.250 $\pm$ 0.062	0.225 $\pm$ 0.024	0.760 $\pm$ 0.039	0.763 $\pm$ 0.021
ADNI-A+	2-year	all_features	0.170 $\pm$ 0.143	0.198 $\pm$ 0.096	0.607 $\pm$ 0.059	0.595 $\pm$ 0.064	0.825 $\pm$ 0.062	0.695 $\pm$ 0.088	0.319 $\pm$ 0.064	0.500 $\pm$ 0.000
		proteomic	0.060 $\pm$ 0.162	-0.076 $\pm$ 0.094	0.614 $\pm$ 0.110	0.467 $\pm$ 0.054	0.852 $\pm$ 0.081	0.756 $\pm$ 0.017	0.194 $\pm$ 0.096	0.181 $\pm$ 0.064
	3-year	all_features	0.000 $\pm$ 0.000	0.034 $\pm$ 0.060	0.604 $\pm$ 0.194	0.561 $\pm$ 0.165	0.000 $\pm$ 0.000	0.067 $\pm$ 0.115	1.000 $\pm$ 0.000	0.971 $\pm$ 0.051
		proteomic	0.000 $\pm$ 0.000	0.108 $\pm$ 0.281	0.652 $\pm$ 0.141	0.634 $\pm$ 0.172	0.000 $\pm$ 0.000	0.067 $\pm$ 0.115	1.000 $\pm$ 0.000	0.977 $\pm$ 0.027
	4-year	all_features	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.427 $\pm$ 0.135	0.439 $\pm$ 0.051	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000
		proteomic	0.000 $\pm$ 0.000	0.163 $\pm$ 0.281	0.387 $\pm$ 0.137	0.414 $\pm$ 0.031	0.000 $\pm$ 0.000	0.083 $\pm$ 0.144	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000
	5+ year	all_features	0.136 $\pm$ 0.097	0.164 $\pm$ 0.056	0.695 $\pm$ 0.064	0.729 $\pm$ 0.098	0.311 $\pm$ 0.252	0.467 $\pm$ 0.291	0.794 $\pm$ 0.205	0.674 $\pm$ 0.270
		proteomic	0.113 $\pm$ 0.170	0.044 $\pm$ 0.094	0.631 $\pm$ 0.082	0.528 $\pm$ 0.028	0.133 $\pm$ 0.067	0.289 $\pm$ 0.038	0.929 $\pm$ 0.054	0.753 $\pm$ 0.056
	multiclass	all_features	0.147 $\pm$ 0.087	0.093 $\pm$ 0.039	0.612 $\pm$ 0.079	0.629 $\pm$ 0.082	0.281 $\pm$ 0.020	0.277 $\pm$ 0.020	0.772 $\pm$ 0.018	0.771 $\pm$ 0.018
		proteomic	0.101 $\pm$ 0.123	-0.013 $\pm$ 0.128	0.577 $\pm$ 0.063	0.595 $\pm$ 0.070	0.264 $\pm$ 0.017	0.250 $\pm$ 0.035	0.759 $\pm$ 0.011	0.746 $\pm$ 0.019



(a) LC dataset restricted to the 40 shared proteins (SH40-LC).



(b) ADNI-A+ dataset restricted to the 40 shared proteins (SH40-ADNI-A+).

Figure 4.10: OVA ROC curves across all four time windows and feature sets for the Overlapping ($LC \cap ADNI-A+$) cohort, using Random Forest. **(a)** LC dataset restricted to the 40 shared proteins. **(b)** ADNI-A+ dataset restricted to the 40 shared proteins. Clinical and c-CSF feature sets are taken from the respective full-panel runs (same outer fold partitions); proteomic and both feature sets use only the 40 proteins shared between the two panels.

4.1.4 Summary of Main Findings

Did the machine learning pipeline achieve meaningful predictive performance for time-to-conversion prediction? Meaningful predictive performance was achieved in both cohorts, but limited to specific configurations. In the LC cohort, only the 2-year OVA window yielded consistently positive MCC values, while all other time windows, the multiclass formulation, and the regression task produced near-zero or negative MCC across all feature sets and both models. In the ADNI-A+ cohort, the 5+ year OVA window was the only configuration yielding relevant performance, with SVM the `all_features` feature set. Neither cohort showed any meaningful gain from the EML-TW formulation or the regression approach, both of which were discontinued from further analysis.

Under which cohort, time window, and feature set configuration were the best results obtained? For LC, the best configuration was the 2-year OVA window with RF and the `all_features` feature set (MCC 0.448 ± 0.072 , AUC 0.732 ± 0.067). For ADNI-A+, the best configuration was the 5+ year OVA window with SVM and the `all_features` feature set (MCC 0.328 ± 0.018). Restricting the proteomic panel to the 40 shared proteins (SH40) caused a severe loss of performance for the LC 2-year window, while the impact on ADNI-A+ was more contained, as the predictive signal for that cohort was already driven primarily by `c-CSF` features rather than proteomic ones.

Did proteomic features provide a genuine discriminative contribution beyond standard clinical and biochemical measures? In the LC cohort, proteomic features provided a clear and statistically supported advantage over `clinical` and `c-CSF` features alone for the 2-year window, confirmed by a Friedman test ($p < 0.05$ for both RF and SVM). In the ADNI-A+ cohort, by contrast, the most informative contribution came from `c-CSF` features, particularly P-TAU and T-TAU for the 5+ year window, with proteomic features playing a complementary but secondary role. A possible explanation of this outcome could be related with the different biological composition of the two cohorts and the technical limitations of the SOMAscan platform, and will be discussed further in Chapter 5.

4.2 Selected Features

This section presents the feature subsets selected by the FSE-StabPred ensemble, with primary focus on the configurations that yielded meaningful predictive performance as identified in Section 4.1, addressing the following questions:

- (a) Which proteins were consistently selected by the FSE-StabPred ensemble across folds for the most informative configurations identified in Section 4.1?
- (b) Is the feature selection consistent across predictive models within the same cohort?
- (c) Do the two cohorts identify a common proteomic signature within the shared panel?

Feature selections from additional configurations are reported selectively where they provide interesting context for the Discussion. Only features appearing in at least 2 out of 3 outer folds are shown, as these represent the most stable selections across data partitions. The optimal feature subset size k^* selected by the RPT- β criterion for each outer fold, cohort, task, model, and feature set is reported in Table 4.11. A deeper biological characterisation of the selected proteomic candidates is deferred to Section 4.3.

Results are organised into three subsections, one per cohort. For the LC and ADNI-A+ cohorts, each subsection opens with the feature selections for the primary configuration identified in Section 4.1, followed by a selective discussion of additional configurations of interest, including the multiclass formulation and the clinical feature set. For the Shared Proteomic Panel (SH40) subsection, the analysis is restricted to a cross-cohort comparison of the feature selections on the shared panel, assessing whether the two cohorts identify consistent proteomic candidates within the common feature space. As already anticipated in the previous section, feature selections under the EML-TW formulation are not discussed further, as this approach did not yield any meaningful improvement over the 4-TW scheme in either cohort (Section 4.1); however the corresponding tables are provided in the Appendix (Tables 8.3 and 8.5).

4.2.1 LC Cohort

This subsection reports the stable feature selections produced by the FSE-StabPred ensemble for the LC cohort. Table 4.8 reports the features selected in at least 2 out of 3 outer folds for all tasks and feature sets under the 4-TW formulation, listed in decreasing order of mean ensemble ranking score; Table 8.3 reports the corresponding selections under the EML-TW formulation. Where more than 10 stable features exist, only the top 10 are shown; complete lists are provided in the Appendix (Tables 8.2 and 8.7).

As established in Section 4.1, the only configuration yielding meaningful predictive performance on the LC cohort is the 2-year OVA window, with RF as the primary model and `proteomic` and `all_features` as the most informative feature sets. The feature selection analysis in the following paragraphs therefore focuses primarily on

this configuration, with selective reference to additional configurations where the selected features provide interesting context for the Discussion.

Proteomic and all_features feature sets — 2-year OVA. The proteomic and all_features feature sets yield richer and more variable selection patterns, as expected given the larger number of candidate features (Table 4.8). For the 2-year OVA window, the ensemble consistently identifies a small stable core of proteins across both models and feature sets, with several proteins selected in all three outer folds by both RF and SVM. This core selection remains largely consistent when clinical and c-CSF features are introduced in the all_features feature set. In this one, infact, the proteomic selections are largely preserved relative to the proteomic-only case, with the occasional addition of the MMSE score, probably sustaining the majority of informativeness inside the proteomic panel.

A degree of instability in k^* is present even for this configuration (Table 4.11): the third outer fold consistently selects a much smaller subset than the first two for RF, and a more pronounced instability is observed for SVM, where k^* drops markedly in folds 2 and 3 for the all_features feature set. For all remaining OVA windows, proteomic selections show considerably higher variability across folds and between models.

Clinical and c-CSF feature sets. For the clinical feature set, the MMSE score is the most consistently selected feature across all tasks and both models, with the sole exception of the intermediate time windows (3-year and 4-year OVA), where gender emerges as the most selected feature. For the c-CSF feature set, A β 42 and T-TAU are selected consistently across most tasks in both models, while P-TAU does not emerge as a stable selection in any configuration.

Multiclass overlap. For the multiclass task, a substantially larger number of proteins are selected stably across folds relative to the OVA configurations (over 70 in some configurations), with the most frequently selected proteins reported in Table 4.8. A comparison with the OVA selections reveals a substantial overlap across all time horizons, as summarised below:

Window	RF		SVM	
	proteomic	all_features	proteomic	all_features
2-year	78.0% (39.0%)	88.2% (36.1%)	70.0% (39.4%)	100.0% (8.2%)
3-year	70.3% (31.7%)	79.2% (22.9%)	64.5% (28.2%)	76.2% (18.8%)
4-year	90.9% (12.2%)	100.0% (8.4%)	75.0% (12.7%)	84.6% (12.9%)
5+ year	88.2% (18.3%)	81.8% (21.7%)	56.9% (40.8%)	77.1% (31.8%)

Values represent the percentage of OVA stable features recovered within the multiclass selection; values in parentheses indicate the percentage they constitute of the total multiclass selection. The consistently high recovery rates confirm that the multiclass model captures window-specific proteomic signal rather than a qualitatively different global signature. For the **all_features** feature set, proteomic selections are largely preserved relative to the proteomic-only case.

Table 4.8: Features selected by FSE-StabPred in at least 2 out of 3 outer folds for the LC cohort (4-TW formulation), listed in decreasing order of mean ensemble ranking score across outer folds. **Bold** entries were selected in all three folds. For the **proteomic** and **all_features** feature sets, only the top 10 stable features are shown; complete lists are in Appendix (Table 8.2).

Task	RF				SVM			
	clinical	c-CSF	proteomic	all_features	clinical	c-CSF	proteomic	all_features
Multiclass	MMSE, AGE	A β 42, T-TAU	P00738, Q06828, P02746, Q86UX2, P0C0L4, Q9UBX5, Q96KN2, Q12860, P02452, P18065 [+72]	P00738, Q06828, P02746, Q86UX2, P0C0L4, Q9UBX5, Q96KN2, P02452, MMSE, P42785, P07225 [+73]	AGE	A β 42, T-TAU	P00738, Q06828, P02746, Q86UX2, P0C0L4, Q9UBX5, Q96KN2, Q12860, P02452, MMSE, P18065 [+61]	P00738, Q06828, P02746, Q86UX2, P0C0L4, Q9UBX5, Q96KN2, P02452, MMSE, P42785, P07225 [+75]
2-year OVA	MMSE, AGE	A β 42, T-TAU	P00738, P02746, P02452, Q15582, P06733, Q12860, Q96KN2, Q15582, P20774, P55058, P19021 [+31]	P00738, P02746, P02452, Q15582, P06733, Q12860, Q96KN2, Q12860, P19021, P55058, P20774 [+24]	MMSE, AGE	A β 42, T-TAU	P00738, P02746, P02452, P06733, Q12860, Q96KN2, Q15582, P20774, P55058, P19021 [+30]	P00738, P02746, P02452, Q15582, P06733, Q96KN2, P55058
3-year OVA	GENDER	T-TAU, A β 42	P07225, P42785, P0C0L5, P42785, Q86UX2, Q06828, P0C0L4, Q12259, Q9P121, Q86UX2, P00748, Q8TEU8 [+27]	P0C0L5, P07225, P42785, Q86UX2, Q06828, P12259, P0C0L4, P12109, Q8TEU8, Q9P121 [+14]	GENDER, MMSE	T-TAU	P07225, P42785, P0C0L5, Q06828, P0C0L4, P12259, Q9P121, Q9UBX5, Q9P121, Q86UX2, P00748, Q8TEU8 [+21]	P0C0L5, P07225, P42785, Q86UX2, P12259, P12109, Q9UBX5, Q9P121, Q9UBX5, Q9P121, P00748, P40925 [+11]
4-year OVA	GENDER, AGE	A β 42	P01599, P01861, P18065, Q969P0, Q9UBX5, Q9NT99, Q86UX2, P01834, P07225, P16035 [+41]	P01599, P01861, Q9UBX5, Q969P0, P18065, Q9NT99, P16035	GENDER, AGE	A β 42	P01599, P01861, P18065, Q969P0, P07711, Q9UBX5, Q9NT99, Q86UX2, P10451, P07225 [+2]	P01599, P01861, P07711, Q9UBX5, Q969P0, P18065, Q86UX2, Q06828, Q9NT99, P10451 [+3]
5+ year OVA	MMSE	T-TAU	P00738, Q14624, P01011, P05546, P0C0L4, P09972, P02649, P00450, P00748, P19021 [+7]	P00738, MMSE, Q14624, P01011, P0C0L4, P00748, P05546, P05452, P02649, P19021 [+12]	MMSE	T-TAU	P00738, Q14624, P01011, P05546, P0C0L4, P09972, P02649, P00450, P00748, P02745 [+41]	P00738, MMSE, Q14624, P01011, P0C0L4, P00748, P05546, P05452, P02649, P02745 [+25]
Regression	MMSE	A β 42, T-TAU	P00738, Q96KN2, P02746, P02452	P00738, MMSE, P02746, P02452, P36222	MMSE	A β 42, T-TAU	P00738, Q96KN2, P02746, P02452	P00738, MMSE, P02746, P02452

4.2.2 ADNI-A+ Cohort

The selected feature subsets for the ADNI-A+ cohort under the 4-TW formulation are reported in Table 4.9 for both RF and SVM models; where more than 10 stable features exist, only the top 10 are shown, with complete lists provided in the Appendix (Table 8.4). As established in Section 4.1, the only configuration that yield meaningful predictive performance is the 5+ year OVA window with SVM and the `all_features` feature set. The feature selection analysis therefore focuses primarily on this configuration, with selective reference to additional configurations where relevant. As discussed in Section 4.1.2, the structural differences between the two cohorts, biological composition (A + T- vs predominantly A + T+) and proteomic platform (SOMAscan 7K vs Short-GeLC LC-SWATH-MS), limit the direct comparability of feature selections between LC and ADNI-A+.

5+ year OVA — SVM, `all_features`. Table 4.9 reports the stable feature selections for the 5+ year OVA window under SVM, the primary configuration of interest for the ADNI-A+ cohort. The `all_features` feature set reveals a striking difference from the LC cohort: rather than being dominated by proteomic features, the most stably selected features are MMSE and P-TAU, which appear in the top positions across the folds in which a meaningful selection is produced. This is consistent with the complementary role of `c-CSF` and `clinical` features discussed in Section 4.1.2, where these modalities drove the sensitivity gain that proteomic features alone could not achieve.

Clinical and `c-CSF` feature sets. For the `clinical` feature set, the MMSE score is the most consistently selected feature across multiclass, 2-year OVA, 5+ year OVA, and regression tasks in both models, SVM and RF, mirroring the pattern observed in LC. Gender and age appear selectively for the 3-year and 4-year OVA tasks respectively. Notably, the informativeness of MMSE is further confirmed by its appearance at the top of the `all_features` selection for the 5+ year window, where it outranks the proteomic features, a pattern not observed in LC.

For the `c-CSF` feature set, a notable difference from LC emerges: P-TAU is the dominant stable selection across most tasks in both models, the inverse of the pattern observed in LC. For the 5+ year window, both P-TAU and T-TAU are selected stably in both models, SVM and RF, suggesting a complementary contribution of the two TAU features at longer conversion horizons. The dominance of P-TAU is further reinforced by its appearance alongside MMSE at the top of the `all_features` selection for the 5+ year window, confirming its role as the most informative biochemical feature for this cohort and conversion horizon.

Multiclass overlap. A comparison of the multiclass stable selections with those of each OVA window reveals a pattern broadly consistent with that observed for LC: the majority of OVA stable features for each window are recovered within the multiclass selection, with the 5+ year window showing the highest recovery rate under both models and feature sets. The overlap figures for the RF and SVM models are summarised below:

Window	RF		SVM	
	proteomic	all_features	proteomic	all_features
2-year	75.4% (33.1%)	52.0% (39.4%)	73.1% (38.9%)	56.5% (39.4%)
3-year	52.1% (37.7%)	19.0% (45.5%)	67.4% (24.6%)	27.9% (36.4%)
4-year	47.2% (19.2%)	22.9% (24.2%)	71.4% (11.9%)	26.7% (12.1%)
5+ year	91.9% (26.2%)	100.0% (12.1%)	87.8% (28.6%)	100.0% (12.1%)

Values represent the percentage of OVA stable features recovered within the multiclass selection; values in parentheses indicate the percentage they constitute of the total multiclass selection. This confirms that the multiclass model captures window-specific proteomic signal rather than a qualitatively different global signature, consistent with the findings discussed in Section 4.1.

Table 4.9: Features selected by FSE-StabPred in at least 2 out of 3 outer folds for the ADNIA+ cohort (4-TW formulation), listed in decreasing order of mean ensemble ranking score. **Bold** entries were selected in all three outer folds. Where more than 10 stable features exist, only the top 10 are shown; complete lists are in Appendix (Table 8.4).

Task	RF				SVM			
	clinical	c-CSF	proteomic	all_features	clinical	c-CSF	proteomic	all_features
Multiclass	MMSE	P-TAU, T-TAU	Q96MU8, P61981, P09603, Q9NZK5, P16442, Q14515, P02794 P02792, Q9NZR1, P10747, O43390 [+118]	MMSE, Q96MU8, P-TAU, P09603, P61981, P60510, Q9NZK5, P39059, P16442, Q14515, P16442 [+25]	MMSE	P-TAU	Q96MU8, P61981, P09603, Q9NZK5, P16442, Q14515, P02794 P02792, Q9NZR1, P10747, O43390 [+115]	MMSE, Q96MU8, P-TAU, P09603, P61981, P60510, Q9NZK5, P39059, Q14515, P16442 [+25]
2-year OVA	MMSE	P-TAU, T-TAU	P61981, P63104, P09603, P31946, Q99497, Q96MU8, P10092, P10092, O60462, Q9H492, P06744 [+45]	MMSE, P61981, P63104, P09603, P-TAU, P31946, Q99497, Q96MU8, P10092, Q99497, O60462 [+15]	MMSE	P-TAU	P61981, P63104, P09603, P31946, Q99497, Q96MU8, P10092, P10092, O60462, Q9H492, P06744 [+55]	MMSE, P61981, P63104, P09603, P-TAU, P31946, Q96MU8, P10092, Q99497, O60462 [+14]
3-year OVA	GENDER	T-TAU	P09603, P02794 P02792, P10451, Q8N8U9, Q9UKL6, Q9NZC2, P14174, Q63HQ2, P49913, P10747 [+82]	MMSE, P09603, P02794 P02792, P10451, P12829, Q8N8U9, Q9UKL6, P14174, Q9Y2I2, Q9NZC2, P0C0S8 [+68]	MMSE, DER	GEN- T-TAU	P09603, P02794 P02792, P10451, P12829, Q8N8U9, Q9UKL6, Q9NZC2, P14174, Q63HQ2, P10747 [+37]	P09603, P02794 P02792, P10451, P12829, Q8N8U9, Q9UKL6, P14174, Q9Y2I2, Q9NZC2, P0C0S8 [+34]
4-year OVA	GENDER	A β 42	Q49961, Q9NZK5, P31994, P10321, O14896, P08833, P09603, Q9BQJ7, Q9GZQ8, O43915 [+46]	Q9NZK5, P49961, P31994, O14896, P09603, P10321, P08833, Q9BQJ7, Q9BX59, P12109 [+27]	GENDER, MMSE	A β 42	Q49961, Q9NZK5, P31994, O14896, P10321, O14896, P08833, P09603, Q9GZQ8, O43915, Q9H492 [+11]	Q9NZK5, P49961, P31994, O14896, P09603, P08833, P16442, Q70626, P09681, Q96CS3 [+5]
5+ year OVA	MMSE	P-TAU, T-TAU	Q96MU8, P61981, P16442, P41217, Q969P0, Q9H320, O14737, O60462, Q13253, Q14515 [+28]	MMSE, P-TAU, O60462, P10747, P31946	MMSE	P-TAU	Q96MU8, P61981, P16442, P41217, Q969P0, Q9H320, O14737, O60462, Q13253, Q14515 [+32]	MMSE, P-TAU, O60462, P10747, P31946
Regression	MMSE	P-TAU	Q96MU8, P10092, Q14515, P10747, O75023, O60462, P60510, Q969P0, P41217 [+104]	MMSE, P-TAU, Q96MU8, P-TAU, P10092, P10092, P10747, P61981, O60462, P60510, P60510, O75023 [+22]	MMSE	P-TAU	Q96MU8, P10092, Q14515, P10747, O75023, Q969P0, P41217 [+100]	MMSE, Q96MU8, P-TAU, P10092, P10747, P61981, Q14515, O60462, P60510, O75023 [+17]

4.2.3 Shared Proteomic Panel between LC and ADNI-A+ Cohorts (SH40)

The selected feature subsets for the SH40 panel are reported in Table 4.10 for both datasets using the Random Forest model; where more than 10 stable features exist, only the top 10 are shown, with complete lists provided in the Appendix (Table 8.6). As established in Section 4.1.3, the analysis is restricted to the `proteomic` and `all_features` feature sets, for the tasks that gave the most meaningful performances.

LC dataset on the shared panel. For the 2-year OVA window, for LC, a small core of proteins is selected stably across all three outer folds under both feature sets (Table 4.10). Notably, none of these proteins appeared among the top selections in the full LC panel for the same window, confirming that the most informative proteomic signal for this task lies largely outside the shared panel, consistent with the severe performance drop reported in Section 4.1.3. For the `all_features` feature set, the MMSE score appears stably alongside the proteomic selections, as observed in the full panel. For the remaining OVA windows and the multiclass task, the selections are sparse and variable across folds, reflecting the absence of a learnable discriminative signal already observed for those configurations on the full panel.

ADNI-A+ dataset on the shared panel. For the 5+ year OVA window, for ADNI-A+, the proteomic selection on the SH40 panel is markedly impoverished relative to the full panel, with only a small number of proteins selected stably under the `proteomic` feature set. Under the `all_features` feature set, MMSE emerges as the dominant stable selection, confirming once more that the most informative proteomic features for this window lie largely outside the shared panel, and that clinical carry more discriminative weight than the restricted proteomic signal in this cohort. For the 2-year OVA window, the proteomic selection is comparatively richer, with several proteins selected stably across all three folds; however, the `all_features` feature set collapses to MMSE and P-TAU alongside a minimal proteomic core, again reflecting the dominant contribution of `c-CSF` features. For the remaining windows and the multiclass task, MMSE and P-TAU appear consistently at the top of the `all_features` selections, further reinforcing their role as the most stable predictors for this cohort.

Cross-cohort consistency. The proteomic selections across the two cohorts within the shared panel are markedly different. While feature selections remain broadly consistent across RF and SVM within the same cohort and window, switching cohort reveals a substantially different proteomic signature, suggesting that the two cohorts

are too different, both technically and biologically, to support a meaningful cross-cohort comparison of proteomic biomarkers within the shared panel. This finding will be discussed further in Chapter 5.

Table 4.10: Features selected by FSE-StabPred in at least 2 out of 3 outer folds for the shared proteomic panel feature set, using Random Forest, listed in decreasing order of mean ensemble ranking score. Results are shown separately for the LC and ADNI-A+ datasets restricted to the 40 shared proteins. **Bold** entries were selected in all three outer folds. Where more than 10 stable features exist, only the top 10 are shown; complete lists are in Appendix (Table 8.6).

Task	LC		ADNI-A+	
	proteomic	all_features	proteomic	all_features
Multiclass	P18065, P19021 , P01024, P01859, P02649, P00747 , P01011, P05090, P12109, Q14624 [+10]	P18065, P19021, MMSE, P01859, AGE, P24592	Q92859 , Q969P0 , P10451, Q14515, P13591, P12109, P02765, Q14118, P01859	MMSE, P-TAU, Q92859, Q14515, P10451, P13591
2-year OVA	P01024, P19021 , P02649, P18065 , P01859, P36222, P12109	P01024 , P02649, P19021, P18065, P01859, P36222	Q969P0 , P10451, P24592 , Q14515, Q14118 , P05090, P29622, P01023 , P02765, P01033 [+5]	MMSE, P-TAU, P02765
3-year OVA	Q9P121, P12109, Q8TEU8	Q9P121, P12109, Q8TEU8	P10451 , P13591, P29622	P10451, P13591 , P29622, Q9P121, GENDER, AGE
4-year OVA	P18065 , Q969P0, P06396, P00746, P00747, P02765	P18065 , Q969P0, P10451, P06396, P00746, P04004, P02760, P02765, P00747	Q92859, P12109 , P25311, P02765, P02760, P01859, P01011, Q9P121	P12109, Q92859 , GENDER , Q14515, P01024, P01859, P02760, P02765, P01011
5+ year OVA	P01011, Q14624 , P19021, P02649, P05090, P00747, P01031	P01011 , MMSE, Q14624, P02649 , P19021, P05090 , Q969P0, Q14118 , P35858, P00747 [+7]	Q969P0 , P12109	MMSE

Table 4.11: Optimal feature subset size (k^*) selected by RPT- β for each outer fold (F1, F2, F3), task, cohort, model, and feature set. SH40-LC and SH40-ADNI-A+ rows report k^* for the respective datasets restricted to the 40 shared proteins (RF only); A dash (—) indicates that the configuration was not evaluated.

Task	Cohort	Model	clinical			c-CSF			proteomic			all_features		
			F1	F2	F3	F1	F2	F3	F1	F2	F3	F1	F2	F3
4-TW formulation														
2-year OVA	LC	RF	1	2	1	1	2	2	69	63	7	85	53	8
		SVM	1	2	1	1	2	2	70	62	6	88	1	8
	ADNI-A+	RF	1	1	1	2	1	2	105	98	51	118	19	29
		SVM	1	1	1	2	1	1	136	100	50	109	12	30
	SH40-LC	RF		—			—		6	8	20	15	4	7
	SH40-ADNI-A+	RF		—			—		8	21	27	8	2	3
3-year OVA	LC	RF	1	2	1	1	2	2	48	78	46	58	21	39
		SVM	2	2	1	2	1	1	51	31	58	66	3	41
	ADNI-A+	RF	1	1	2	1	1	1	120	125	152	86	143	146
		SVM	2	2	2	2	1	1	29	140	103	30	144	86
	SH40-LC	RF		—			—		14	4	3	7	5	4
	SH40-ADNI-A+	RF		—			—		4	8	2	5	27	2
4-year OVA	LC	RF	1	2	2	1	1	1	12	64	17	11	63	9
		SVM	2	2	2	1	2	2	11	52	30	8	66	31
	ADNI-A+	RF	1	2	1	1	1	1	9	248	102	11	170	73
		SVM	2	1	1	1	1	1	7	127	75	9	176	23
	SH40-LC	RF		—			—		7	17	6	11	14	11
	SH40-ADNI-A+	RF		—			—		11	13	12	12	16	8
5+ year OVA	LC	RF	1	1	1	1	2	1	24	3	88	41	3	81
		SVM	1	1	1	1	2	1	92	4	93	57	3	102
	ADNI-A+	RF	1	1	1	1	2	2	102	83	3	80	1	4
		SVM	1	1	1	1	2	1	91	102	8	71	1	4
	SH40-LC	RF		—			—		12	19	2	16	24	19
	SH40-ADNI-A+	RF		—			—		3	1	11	7	1	1
Multiclass	LC	RF	1	2	1	1	2	2	51	124	93	47	133	86
		SVM	1	1	1	2	1	2	40	124	83	47	133	93
	ADNI-A+	RF	1	1	1	1	2	2	149	202	149	128	1	57
		SVM	1	1	1	1	2	1	149	142	179	139	1	57
	SH40-LC	RF		—			—		12	25	22	9	19	1
	SH40-ADNI-A+	RF		—			—		11	3	12	15	1	7
Regression	LC	RF	1	1	1	1	2	1	78	1	4	64	2	4
		SVM	1	1	1	1	2	1	78	1	4	64	1	4
	ADNI-A+	RF	1	1	1	1	1	1	119	103	231	64	66	22
		SVM	1	1	1	1	1	1	112	103	231	64	66	9
EML-TW formulation														
Mid OVA	LC	RF	2	2	1	2	2	2	14	1	13	35	1	15
		SVM	2	2	1	1	2	2	15	3	11	13	1	11
	ADNI-A+	RF	2	2	2	1	1	1	49	205	4	92	207	3
		SVM	2	1	2	2	1	1	17	201	3	3	200	3
Multiclass	LC	RF	1	1	1	2	2	1	62	1	25	96	1	22
		SVM	1	1	1	2	2	1	65	1	25	65	1	15
	ADNI-A+	RF	1	1	1	1	2	2	90	143	5	77	1	6
		SVM	1	1	1	1	1	1	90	168	5	81	1	6

4.2.4 Summary of Main Findings

Which proteins were consistently selected by the FSE-StabPred ensemble across folds for the most informative configurations? For the LC cohort, a small but stable core of proteins was identified for the 2-year OVA window under both RF and SVM, with several proteins confirmed in all three outer folds under both `proteomic` and `all_features` feature sets (Table 4.8). These proteins represent the most stable proteomic selections for this configuration and will be further examined in Section 4.3, where they are integrated with Gini importance scores and Cox proportional hazards survival analysis to identify the most robust candidate biomarkers. For the ADNI-A+ cohort, the 5+ year OVA window under SVM and `all_features` yielded a stable but small proteomic core, dominated by MMSE and P-TAU at the top of the ranking, reflecting the predominant contribution of clinical and `c-CSF` features over proteomic ones for this cohort. For all other configurations in both cohorts, feature selections were sparse, highly variable across folds, or inconsistent between models, consistent with the absence of a learnable discriminative signal reported in Section 4.1.

Is the feature selection consistent across predictive models within the same cohort? Within the same cohort and time window, RF and SVM showed broadly consistent proteomic selections for the primary configurations of interest, with the most informative proteins appearing stably in the top selections of both models. This cross-model consistency strengthens the candidacy of the recurrently selected proteins as robust predictive candidates, as their identification does not depend on the specific inductive bias of a single model. For non-performant configurations, consistency between models was lower and more variable, as expected in the absence of a reliable discriminative signal.

Do the two cohorts identify a common proteomic signature within the shared panel? The cross-cohort comparison on the SH40 panel (Section 4.2.3) reveals that the two cohorts identify largely non-overlapping sets of discriminative proteins, even within the constrained shared panel. The Gini importance profiles for the 40 shared proteins diverge substantially between LC and ADNI-A+, with only complement component C3 (P01024) showing comparable importance in both cohorts. These findings confirm that the two datasets do not share a common proteomic signature within the SH40 panel, and that the observed divergence reflects both the different biological composition of the cohorts and the technical differences between the two proteomic platforms.

4.3 Biomarkers Evaluation

This section presents the biomarker analysis performed on the feature subsets selected by the FSE-StabPred ensemble and addresses the following questions:

- (a) Which proteins emerge as the most robust candidates for MCI-to-AD conversion, supported by convergent evidence from feature selection, Gini importance, and survival analysis?
- (b) What biological processes are associated with the identified candidate biomarkers, and are these consistent with known pathophysiological mechanisms of Alzheimer’s disease?

The analysis is organised into three subsections. For the LC cohort (Section 4.3.1), the analysis is the most comprehensive, structured around three analytical paragraphs: a **Cox Proportional Hazards Survival Analysis**, assessing the univariate association between baseline protein abundance and time to conversion; a **Gini Importance Analysis**, extracting and aggregating the intrinsic feature importance scores from the Random Forest models across outer folds; and a **Strong Candidate Biomarkers Analysis**, integrating the two preceding frameworks into a single summary of the most robust proteomic candidates, complemented by a pathway enrichment analysis restricted to AD-relevant biological processes. For the ADNI-A+ cohort (Section 4.3.2), the analysis is limited to the Cox proportional hazards survival analysis, reported as a descriptive finding only: as discussed in Section 4.1.2, the proteomic panel did not provide a meaningful predictive advantage over c-CSF features alone in this cohort, and the deeper biomarker characterisation performed on LC is therefore not pursued. Finally, the Cross-cohort Overlap subsection (Section 4.3.3) presents a Gini importance comparison restricted to the 40 shared proteins of the SH40 panel, with the aim of assessing whether the two cohorts assign comparable discriminative importance to the same features within the shared proteomic space.

4.3.1 LC cohort

Cox Proportional Hazards Survival Analysis To assess whether individual baseline protein levels are statistically associated with time to progression to dementia, a Cox proportional hazards survival analysis was performed on the LC cohort, as described in Section 3.4. Eight proteins reached statistical significance at the $p < 0.05$ threshold (Table 4.12), of which five showed a positive $\log(\text{HR})$ indicating an association with faster progression, and three a negative $\log(\text{HR})$ indicating a protective association. The results are visualised in Figure 4.11. Among the risk-associated proteins, haptoglobin (HP, P00738) was the most strongly significant

($\log(\text{HR}) = 0.386$, $p = 0.0013$), while apolipoprotein E (APOE, P02649; $\log(\text{HR}) = -0.376$, $p = 0.008$) showed the strongest protective association. The complete set of significant proteins and their effect sizes are reported in Table 4.12. The set of significant proteins was subsequently used to perform a pathway enrichment analysis restricted to AD-relevant biological processes.

Gini Importance Analysis Gini importance scores were extracted from the Random Forest models trained on the selected feature subsets at each outer fold, as described in Section 3.4. The aggregated bar plots showing mean Gini importance $\pm$ SD across the three outer folds are shown in Figure 4.12.

For the `proteomic` feature set, haptoglobin (HP, P00738) ranks first across all three outer folds, followed by a set of hard-selected proteins (3/3 folds) including COL1A1 (P02452), TGFBI (Q15582), and ENO1 (P06733). A second tier of soft-selected features (2/3 folds) is visible in the figure. Among the top 20, two proteins overlap with the Cox significant set: HP (P00738, risk) and C1QB (P02746, protective), providing convergent support from two independent analytical frameworks. The large majority of the top 20 features were selected by FSE-StabPred in at least one fold, with only one protein not selected in any fold.

For the `all_features` feature set, the ranking is broadly consistent with the `proteomic`-only case, with HP, COL1A1, and TGFBI again leading the ranking as hard-selected features. However, a reshuffling of the Gini importance ranking occurs: several proteins prominent in the `proteomic`-only top 20 are replaced by different features, not because they are less informative per se, but because their discriminative contribution is subsumed at the level of individual decision trees by the clinical and c-CSF variables when these are included in the feature space. FSE-StabPred is less susceptible to this bias, as it combines multiple statistical ranking methods beyond decision trees alone, and consistently reselects the proteomic features identified under the `proteomic`-only setting regardless of the expansion of the feature space. Notably, no clinical or c-CSF features appear among the top 20 Gini importance scores under either feature set, confirming that the performance gain associated with the `all_features` configuration is driven by robust proteomic signal rather than by clinical covariates.

Strong Candidate Biomarkers Analysis. To identify the most robust proteomic candidates, the results of the FSE-StabPred ensemble feature selection, the Gini importance analysis, and the Cox proportional hazards survival analysis were integrated into a single summary. A protein was considered a strong candidate if it was selected by FSE-StabPred in at least 2 out of 3 outer folds under the `proteomic` feature set, maintained its importance in the `all_features` feature set, and appeared among the

top 20 features by mean Gini importance under at least one of the two feature sets. The Cox direction and significance are reported as an additional layer of biological evidence, but were not used as a selection criterion. The resulting set of 10 strong candidate biomarkers is reported in Table 4.13.

Two candidates are supported by all three analytical frameworks: HP (P00738), which ranks first by mean Gini importance under both feature sets and shows the strongest Cox association ($p = 0.0013$, risk direction), and C1QB (P02746), which is hard-selected in all three folds under both feature sets and is Cox-significant in the protective direction ($p = 0.0281$). These two proteins represent the strongest candidates for further investigation. The remaining eight candidates are supported by FSE-StabPred and Gini importance but do not reach statistical significance in the Cox model; two of these (IGHG2 (P01859, $p = 0.055$) and CNDP1 (Q96KN2, $p = 0.085$)) show borderline p -values.

A pathway enrichment analysis was subsequently performed on the strong candidate set (Figure 4.13). On the risk side, the enriched pathways fall into three main categories: intracellular transport, neural development and regeneration, and oxidative stress and mitochondrial function. On the protective side, three categories emerge: extracellular matrix and collagen organisation, neural development and regeneration, and neuroinflammation. The biological interpretation of these pathways and of the individual candidate proteins is deferred to Chapter 5.

Table 4.12: Proteins statistically significant in the Cox proportional hazards survival analysis ($p < 0.05$), ordered by increasing p -value. $\log(\text{HR})$ values are per one standard deviation change in normalised protein abundance. A positive $\log(\text{HR})$ indicates an association with faster conversion (risk); a negative $\log(\text{HR})$ indicates an association with slower conversion (protective).

UniProt	Gene	$\log(\text{HR})$	p -value	Direction
P00738	HP	0.386	0.0013	Risk
P02649	APOE	-0.376	0.0076	Protective
P02746	C1QB	-0.327	0.0281	Protective
Q14624	ITIH4	0.306	0.0329	Risk
P01024	C3	0.301	0.0379	Risk
P09972	ALDOC	-0.260	0.0403	Protective
P00747	PLG	0.273	0.0458	Risk
P01599	IGKV1-17	0.308	0.0474	Risk

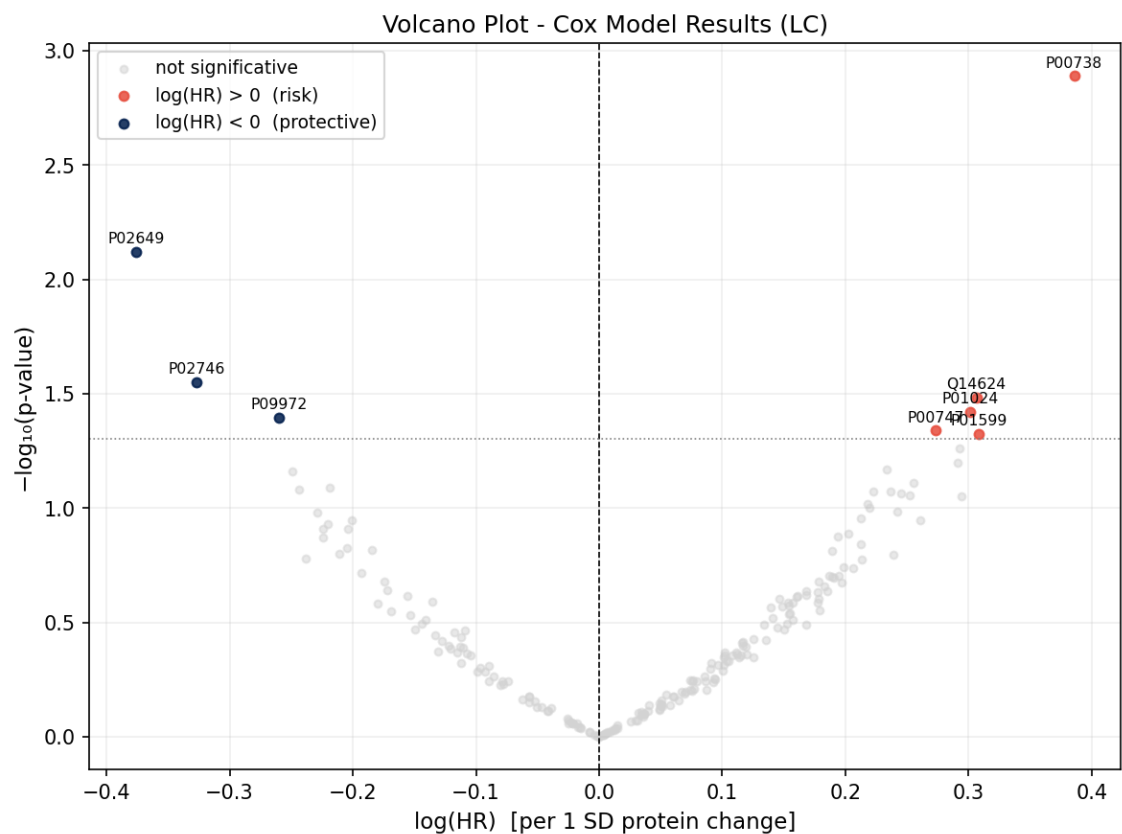
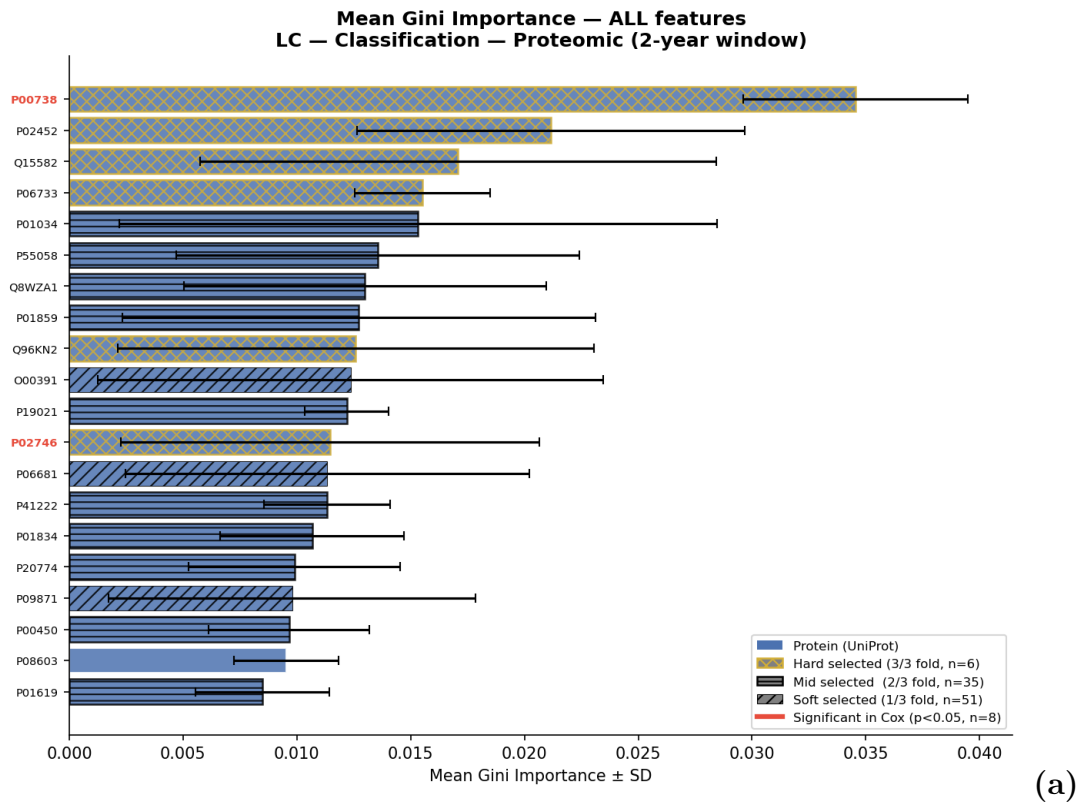
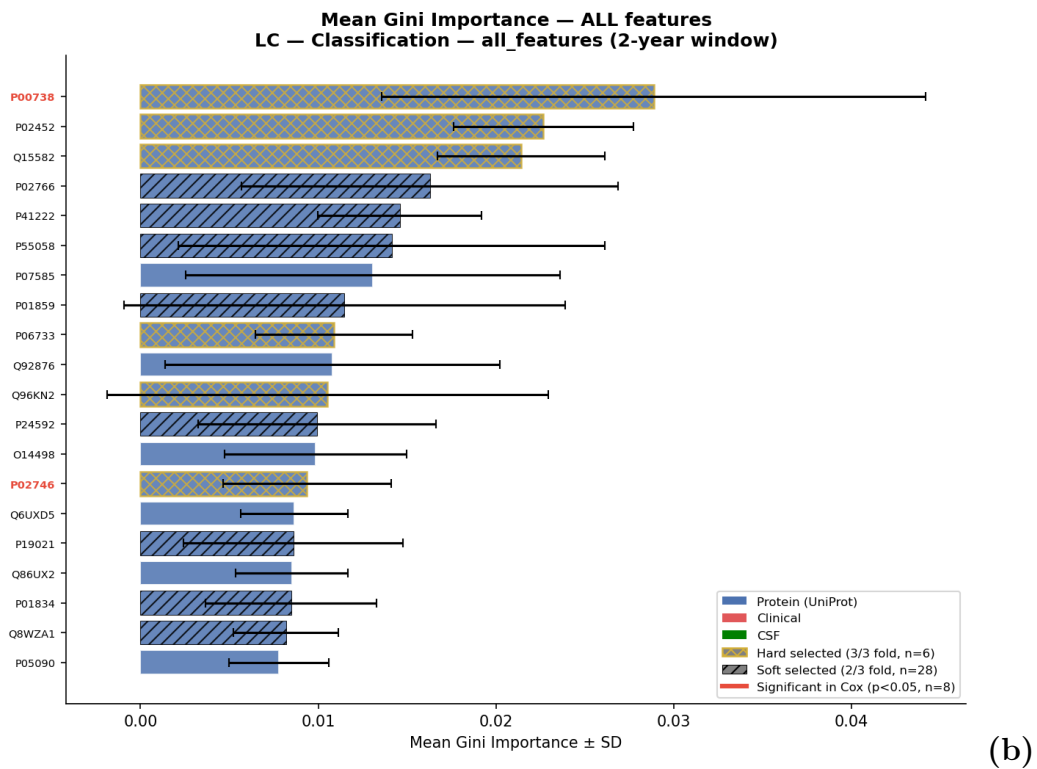


Figure 4.11: Volcano plot of Cox proportional hazards survival analysis results for the LC cohort. The x-axis shows $\log(\text{HR})$ per one standard deviation change in normalised protein abundance; the y-axis shows $-\log_{10}(\text{p-value})$. Red points indicate risk-associated proteins ($\log(\text{HR}) > 0$); blue points indicate protective proteins ($\log(\text{HR}) < 0$); grey points are not statistically significant ($p \geq 0.05$). The dashed horizontal line marks the $p = 0.05$ threshold.



proteomic feature set.



all_features feature set.

Figure 4.12: Mean Gini importance ($\pm$ SD across 3 outer folds) for the top 20 features in the LC cohort, 2-year OVA window. **(a)** proteomic feature set. **(b)** all_features feature set. Hatch patterns encode FSE-StabPred selection stability: gold cross-hatch for features selected in all 3 folds; diagonal lines for features selected in 2 out of 3 folds; no hatch for features not selected by FSE-StabPred. Feature labels in red indicate proteins that are also statistically significant in the Cox proportional hazards survival analysis ($p < 0.05$).

Table 4.13: Strong candidate biomarkers for the LC cohort, 2-year OVA window, identified by convergent evidence from FSE-StabPred ensemble feature selection, Gini importance analysis, and Cox proportional hazards survival analysis. RPT_{prot} and $RPT_{\text{all_features}}$: number of outer folds (out of 3) in which the protein was selected by FSE-StabPred under the **proteomic** and **all_features** feature sets respectively. $Gini_{\text{prot}}$ and $Gini_{\text{all_features}}$: mean Gini importance across the three outer folds. Cox direction and p -value are from the univariate Cox model. Bold rows indicates proteins with significant Cox HR value, considering $p < 0.05$.

UniProt	Gene	RPT_{prot}	$RPT_{\text{all_features}}$	$Gini_{\text{prot}}$	$Gini_{\text{all_features}}$	Cox dir.	Cox p
P00738	HP	3	3	0.0346	0.0289	Risk	0.0013
Q15582	TGFBI	3	3	0.0171	0.0214	Risk	0.2585
P06733	ENO1	3	2	0.0155	0.0109	Risk	0.7632
P02746	C1QB	3	3	0.0115	0.0094	Protective	0.0281
P02452	COL1A1	2	3	0.0212	0.0227	Risk	0.3030
P55058	PLTP	2	2	0.0136	0.0141	Risk	0.1294
Q8WZA1	POMGNT1	2	2	0.0130	0.0082	Protective	0.1131
P01859	IGHG2	2	2	0.0127	0.0115	Risk	0.0546
Q96KN2	CNDP1	2	2	0.0126	0.0105	Risk	0.0845
P19021	PAM	2	2	0.0122	0.0086	Protective	0.1239

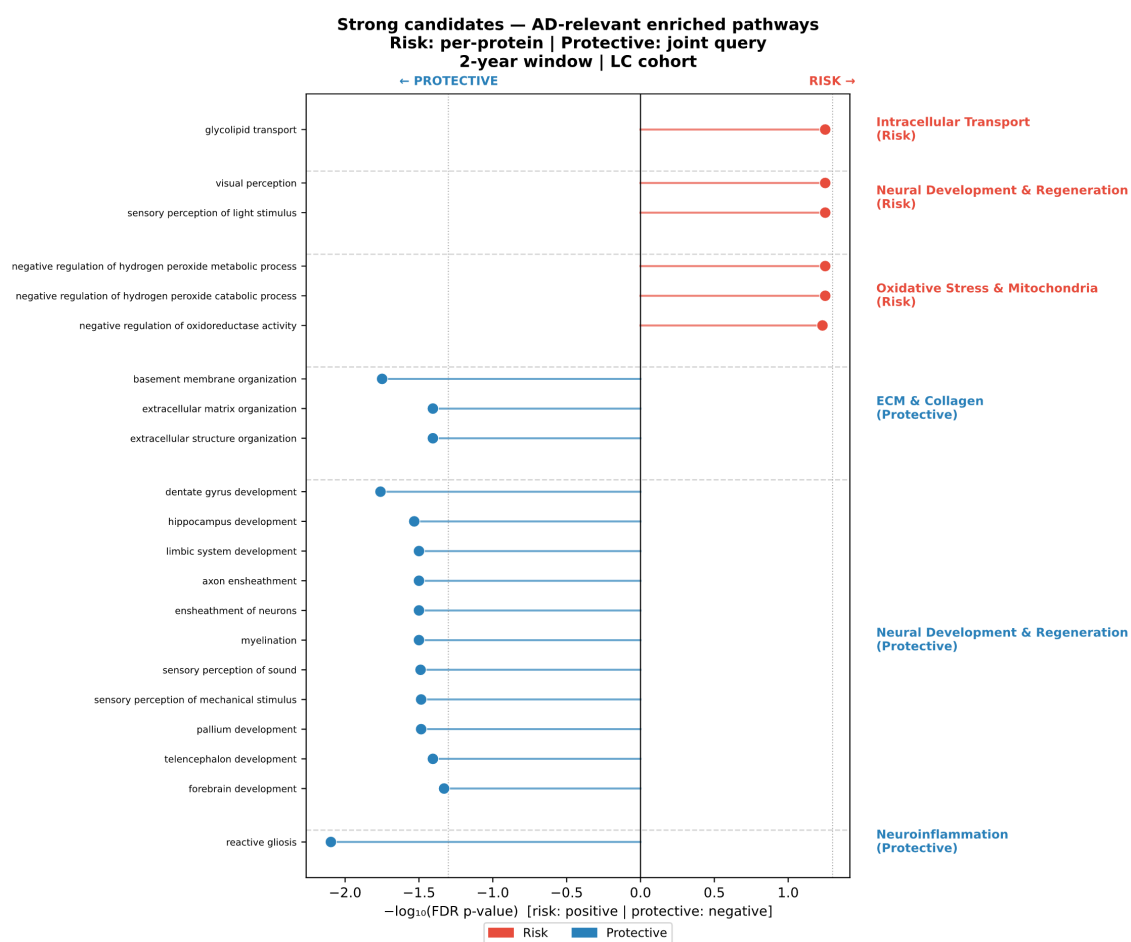


Figure 4.13: AD-relevant enriched pathways for the strong candidate biomarkers of the LC cohort, 2-year OVA window. Risk-associated pathways (red) are derived from per-protein enrichment of risk candidates; protective pathways (blue) are derived from a joint query of protective candidates. The x-axis shows $-\log_{10}(\text{FDR } p\text{-value})$, with positive values indicating risk-associated pathways and negative values indicating protective pathways. Pathways are grouped by AD-relevant biological category.

4.3.2 ADNI-A+ Cohort.

This subsection reports the Cox proportional hazards survival analysis for the ADNI-A+ cohort. As discussed in Section 4.1.2, the proteomic panel did not provide a meaningful predictive advantage over c-CSF features alone in this cohort; the biomarker analysis is therefore limited to a descriptive survival analysis, without the integration of Gini importance and feature selection pursued for LC.

Cox Proportional Hazards Survival Analysis. A Cox proportional hazards survival analysis was performed on the ADNI-A+ cohort following the same procedure described in Section 3.4, applied to all 570 proteins in the SOMAscan panel after preprocessing. A total of 51 proteins reached statistical significance at the $p < 0.05$ threshold, of which 10 showed a positive $\log(\text{HR})$ (risk direction) and 41 a negative $\log(\text{HR})$ (protective direction). The top 20 proteins by significance are reported in Table 4.14; the complete list is provided in Appendix (Table 8.9). The results are visualised in Figure 4.14.

The larger number of significant proteins relative to LC (51 vs. 8) is consistent with the larger sample size ($n = 187$ vs. $n = 56$), which provides greater statistical power to detect univariate associations. However, the effect sizes are considerably smaller than those observed in LC: the maximum $|\log(\text{HR})|$ in ADNI-A+ is approximately 0.24 (SPARCL1, Q14515), compared to 0.39 (HP, P00738) in LC, and the majority of significant proteins show $|\log(\text{HR})|$ values in the range 0.13–0.24. This pattern suggests that while more proteins are detectable as statistically associated with time to conversion in the larger cohort, their individual effect sizes are modest and the overall proteomic survival signal is weaker than in LC. This is consistent with the performance results reported in Section 4.1.2, where proteomic features did not provide a meaningful predictive advantage over c-CSF features alone, and with the entirely A + T– composition of the cohort, where c-CSF measures may already capture the majority of prognostic variability [52]. The biological interpretation of these findings and the implications of the divergence between the two cohorts are addressed in Chapter 5.

Table 4.14: Top 20 proteins statistically significant in the Cox proportional hazards survival analysis on the ADNI-A+ cohort ($p < 0.05$), ordered by increasing p -value. $\log(\text{HR})$ values are per one standard deviation change in normalised protein relative abundance. A positive $\log(\text{HR})$ indicates an association with faster conversion (risk); a negative $\log(\text{HR})$ indicates an association with slower conversion (protective). The complete list of 51 significant proteins is provided in Appendix (Table 8.9).

UniProt	Gene	$\log(\text{HR})$	p -value	Direction
Q14515	SPARCL1	-0.238	0.0017	Protective
P61981	YWHAG	+0.218	0.0019	Risk
Q96MU8	KREMEN1	-0.206	0.0023	Protective
P60510	PPP4C	-0.219	0.0035	Protective
P10092	CALCB	-0.232	0.0036	Protective
O75023	LILRB5	-0.215	0.0037	Protective
O60462	NRP2	-0.207	0.0044	Protective
Q13253	NOG	-0.199	0.0047	Protective
P62258	YWHAE	+0.217	0.0059	Risk
P10747	CD28	-0.195	0.0062	Protective
P02751	FN1	-0.220	0.0066	Protective
P31946	YWHAB	+0.197	0.0078	Risk
P63104	YWHAZ	+0.196	0.0118	Risk
Q96JB5	CDK5RAP3	-0.164	0.0122	Protective
Q9H320	VCX	-0.173	0.0126	Protective
P06276	BCHE	-0.174	0.0151	Protective
Q6UXG2	ELAPOR1	-0.181	0.0153	Protective
Q92859	NEO1	-0.168	0.0154	Protective
Q8NIFY4	SEMA6D	-0.172	0.0171	Protective
P14778	IL1R1	-0.194	0.0176	Protective

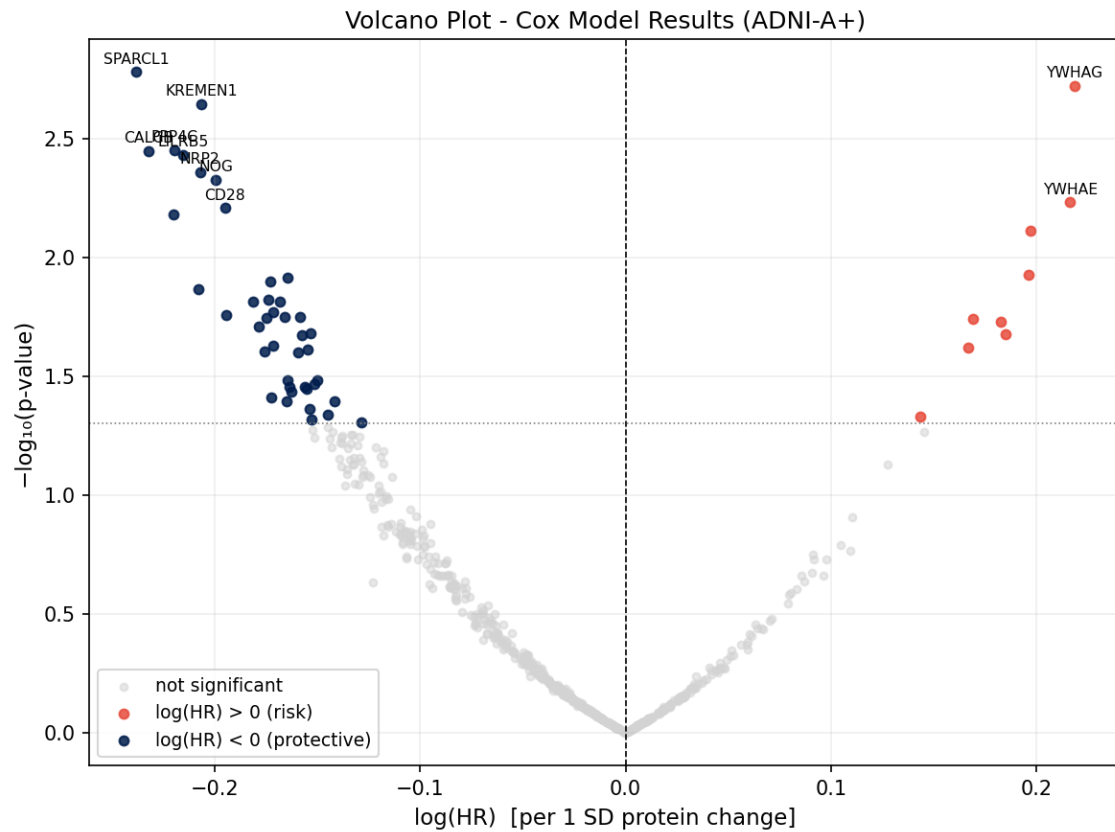


Figure 4.14: Volcano plot of Cox proportional hazards survival analysis results for the ADNI-A+ cohort. The x-axis shows $\log(HR)$ per one standard deviation change in normalised protein relative abundance; the y-axis shows $-\log_{10}(p\text{-value})$. Red points indicate risk-associated proteins ($\log(HR) > 0$); blue points indicate protective proteins ($\log(HR) < 0$); grey points are not statistically significant ($p \geq 0.05$). The dashed horizontal line marks the $p = 0.05$ threshold.

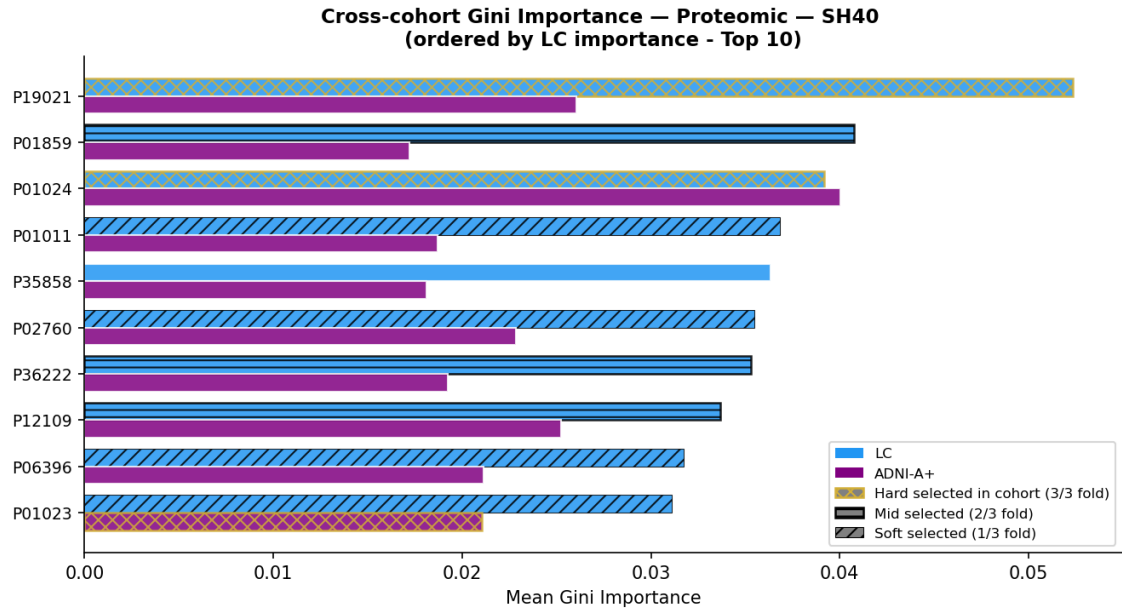
4.3.3 Shared Proteomic Panel between LC and ADNI-A+ Cohorts (SH40)

To complement the feature selection comparison presented in Section 4.2.3, a cross-cohort Gini importance analysis was performed on the SH40 panel, comparing the mean Gini importance of each shared protein under the LC and ADNI-A+ Random Forest models. The results are shown in Figure 4.15 for both the `proteomic` and `all_features` feature sets, ordered by LC importance.

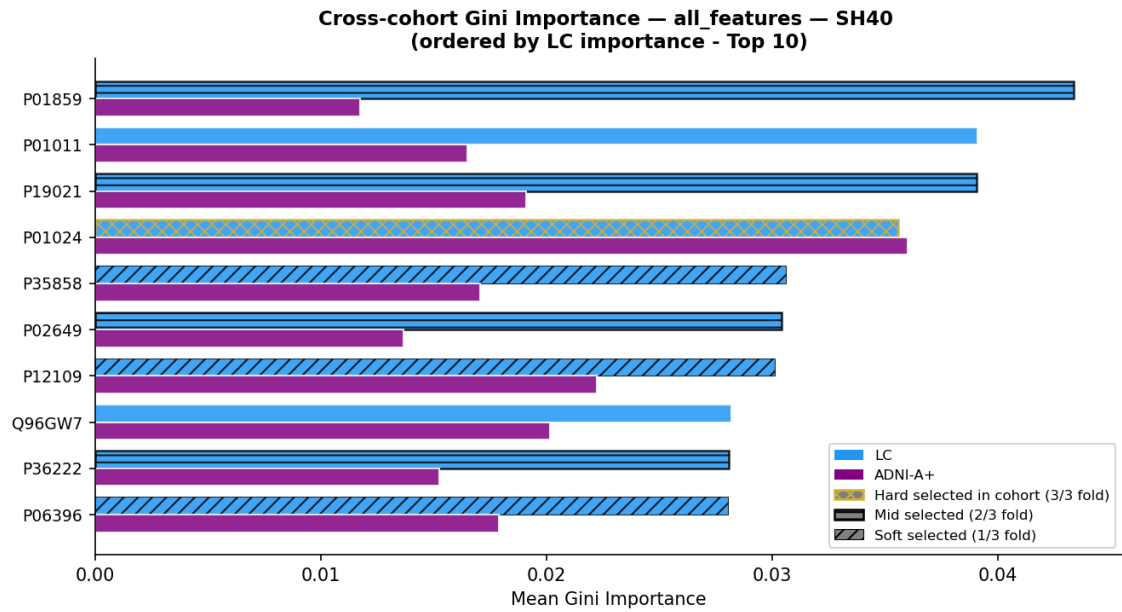
The most striking observation is the substantial divergence in Gini importance profiles between the two cohorts. For the `proteomic` feature set, the proteins ranked highest by LC show considerably lower importance in ADNI-A+, and none of the top 10 LC proteins are among the most discriminative features for the ADNI-A+ model. Conversely, the protein showing the highest Gini importance in ADNI-A+ within the shared panel is hard-selected in all three folds for that cohort, yet shows very low importance in LC. This divergence is consistent with the markedly different feature selections observed for the two cohorts within the SH40 panel (Section 4.2.3), and reinforces the conclusion that the two datasets do not share a common proteomic signature within the shared panel.

A partial exception is represented by complement component C3 (P01024), which shows comparable and relatively high Gini importance in both cohorts under both feature sets, and is highlighted in Figure 4.15 as Cox-significant in the LC cohort. Notably, P01024 was not selected by FSE-StabPred in any outer fold for ADNI-A+ within the SH40 panel, indicating that despite its high RF-intrinsic importance it did not reach the predictability and stability threshold required for selection by the ensemble. For the `all_features` feature set, the pattern is broadly consistent with the `proteomic`-only case.

Overall, the cross-cohort Gini importance comparison confirms that the SH40 panel does not capture a shared proteomic signature between LC and ADNI-A+, and that the two cohorts identify largely non-overlapping sets of discriminative proteins even within the constrained shared panel. The biological and technical reasons underlying this divergence are discussed in Chapter 5.



(a) proteomic feature set.



(b) all_features feature set.

Figure 4.15: Cross-cohort Gini importance comparison on the overlapping 40-protein panel (SH40), showing mean Gini importance for LC (blue) and ADNI-A+ (purple) Random Forest models. **(a)** proteomic feature set. **(b)** all_features feature set. Features are ordered by decreasing LC importance (top 10 shown). Hatch patterns encode FSE-StabPred selection stability within each cohort: gold cross-hatch for features selected in all 3 folds; diagonal lines for features selected in 2 out of 3 folds; no hatch for features selected in 1 fold only.

4.3.4 Summary of Main Findings

Which proteins emerge as the most robust candidates for MCI-to-AD conversion, supported by convergent evidence from multiple analytical frameworks? For the LC cohort, ten strong candidate biomarkers were identified for the 2-year OVA window through the integration of FSE-StabPred feature selection, Gini importance analysis, and Cox proportional hazards survival analysis (Table 4.13). Of these, haptoglobin (HP, P00738) and complement C1q subunit B (C1QB, P02746) are the most robustly supported, being confirmed by all three analytical frameworks. The remaining eight candidates are supported by FSE-StabPred and Gini importance but do not reach statistical significance in the Cox model. For the ADNI-A+ cohort, 51 proteins reached statistical significance in the Cox analysis, but with considerably smaller effect sizes than in LC, consistent with the weaker proteomic signal observed for this cohort in the predictive performance analysis.

What biological processes are associated with the identified candidate biomarkers, and are these consistent with known pathophysiological mechanisms of Alzheimer’s disease? The pathway enrichment analysis performed on the strong candidate set identified processes falling into three main categories on the risk side (intracellular transport, neural development and regeneration, and oxidative stress) and three on the protective side (extracellular matrix and collagen organisation, neural development and regeneration, and neuroinflammation). The biological possible interpretation of these pathways and their relevance to AD pathophysiology is addressed in Chapter 5, where the candidate biomarkers are discussed in the context of the existing literature.

5. Discussion

This chapter interprets the main findings of the present study in light of the four research questions defined in Section 1.1.

5.1 Do CSF Proteomic Features Improve Prognostic Performance?

Proteomic role in predictive performance in LC cohort. In the LC cohort, proteomic features provided a clear and statistically supported improvement in MCC for the 2-year OVA window relative to `clinical` or `c-CSF` features alone (Friedman test: RF $\chi^2 = 7.93$, $p = 0.048$; SVM $\chi^2 = 8.2$, $p = 0.042$), with Gini importance analysis confirming that the performance gain is driven by robust proteomic signal rather than by clinical covariates. The `all_features` feature set achieved the best overall performance (MCC 0.448 ± 0.072), indicating that combining all modalities yields further gains over any single one alone. No meaningful discriminative signal was identified beyond the 2-year horizon, a pattern consistent with a fundamental property of the prediction problem: as the conversion horizon lengthens, the prognostic information contained in a single baseline proteomic measurement becomes progressively less sufficient, as the trajectory of neurodegeneration is increasingly shaped by dynamic biological processes and external factors not captured at a single time point. Such factors include environmental exposures that accelerate biological ageing through epigenetic modifications [21], acute events such as severe sepsis that independently induce long-term cognitive impairment [18], psychiatric conditions such as depression [44], metabolic dysregulation driven by insulin resistance [12], and protective lifestyle factors such as bilingualism that may delay clinical expression of the disease by several years [26]. The 2-year window represents the most proximal horizon, where the baseline proteomic profile is most likely to reflect the biological state immediately preceding conversion and where this external variability has had the least time to accumulate.

Proteomic role in predictive performance in ADNI-A+ cohort. In the ADNI-A+ cohort, by contrast, proteomic features did not provide a meaningful predictive advantage over `c-CSF` features alone, with the most informative contri-

bution for the 5+ year OVA window coming from P-TAU and MMSE rather than from proteomic candidates. The likely explanation is technical: the SOMAscan 7K platform quantifies proteins indirectly through aptamer binding in native conformation, introducing sources of noise absent in mass-spectrometry approaches, where proteins undergo conformational changes, post-translational modifications, or protein-protein interactions that may occlude the aptamer binding epitope, causing SOMAscan to reflect epitope availability rather than actual protein concentration [3]. This difference in platform quality may partly explains why proteomic features provided a clear predictive advantage in LC but not in ADNI-A+. The observation that a larger cohort ($n = 187$ vs $n = 56$) did not translate into improved proteomic performance underscores a fundamental principle: predictive power depends not only on sample size but critically on the informativeness of the features measured.

The relevance of neuropsychological data over the proteomic features for long-term time-to-conversion. Comparing with Pereira et al. [35], neuropsychological measures substantially outperform proteomic features for this prediction task, achieving AUC above 0.87 across multiple time windows. Neuropsychological measures such as MMSE are more directly connected to the outcome of interest, as cognitive decline is by definition the variable being predicted, whereas proteomic profiles capture an upstream biological substrate susceptible to the environmental and biological variability discussed above [21, 18, 44, 12, 26]. The consistent emergence of MMSE among the top selections for the 5+ year window in both cohorts is directly aligned with Pereira et al. [34], who demonstrated that neuropsychological measures carry meaningful prognostic information particularly for longer time windows. The advantage of proteomic biomarkers, however, lies in their independence from cognitive assessment and their potential to reveal molecular mechanisms of disease progression preceding the functional decline captured by neuropsychological tests.

5.2 Which Modelling Configuration is Best Suited to Time-to-Conversion Prediction?

Regression approach. The regression approach was consistently underpowered across both cohorts, with RMSE values close to the range of the target variable, indicating that neither model learned a reliable continuous mapping between proteomic profiles and exact time to conversion. The discretisation of the prediction problem into time windows proved a more tractable approach under the present data constraints.

Multiclass approach. The multiclass formulation did not yield any advantage over the OVA approach in either cohort. In LC, the per-class analysis revealed that the multiclass model recovers slightly higher sensitivity at the cost of a substantial specificity drop, a pattern attributable to the redistribution of the aggregated negative class across time windows rather than to genuine discriminative capacity. The high overlap between multiclass and OVA feature selections confirms that the multiclass model captures window-specific signal without consolidating it into a more powerful global model. These limitations are more likely attributable to the small and imbalanced datasets than to a fundamental limitation of the formulation, and a multiclass approach on a larger and more balanced cohort remains an interesting direction for future work.

EML-TW formulation. To reduce class imbalance, a merging procedure was applied to the intermediate classes into a single mid class (EML-TW). However, this alternative formulation did not bring any consistent improvement. A possible reason is that the merged class did not become sufficiently large to gain meaningful predictive power, while the fusion of two temporally adjacent but biologically distinct groups likely introduced additional noise rather than a cleaner learning target.

SVM vs RF modeling approach. Regarding the two models, RF and SVM showed complementary behaviours: in LC, RF’s more conservative prediction behaviour made it the more reliable choice, whereas in ADNI-A+ SVM’s higher sensitivity for the 5+ year window proved more appropriate given the larger sample size. Importantly, both models independently identified the same window-specific patterns, providing cross-model confirmation of the identified signals.

In summary, the OVA classification with the 4-TW scheme emerged as the most appropriate framework for both cohorts, with meaningful predictive performances relative to OVA with RF and `all_features` configuration for 2-year time-to-conversion task in LC cohort (MCC 0.448 ± 0.072 , AUC 0.732 ± 0.067), and OVA with SVM and `all_features` configuration for 5+ year task in ADNI-A+ (MCC 0.328 ± 0.018).

5.3 Which Proteomic Biomarkers Emerge as the Most Informative and What is Their Biological Relevance?

This section interprets the ten strong candidate biomarkers identified for the LC cohort 2-year OVA window (Table 4.13). Their consistent selection by FSE-StabPred across folds and models indicates predictive relevance within the present

pipeline; literature support provides biological plausibility but does not establish causal involvement in disease progression. All interpretations remain exploratory given the limited sample size and the absence of independent replication.

Neuroinflammation and complement activation. The most robustly supported theme is neuroinflammation, represented by HP and C1QB, the only two candidates supported by all three analytical frameworks.

1. **HP (P00738, Risk)** is an acute-phase glycoprotein that binds free haemoglobin and modulates microglia polarisation [28]; elevated HP levels have been reported in AD serum with a positive association with cognitive impairment severity [42]. Its risk association ($\log(\text{HR}) = 0.386$, $p = 0.0013$) and top Gini ranking suggest that its elevation in faster-converting patients may reflect a compensatory response to increased neuroinflammatory burden, though the mechanistic direction cannot be established from the present data.
2. **C1QB (P02746, Protective)** shows a protective association ($\log(\text{HR}) = -0.327$, $p = 0.028$). C1q has been shown to mediate the relationship between amyloid- β and tau pathology through modulation of microglial and astrocytic signalling, via sTREM2-dependent mechanisms [16].[16]; in early and prodromal phases of AD, it exerts neuroprotective effects through complement-independent mechanisms, including promotion of astrocyte-mediated clearance of amyloid- β and cellular debris [41]. Higher C1QB levels in the MCI patients of the LC cohort may therefore reflect a more active early protective response.
3. **IGHG2 (P01859, Risk)** further contributes to this theme (borderline $p = 0.055$). Immunoglobulin G allotypes have been proposed as AD susceptibility factors through epistatic interactions [32], though its specific role in LC population remains unclear.

Extracellular matrix organisation and structural remodelling.

1. **TGFBI (Q15582, Risk)** is induced by TGF- β signalling, a pathway whose dysregulation in ageing impairs microglial activation and promotes neuroinflammation in AD [50]; its hard-selection in all three folds despite the absence of a significant Cox association suggests predictive relevance.
2. **COL1A1 (P02452, Risk)** encodes the most abundant structural collagen in the human body [30], ranking second by Gini importance and hard-selected under `all_features`. No direct functional association with AD pathophysiology has been established in the literature to date; further targeted analyses

would be useful to characterise its potential relevance and whether a genuine link with time-to-conversion exists.

Amyloid metabolism and lipid signalling.

1. **ENO1 (P06733, Risk)**, beyond its canonical glycolytic role, interacts directly with A β peptides, inhibiting fibril formation and promoting A β degradation, with intracranial ENO1 infusion reducing amyloid deposits in transgenic AD models [17]; its high concentration and selection as predictive feature in late time-to-conversion patient may reflect compensatory upregulation in response to amyloid burden.
2. **PLTP (P55058, Risk)** is involved in lipid and cholesterol metabolism, and PLTP deficiency in aged mice impairs memory and increases A β 42 levels [55]; its risk-associated selection should be related to a compensatory mechanism, but further analysis would be needed to assess its actual role as a predictive biomarker.

Neuropeptide biosynthesis and glycosylation.

1. **PAM (P19021, Protective)** catalyses the α -amidation of neuropeptide precursors, and its activity has been found reduced in the CSF and temporal cortex of AD patients with a decline of approximately 16% per year [53]; higher baseline PAM levels in slower-converting patients may reflect greater residual neuropeptide biosynthesis capacity.
2. **POMGNT1 (Q8WZA1, Protective)** overexpression ameliorated A β accumulation and attenuated tau hyperphosphorylation in preclinical models [11], consistent with the protective direction observed here.
3. **CNDP1 (Q96KN2, Risk)** encodes serum carnosinase with antioxidant properties (borderline $p = 0.085$); evidence linking it specifically to AD pathology remains preliminary [2].

Pathway enrichment. The risk-associated pathways cluster around oxidative stress and mitochondrial dysfunction, neural signalling disruption, and lipid transport, consistent with the selection of HP and ENO1 and with established mechanisms linking amyloid accumulation to reactive oxygen species generation [43, 8]. More broadly, lipid dyshomeostasis has been proposed as a central mechanism in AD pathology, contributing to abnormal APP processing [8]. The protective pathways cluster around neural development and regeneration, ECM organisation, and neuroinflammation modulation via reactive gliosis, consistent with the neuroprotective role of

early complement activation (C1QB) and ECM remodelling in A β clearance [41, 52]. Given that the LC cohort is predominantly A+T+ (87.5%), these findings are most directly interpretable in the context of that subgroup, where synaptic plasticity and immune processes predominate [52], and should be regarded as exploratory pending independent replication.

5.4 To What Extent Do Dataset Size and Panel Composition Influence Generalisability?

Cross-Cohort Analysis. The two cohorts differ both technically and biologically in ways that fundamentally limit cross-cohort generalisability.

- (a) Technically, the SOMAScan platform introduces aptamer-related noise absent in LC-SWATH-MS, reducing the discriminative quality of the ADNI-A+ proteomic signal as discussed in Section 5.1.
- (b) Biologically, the predominantly A+T+ composition of LC (87.5%) versus the entirely A+T− composition of ADNI-A+ implies fundamentally different underlying pathophysiological processes [52], making a direct comparison of proteomic biomarker selections between the two cohorts harder.

The SH40 analysis confirmed these points: even within the constrained shared panel, the two cohorts identified largely non-overlapping discriminative proteins, with only complement component C3 (P01024) showing comparable Gini importance in both, but failing to meet the FSE-StabPred stability and predictability threshold for ADNI-A+.

Dataset size and class imbalance. Within each cohort, feature selections were broadly consistent between RF and SVM, strengthening the candidacy of the recurrently selected proteins as predictively relevant features. However, the instability of k^* in non-performant configurations (Table 4.11) serves as a diagnostic indicator of the absence of a learnable proteomic signature: when predictability does not improve consistently with k , the RPT- β criterion collapses k^* to minimal values for stability reasons rather than discriminative ones. The high standard deviations of Gini importance scores further reflect patient variability between folds under small-sample partitioning.

6. Conclusions and Future Directions

6.1 Conclusions

This thesis evaluated the predictive capacity of baseline CSF proteomic profiles for time-to-conversion from MCI to Alzheimer’s dementia, applying an extended FSE-StabPred ensemble feature selection framework to two independent cohorts profiled on different proteomic platforms.

Do CSF proteomic features improve prognostic performance beyond conventional biomarkers? The answer is affirmative, but only under specific conditions. In the LC cohort, proteomic features provided a statistically supported improvement over `clinical` and `c-CSF` features alone for the 2-year OVA window, with the `all_features` combination yielding the best overall performance. In ADNI-A+, by contrast, proteomic features did not provide a meaningful predictive advantage, with the most informative signal for the 5+ year window coming from `c-CSF` features and MMSE. The reason for this divergence probably arise from the combined effect of biological heterogeneity between cohorts, platform-specific noise in the SOMAscan data, and class imbalance, underscoring that predictive power depends not only on sample size but critically on the quality and biological relevance of the measured features.

Which modelling configuration is best suited to time-to-conversion prediction? Overall predictive performance was poor across most configurations, with small but meaningful results achieved through binary One-vs-All classification. The OVA scheme clearly outperformed the multiclass and regression formulations under the present data constraints. An attempt to recover performance for the minority intermediate windows through the EML-TW merging strategy did not yield any consistent gain. Meaningful performance was therefore limited to the 2-year window in LC and the 5+ year window in ADNI-A+, with the best configurations being the 2-year OVA with RF and the 5+ year OVA with SVM respectively, both using the `all_features` feature set.

Which proteomic biomarkers emerge as the most informative and what is their biological relevance? Ten strong candidate biomarkers were identified for the LC cohort 2-year conversion horizon through convergent evidence from FSE-StabPred selection, Gini importance, and Cox proportional hazards analysis. Haptoglobin (HP) and complement C1q subunit B (C1QB) emerged as the strongest candidates, supported by all three frameworks and by the existing literature, and are proposed as priority targets for experimental validation. The pathway enrichment analysis linked risk candidates to oxidative stress, intracellular lipid transport, and neural signalling disruption, and protective candidates to extracellular matrix organisation, myelination, and early neuroinflammatory regulation, providing a biologically coherent picture consistent with the A+T+ composition of the LC cohort. These results should nonetheless be interpreted with caution, as the enrichment analysis was performed on a small set of candidate proteins, and independent replication is needed to confirm their biological relevance.

To what extent do dataset size and panel composition influence generalisability? The identified proteomic signals are specific to their respective cohorts, platforms, and biological compositions, and cannot be assumed to generalise across settings. The SH40 cross-cohort analysis demonstrates limited cross-cohort consistency, with the two cohorts identifying largely non-overlapping discriminative proteins even within the shared panel; the reasons for this divergence may reflect differences in biological composition, proteomic platform, and sample size, though these can only be proposed as possible explanations. In this direction, independent replication in larger, well-characterised cohorts with matched biological profiles and harmonised proteomic platforms is necessary before any of the identified candidates can be considered for clinical translation.

Taken together, these findings provide a proof of concept that baseline CSF proteomics, when analysed through a robust and stability-aware feature selection framework, can identify prognostic signatures for short-term MCI-to-AD conversion that complement and extend standard clinical and biochemical biomarkers. The utility of this approach remains limited by the current inability to identify reliable biomarkers for longer conversion horizons, which represent a more clinically relevant target for early intervention. However, the methodological pipeline developed here, and the candidate biomarkers it identified, provide a foundation for future work aimed at building more powerful and generalisable prognostic models for Alzheimer’s disease.

6.2 Future Research Directions

Several directions for future work emerge from these findings:

1. The most immediately promising is the replication of the LC analysis on a larger dataset profiled with the same SWATH-MS platform or a methodologically comparable mass-spectrometry approach: under those conditions, a larger sample size could genuinely improve predictive performance, particularly for the intermediate and longer-term conversion windows that were underpowered in the present study, and could support a stratified analysis by tau status that was not feasible here due to the limited number of T− patients.
2. A complementary direction would be to exploit the longitudinal structure of available datasets to model proteomic trajectories over time rather than relying solely on baseline concentrations; longitudinal proteomic data could reveal dynamic biomarker patterns that carry substantially more prognostic information than static baseline profiles, and could facilitate the identification of novel pathways involved in disease progression.
3. A third direction concerns the integration of multi-modal data: given that proteomic features were shown to carry a small but present predictive signal in the LC cohort, combining them with neuropsychological measures [35], and with alternative biological modalities such as metabolomics and transcriptomics, could capture complementary aspects of the disease trajectory and improve performance across all time windows, providing a less noisy and more informative substrate for time-to-conversion prediction, particularly at longer horizons.
4. Finally, the evaluation of gradient boosting methods and other advanced ensemble approaches well-suited to high-dimensional numerical feature spaces represents a natural methodological extension that may yield performance gains in the LC proteomic setting where the existence of a meaningful discriminative signal has been established.

7. Acknowledgments

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8. Appendix

This chapter contains supplementary material referenced throughout the thesis. Table 8.1 reports the complete list of proteomic features retained after preprocessing for both cohorts, with underlined proteins belonging to the shared SH40 panel. Table 8.2 reports the complete feature selections produced by FSE-StabPred for the LC cohort under the 4-TW formulation, for the **proteomic** and **all_features** feature sets. Table 8.3 reports the corresponding complete selections under the EML-TW formulation, across all feature sets. Table 8.4 and Table 8.5 report the same information for the ADNI-A+ cohort, under the 4-TW and EML-TW formulations respectively. Table 8.6 reports the complete feature selections for the **proteomic** and **all_features** feature sets on the shared SH40 panel, separately for the LC and ADNI-A+ cohorts. Tables 8.7 and 8.8 report the classification performance of the EML-TW formulation for the LC and ADNI-A+ cohorts respectively. Table 8.9 reports the complete list of proteins reaching statistical significance in the Cox proportional hazards survival analysis on the ADNI-A+ cohort.

Table 8.1: Complete list of proteomic features retained after preprocessing for the LC cohort (209 proteins) and the ADNI-A+ cohort (570 proteins). Underline entries are part of the SH40 shared panel.

LC Cohort	ADNI-A+ Cohort
A0A0A0MS15, A0A0B4J1X8, A0A0C4DH38, O00391, O00533, O14498, O14594, O14773, O15394, O43505, O75326, O94856, O94985, P00450, P00734, P00736, P00738, P00746, P00747, P00748, P00751, P01008, P01009, <u>P01011</u> , P01019, <u>P01023</u> , <u>P01024</u> , <u>P01031</u> , <u>P01033</u> , P01034, P01042, P01599, P01619, P01834, P01857, <u>P01859</u> , P01860, P01861, P01871, P01876, P02452, P02647, <u>P02649</u> , P02671, P02675, P02679, P02745, P02746, P02747, P02748, P02749, P02750, <u>P02751</u> , P02753, P02760, P02763, <u>P02765</u> , P02766, P02768, P02774, P02787, P02790, <u>P04004</u> , P04066, P04075, P04180, <u>P04196</u> , P04216, P04217, P04264, P05060, P05067, <u>P05090</u> , <u>P05154</u> , P05155, <u>P05156</u> , P05452, P05543, P05546, <u>P06312</u> , <u>P06396</u> , P06681, P06727, P06733, P07195, P07225, P07333, P07339, P07357, P07585, P07711, P07998, P08123, P08174, P08185, P08253, P08294, P08571, P08603, P08697, P09871, P09972, P0C0L4, P0C0L5, <u>P10451</u> , <u>P10643</u> , P10645, P10909, <u>P12109</u> , P12111, P12259, P13473, <u>P13591</u> , P13611, P13645, <u>P14618</u> , P16035, P16070, P16870, P17174, <u>P18065</u> , <u>P19021</u> , P19652, P19823, <u>P19827</u> , P20774, P22352, P23142, P23468, P23471, P23515, <u>P24592</u> , <u>P25311</u> , P27169, <u>P29622</u> , P33908, P35527, P35858, <u>P36222</u> , P36955, P39060, P40925, P41222, P42785, P43121, P43251, P43652, P49641, P51693, P51884, P54289, P54764, P55058, P61916, P78324, P78509, P98160, Q02246, Q06828, Q08380, Q12805, Q12860, Q13228, Q13332, Q13449, Q13740, Q13822, Q14118, Q14515, Q14624, Q14982, Q15113, Q15582, Q15818, <u>Q16270</u> , <u>Q16610</u> , <u>Q562R1</u> , <u>Q6EMK4</u> , <u>Q6MZW2</u> , <u>Q6UX73</u> , <u>Q6UXD5</u> , <u>Q7Z3B1</u> , <u>Q7Z7M0</u> , <u>Q86UX2</u> , <u>Q8N126</u> , <u>Q8NfZ8</u> , <u>Q8TEU8</u> , <u>Q8WZA1</u> , <u>Q92520</u> , <u>Q92823</u> , <u>Q92859</u> , <u>Q92876</u> , <u>Q969P0</u> , <u>Q96GW7</u> , <u>Q96KN2</u> , <u>Q96PD5</u> , <u>Q99435</u> , <u>Q99574</u> , <u>Q9NQT7</u> , <u>Q9NRN5</u> , <u>Q9NT99</u> , <u>Q9P121</u> , <u>Q9P252</u> , <u>Q9UBP4</u> , <u>Q9UBX5</u> , <u>Q9ULB1</u> , <u>Q9Y4C0</u> , <u>Q9Y646</u> , <u>Q9Y6R7</u>	P51814, P41235, Q9Y5C1, Q15327, Q3SXP7, P15509, Q9H6B4, Q9H6X2, Q9HC56, Q7Z3D4, P26572, O60676, Q5VY80, Q9HDC9, Q6UXG2, P43405, P09603, O43921, Q9UKJ1, Q9P109, P0C2L3, <u>Q9P121</u> , O14610, P19256, Q9UL19, P02765, Q8WVF2, P02818, P01876 P01877, <u>P36222</u> , Q8WUY8, Q2TAL6, <u>P12109</u> , Q14494, Q9GZM7, Q9NZR2, Q6GTx8, Q96EP1, P67870, Q14508, Q03426, Q9NRY6, O14543, O14791, P13987, Q13247, P51815, Q9GZQ8, P10242, O43665, Q8N6H7, P57771, Q6S5L8, Q06124, Q9NZC2, Q9Y5G3, Q8IVU1, Q63HQ2, Q15057, P16152, P31946, Q6PHR2, O00482, O14818, Q99962, Q9ULZ2, Q9NZD2, O14737, P36404, Q9HIY0, P20618, Q92833, P40926, Q9Y2T3, Q9BPU6, P25325, Q92870, Q15633, Q09FC8, P41252, Q86VK4, Q92974, Q9NZR1, Q9NYL9, O14683, P39687, P41208, P06703, O43915, Q12841, <u>P10451</u> , <u>Q4004</u> , <u>Q02487</u> , <u>Q13591</u> , <u>Q8N2S1</u> , <u>P01859</u> , <u>P41221</u> , <u>O43761</u> , <u>Q8TEU8</u> , <u>Q9H2X0</u> , <u>Q5BIV9</u> , <u>Q9ULK0</u> , <u>Q8N5B7</u> , <u>O43390</u> , <u>Q96P16</u> , <u>Q5J8M3</u> , <u>Q5VZY2</u> , <u>Q7Z6M1</u> , <u>Q14449</u> , <u>Q9BQ69</u> , <u>Q9NRA1</u> , <u>P00746</u> , <u>Q49AH0</u> , <u>P42701</u> , <u>Q14515</u> , <u>P08833</u> , <u>Q9TBY0</u> , <u>Q16881</u> , <u>Q92888</u> , <u>Q99728</u> , <u>Q96C10</u> , <u>Q8N474</u> , <u>Q9NPH3</u> , <u>O15444</u> , <u>Q13478</u> , <u>P24592</u> , <u>P29622</u> , <u>P01137</u> , <u>P09486</u> , <u>Q9HBX9</u> , <u>Q16778</u> , <u>P62258</u> , <u>Q16667</u> , <u>Q8N4E4</u> , <u>O00213</u> , <u>Q7Z628</u> , <u>P48147</u> , <u>P35243</u> , <u>Q15053</u> , <u>O76064</u> , <u>Q8N554</u> , <u>P02031</u> , <u>Q8TDL5</u> , <u>P15086</u> , <u>O60462</u> , <u>P08637</u> , <u>Q14393</u> , <u>P00338</u> , <u>Q99519</u> , <u>P02760</u> , <u>Q13444</u> , <u>Q86VZ4</u> , <u>O95498</u> , <u>O75882</u> , <u>O00292</u> , <u>P06276</u> , <u>P35542</u> , <u>Q9BQR3</u> , <u>P20062</u> , <u>P02458</u> , <u>Q6UWR7</u> , <u>P14207</u> , <u>Q14982</u> , <u>Q01995</u> , <u>P42957</u> , <u>P09681</u> , <u>P51692</u> , <u>Q7Z4F1</u> , <u>Q7Z4P5</u> , <u>P20337</u> , <u>Q92804</u> , <u>O00584</u> , <u>Q9H5Y7</u> , <u>Q9ULL4</u> , <u>P10092</u> , <u>P52564</u> , <u>Q9BVB7</u> , <u>Q96MU8</u> , <u>Q96GG9</u> , <u>Q13541</u> , <u>Q9UGI8</u> , <u>Q99426</u> , <u>Q92598</u> , <u>O75575</u> , <u>Q9Y492</u> , <u>O95749</u> , <u>O00560</u> , <u>P35558</u> , <u>Q8N6N6</u> , <u>Q12962</u> , <u>Q69YI7</u> , <u>Q9Y5X3</u> , <u>P28070</u> , <u>P24043</u> <u>P07942</u> <u>P11047</u> , <u>P35241</u> , <u>O75832</u> , <u>P50502</u> , <u>Q9H0F7</u> , <u>Q6T311</u> , <u>O14796</u> , <u>P02461</u> , <u>Q9Y653</u> , <u>P0DMV9</u> , <u>P43629</u> , <u>Q96PB8</u> , <u>Q9NQG5</u> , <u>Q15004</u> , <u>O43427</u> , <u>Q13442</u> , <u>O14908</u> , <u>Q9UBV8</u> , <u>P51153</u> , <u>Q9P021</u> , <u>Q6UWS5</u> , <u>Q9Y3A3</u> , <u>O43570</u> , <u>P51809</u> , <u>Q99497</u> , <u>Q96RU2</u> , <u>P24593</u> , <u>P31371</u> , <u>P35247</u> , <u>Q15700</u> , <u>P08476</u> , <u>O76070</u> , <u>Q14314</u> , <u>P01213</u> , <u>P69892</u> , <u>P12829</u> , <u>Q9H910</u> , <u>P13591</u> , <u>Q9BW30</u> , <u>Q63HQ0</u> , <u>P10747</u> , <u>Q9NZN1</u> , <u>Q8NFY4</u> , <u>P0DPA2</u> , <u>Q5H8C1</u> , <u>Q6ZMI3</u> , <u>P13984</u> , <u>O15194</u> , <u>Q96DE0</u> , <u>Q9BLD8</u> , <u>Q5SZJ8</u> , <u>Q7Z5W3</u> , <u>P30047</u> , <u>Q13572</u> , <u>Q63HM9</u> , <u>Q13057</u> , <u>Q86UA6</u> , <u>Q96JB5</u> , <u>Q96S95</u> , <u>Q9GZZ9</u> , <u>Q9BWK5</u> , <u>Q6P5E7</u> , <u>P52888</u> , <u>P47712</u> , <u>O15354</u> , <u>Q9Y4L1</u> , <u>Q96LC9</u> , <u>Q8IXL6</u> , <u>O95968</u> , <u>Q9NSA3</u> , <u>O60888</u> , <u>Q92185</u> , <u>Q8N4A0</u> , <u>O94966</u> , <u>O75191</u> , <u>P10321</u> , <u>P0C0L4</u> <u>P0C0L5</u> , <u>Q16633</u> , <u>P40306</u> , <u>P22003</u> , <u>Q16552</u> <u>Q96PD4</u> , <u>Q9Y4X3</u> , <u>Q12778</u> , <u>Q5PSV4</u> , <u>Q9H8W3</u> , <u>Q8N7N1</u> , <u>Q53H82</u> , <u>O75364</u> , <u>P0C0S8</u> , <u>Q8IYA7</u> , <u>Q92802</u> , <u>P61599</u> , <u>Q8IWS0</u> , <u>Q99697</u> , <u>P60510</u> , <u>Q9UP38</u> , <u>Q9HBM1</u> , <u>Q8IXJ6</u> , <u>P01033</u> , <u>Q96JH7</u> , <u>Q96N06</u> , <u>Q96AX2</u> , <u>P61968</u> , <u>Q9UH92</u> , <u>Q6P1K2</u> , <u>Q9UKL6</u> , <u>Q14565</u> , <u>Q96S82</u> , <u>Q9GZT8</u> , <u>O43903</u> , <u>Q9NV23</u> , <u>Q9H320</u> , <u>Q9P2A4</u> , <u>Q8IYB1</u> , <u>Q49923</u> , <u>Q9BQC3</u> , <u>Q96HC4</u> , <u>P01031</u> , <u>P54821</u> , <u>Q9H7C9</u> , <u>Q96NG3</u> , <u>P11216</u> , <u>Q86W54</u> , <u>Q9Y216</u> , <u>Q9BW62</u> , <u>O95832</u> , <u>Q96HZ4</u> , <u>Q6ICB4</u> , <u>Q5HYN5</u> , <u>O15021</u> , <u>Q9Y5W9</u> , <u>Q8NC06</u> , <u>Q4G0F5</u> , <u>Q10713</u> , <u>Q14764</u> , <u>Q96KC8</u> , <u>Q6P582</u> , <u>Q9H334</u> , <u>Q9BXP5</u> , <u>Q7L513</u> , <u>P55809</u> , <u>O00401</u> , <u>O95260</u> , <u>Q9Y6A9</u> , <u>Q9BTV6</u> , <u>Q5T447</u> , <u>Q9BVT8</u> , <u>Q9BQA9</u> , <u>O15381</u> , <u>Q9UGP4</u> , <u>P49354</u> , <u>Q12857</u> , <u>P05156</u> , <u>Q13905</u> , <u>O15123</u> , <u>P28908</u> , <u>P52803</u> , <u>P21860</u> , <u>P10809</u> , <u>Q16674</u> , <u>Q12904</u> , <u>Q9Y286</u> , <u>Q15465</u> , <u>P02745</u> <u>P02746</u> <u>P02747</u> , <u>P02788</u> , <u>O75173</u> , <u>Q15389</u> , <u>Q92838</u> , <u>Q9ULZ9</u> , <u>P10643</u> , <u>P05019</u> , <u>P22894</u> , <u>P12272</u> , <u>P18075</u> , <u>P14778</u> , <u>Q9Y336</u> , <u>Q969D9</u> , <u>P01215</u> <u>P01225</u> , <u>P21246</u> , <u>P04275</u> , <u>P00742</u> , <u>P49763</u> , <u>Q9BZM6</u> , <u>Q9HAV5</u> , <u>Q9NR28</u> , <u>P22748</u> , <u>Q76M96</u> , <u>P16112</u> , <u>Q9NS85</u> , <u>Q8IWW2</u> , <u>P31994</u> , <u>O75015</u> , <u>P10646</u> , <u>Q9BU40</u> , <u>P20718</u> , <u>P02751</u> , <u>Q96GW7</u> , <u>O75716</u> , <u>P07996</u> , <u>P61769</u> , <u>P01563</u> , <u>P48061</u> , <u>Q96IY4</u> , <u>P35968</u> , <u>Q8N8U9</u> , <u>P11717</u> , <u>Q9BY79</u> , <u>P01023</u> , <u>P00747</u> , <u>P22607</u> , <u>Q9NQU5</u> , <u>P24666</u> , <u>P01011</u> , <u>P61981</u> , <u>Q01638</u> , <u>Q9HAP6</u> , <u>P06744</u> , <u>P02741</u> , <u>Q99988</u> , <u>O43320</u> , <u>Q86UE6</u> , <u>P62979</u> , <u>P01031</u> <u>P13671</u> , <u>P06396</u> , <u>Q70626</u> , <u>Q14624</u> , <u>P51654</u> , <u>Q9UK05</u> , <u>P01189</u> , <u>P01024</u> , <u>P02671</u> <u>P02675</u> <u>P02679</u> , <u>P04196</u> , <u>P30044</u> , <u>P09936</u> , <u>P41217</u> , <u>Q9NS68</u> , <u>P04035</u> , <u>P20393</u> , <u>P00740</u> , <u>P02649</u> , <u>Q92851</u> , <u>Q99829</u> , <u>Q99983</u> , <u>O75509</u> , <u>P54826</u> , <u>P05129</u> , <u>Q96B86</u> , <u>Q14696</u> , <u>Q7Z5A7</u> , <u>P19021</u> , <u>Q9Y2I2</u> , <u>A4D1T9</u> , <u>P81534</u> , <u>Q9NPH6</u> , <u>Q6PB30</u> , <u>P13385</u> , <u>P63104</u> , <u>P19429</u> , <u>P02794</u> <u>P02792</u> , <u>P01375</u> , <u>P17181</u> , <u>P07686</u> , <u>Q9NZK5</u> , <u>P46734</u> , <u>Q9BX59</u> , <u>Q86SI9</u> , <u>P0C0P6</u> , <u>O76076</u> , <u>O75973</u> , <u>P22792</u> , <u>Q9H972</u> , <u>P26022</u> , <u>Q9BXN1</u> , <u>P83859</u> , <u>P08493</u> , <u>O60235</u> , <u>Q5TAT6</u> , <u>P35858</u> , <u>P60022</u> , <u>Q15063</u> , <u>P0CG47</u> , <u>P06307</u> , <u>Q13596</u> , <u>O95295</u> , <u>Q969P0</u> , <u>Q9Y662</u> , <u>O75023</u> , <u>Q7Z3B1</u> , <u>Q8N807</u> , <u>Q96DB5</u> , <u>P0DMV8</u> , <u>Q07065</u> , <u>Q99584</u> , <u>Q8NCHO</u> , <u>P27348</u> , <u>Q96HD1</u> , <u>P16860</u> , <u>Q9UJ37</u> , <u>Q9NYZ4</u> , <u>Q9BXR6</u> , <u>P10606</u> , <u>A2RU67</u> , <u>Q06481</u> , <u>Q9C0C4</u> , <u>P19827</u> , <u>P09105</u> , <u>Q14184</u> , <u>P49961</u> , <u>Q8N128</u> , <u>P55808</u> , <u>Q1L6U9</u> , <u>Q8N5K1</u> , <u>P30405</u> , <u>P43307</u> , <u>P14174</u> , <u>Q14956</u> , <u>Q6P988</u> , <u>Q8NBZ7</u> , <u>P05090</u> , <u>Q9Y663</u> , <u>Q5VY43</u> , <u>O00300</u> , <u>P15586</u> , <u>Q6IA17</u> , <u>Q96SL4</u> , <u>Q96PQ1</u> , <u>P01178</u> , <u>Q86XK7</u> , <u>Q14118</u> , <u>P80370</u> , <u>P24310</u> , <u>Q30KQ5</u> , <u>P41159</u> , <u>Q13253</u> , <u>P18065</u> , <u>O15072</u> , <u>Q7LG56</u> , <u>A6NGZ8</u> , <u>A6NM11</u> , <u>Q92859</u> , <u>Q9NPA1</u> , <u>Q96RL6</u> , <u>Q96DZ1</u> , <u>Q6P179</u> , <u>P39059</u> , <u>Q5VZ66</u> , <u>O60476</u> , <u>Q9BQI7</u> , <u>O00182</u> , <u>P37802</u> , <u>Q9Y2E5</u> , <u>P16442</u> , <u>Q8TAE8</u> , <u>O75144</u> , <u>P25311</u> , <u>Q9HC57</u> , <u>Q8N386</u> , <u>Q8N539</u> , <u>P49913</u> , <u>Q9Y646</u> , <u>P25445</u> , <u>O75071</u> , <u>Q13308</u> , <u>Q9Y5Z6</u> , <u>Q9H8J5</u> , <u>Q13753</u> , <u>Q6UXI7</u> , <u>Q9H8L6</u> , <u>Q96CS3</u> , <u>Q8IWW3</u> , <u>P21673</u> , <u>P06753</u> , <u>Q9GZT9</u> , <u>P25815</u> , <u>Q96EC8</u> , <u>Q8N729</u> , <u>O14896</u>

Table 8.3: Features selected by FSE-StabPred in at least 2 out of 3 outer folds for the LC cohort (EML-TW formulation), listed in decreasing order of mean ensemble ranking score. **Bold** entries were selected in all three outer folds.

Task	RF					SVM				
	clinical	c-CSF	proteomic	all_features	clinical	c-CSF	proteomic	all_features		
Multiclass	MMSE	A β 42, T-TAU	P00738 , P02746, P02452, Q96KN2, P0C0L4, Q12860, Q15582, P06733, Q15582, MMSE, P06733, P55058, P01011, P08571, P55058, P01024, P05060, Q14624	P00738 , P02746, P02452, P0C0L4, Q96KN2, Q15582, MMSE, P06733, P08571, P55058, P01011, Q8WZAI, Q14624, P00748, P09871	MMSE	A β 42, T-TAU	P00738 , P02746, P02452, Q96KN2, P0C0L4, Q12860, Q96KN2, Q15582, P06733, Q15582, P55058, P01011, P01024, P05060, Q14624, P09871	P00738 , P02746, P02452, P0C0L4, Q96KN2, Q15582, P06733, P55058, P01011		
Mid OVA	GENDER, AGE	T-TAU, P-TAU	P02746, P06733	P02746, P08571, Q15582, P06733	GENDER, AGE	T-TAU, P-TAU	P06733, Q02246	P06733		

Table 8.5: Features selected by FSE-StabPred in at least 2 out of 3 outer folds for the ADNI-A + cohort (EML-TW formulation), listed in decreasing order of mean ensemble ranking score. **Bold** entries were selected in all three outer folds.

Task	RF				SVM			
	clinical	c-CSF	proteomic	all_features	clinical	c-CSF	proteomic	all_features
Multiclass	MMSE	P-TAU, T-TAU	Q96MU8, P61981, P09603, P09603, P09603, Q9NZK5, P16442, Q14515, O60462 , Q6UXG2, P10092, O43390, Q92859, Q99497, P63104, P41217, P49961, P06276, P08833 , P06744, Q8N7N1, P10747, P31946, Q9H492, Q96JB5, Q969P0, Q8IXJ6, P54826, Q9UUKL6, Q9NZC2, P39059, A6NGZ8, P20062, P35247, Q9UH92, Q13444, Q9ULZ9, P09486, P02741, Q9NYL9, P13385, Q9ULZ2	MMSE, P-TAU, P09603 , O60462, P10747, P31946, P08833	MMSE	P-TAU	Q96MU8, P61981, P09603, Q9NZK5, P16442, Q14515, O60462 , Q6UXG2, P10092, O43390, Q92859, Q99497, P63104, P41217, P49961, P06276, P08833 , P06744, Q8N7N1, P10747, P31946, Q9H492, Q96JB5, Q969P0, Q8IXJ6, Q96B86, O14737, P54826, P55808, Q9UUKL6, Q9NZC2, P39059, A6NGZ8, P20062, P35247, Q9UH92, Q13444, Q9ULZ9, P24592, P09486, P02741, Q9NYL9, Q9UKJ1 , P43307, P14174, O00292, P13385, Q9ULZ2	MMSE, P-TAU, P09603 , O60462, P10747, P31946, P08833
Mid OVA	MMSE, DER	GEN- T-TAU	P09603, P49961, Q9NZK5, P26572, P08833, Q92859, Q9NZC2 , Q9UUKL6, P06276, Q9H492, P24310, Q9BQJ7, Q96B86, Q8IXJ6, P35542, Q9Y4L1, P54826, Q6UXI7, P14174, P02741, O00292, P15086, P09681	P09603, Q9NZK5, P49961, P26572 , Q92859, P08833, Q9NZC2 , Q9UUKL6, Q9H492, Q9BQJ7, Q96B86, Q9Y2I2, Q9GZQ8, Q9P121, Q9Y4L1, A6NGZ8, P21246, Q8IXJ6, P35542, Q6UXI7, P12829, P54826, O00292, P02741, P14174, Q13444, P15086, P09681, Q9Y662, P20062, P31946, P16442, Q15053, P08476	MMSE, GENDER	T-TAU	P09603, P26572, P08833, Q9NZC2, Q9UUKL6, P35542, Q9Y4L1, P54826, Q6UXI7, O00292	P09603 , P26572, P08833, P12829, O00292

Table 8.6: Features selected by FSE-StabPred in at least 2 out of 3 outer folds for the LC-ADNI-A+ shared panel (SH40), using Random Forest, listed in decreasing order of mean ensemble ranking score. Results are shown separately for the LC and ADNI-A+ datasets restricted to the 40 shared proteins. **Bold** entries were selected in all three outer folds.

Task	LC			ADNI-A+		
	proteomic	all_features	proteomic	all_features	proteomic	all_features
Multiclass	P18065 , P01024, P02649 , P01011, P12109, P24592, P06396, Q8TEU8, P36222, P13591, Q9Y646	P19021 , P18065, MMSE, P01859, AGE, P00747 , P24592, P05090, Q14624, Q9P121, P24592, Q969P0, Q8TEU8, P10451, P04004, P13591, Q9Y646	P19021, P18065, P01859, AGE, P02765, P01859	Q969P0 , Q14515, P12109, Q14118, MMSE, Q92859, Q14515, Q92859, P10451, P13591, P12109, P10451, P13591		
	P01024 , P02649 , P01859, P12109	P19021 , P18065 , P36222, P01859, P36222	Q969P0 , P24592 , Q14118 , P02765 , P01033, P13591, P02751, P00746	P10451, Q14515, P05090, P01023 , P01033, P04004 , P19827, P13591, P10451, P29622, Q9P121, GENDER, AGE	P-TAU , Q14515, Q14515, P13591	P-TAU , Q14515, Q14515, P13591
	P01024 , P02649 , P01859, P12109	P19021 , P18065 , P36222, P01859, P36222	Q969P0 , P24592 , Q14118 , P02765 , P01033, P13591, P02751, P00746	P10451, Q14515, P05090, P01023 , P01033, P04004 , P19827, P13591, P10451, P29622, Q9P121, GENDER, AGE	P-TAU , Q14515, Q14515, P13591	P-TAU , Q14515, Q14515, P13591
	P01024 , P02649 , P01859, P12109	P19021 , P18065 , P36222, P01859, P36222	Q969P0 , P24592 , Q14118 , P02765 , P01033, P13591, P02751, P00746	P10451, Q14515, P05090, P01023 , P01033, P04004 , P19827, P13591, P10451, P29622, Q9P121, GENDER, AGE	P-TAU , Q14515, Q14515, P13591	P-TAU , Q14515, Q14515, P13591
	P01024 , P02649 , P01859, P12109	P19021 , P18065 , P36222, P01859, P36222	Q969P0 , P24592 , Q14118 , P02765 , P01033, P13591, P02751, P00746	P10451, Q14515, P05090, P01023 , P01033, P04004 , P19827, P13591, P10451, P29622, Q9P121, GENDER, AGE	P-TAU , Q14515, Q14515, P13591	P-TAU , Q14515, Q14515, P13591
	P01024 , P02649 , P01859, P12109	P19021 , P18065 , P36222, P01859, P36222	Q969P0 , P24592 , Q14118 , P02765 , P01033, P13591, P02751, P00746	P10451, Q14515, P05090, P01023 , P01033, P04004 , P19827, P13591, P10451, P29622, Q9P121, GENDER, AGE	P-TAU , Q14515, Q14515, P13591	P-TAU , Q14515, Q14515, P13591
	P01024 , P02649 , P01859, P12109	P19021 , P18065 , P36222, P01859, P36222	Q969P0 , P24592 , Q14118 , P02765 , P01033, P13591, P02751, P00746	P10451, Q14515, P05090, P01023 , P01033, P04004 , P19827, P13591, P10451, P29622, Q9P121, GENDER, AGE	P-TAU , Q14515, Q14515, P13591	P-TAU , Q14515, Q14515, P13591
	P01024 , P02649 , P01859, P12109	P19021 , P18065 , P36222, P01859, P36222	Q969P0 , P24592 , Q14118 , P02765 , P01033, P13591, P02751, P00746	P10451, Q14515, P05090, P01023 , P01033, P04004 , P19827, P13591, P10451, P29622, Q9P121, GENDER, AGE	P-TAU , Q14515, Q14515, P13591	P-TAU , Q14515, Q14515, P13591
	P01024 , P02649 , P01859, P12109	P19021 , P18065 , P36222, P01859, P36222	Q969P0 , P24592 , Q14118 , P02765 , P01033, P13591, P02751, P00746	P10451, Q14515, P05090, P01023 , P01033, P04004 , P19827, P13591, P10451, P29622, Q9P121, GENDER, AGE	P-TAU , Q14515, Q14515, P13591	P-TAU , Q14515, Q14515, P13591
	P01024 , P02649 , P01859, P12109	P19021 , P18065 , P36222, P01859, P36222	Q969P0 , P24592 , Q14118 , P02765 , P01033, P13591, P02751, P00746	P10451, Q14515, P05090, P01023 , P01033, P04004 , P19827, P13591, P10451, P29622, Q9P121, GENDER, AGE	P-TAU , Q14515, Q14515, P13591	P-TAU , Q14515, Q14515, P13591
2-year OVA	P01024 , P02649 , P01859, P12109	P19021 , P18065 , P36222, P01859, P36222	Q969P0 , P24592 , Q14118 , P02765 , P01033, P13591, P02751, P00746	P10451, Q14515, P05090, P01023 , P01033, P04004 , P19827, P13591, P10451, P29622, Q9P121, GENDER, AGE	P-TAU , Q14515, Q14515, P13591	P-TAU , Q14515, Q14515, P13591
3-year OVA	Q9P121, Q8TEU8	P12109, Q9P121, Q8TEU8	P12109, P29622	P13591, P10451, P29622	P13591 , P10451 , P29622, Q9P121, GENDER, AGE	P13591 , P10451 , P29622, Q9P121, GENDER, AGE
4-year OVA	P18065 , Q969P0, P06396, P00747, P02765	P18065 , Q969P0, P00747, P02765	Q969P0, P06396, P04004, P02765, P00747	Q92859 , P25311, P02765, P01859, P01011, Q9P121	P12109 , P12109 , Q92859 , Q14515 , GENDER , P01024, P02765, P01011	P12109 , P12109 , Q92859 , Q14515 , GENDER , P01024, P02765, P01011
5+ year OVA	P01011 , P19021, P05090, P01031	Q14624 , P01011 , P02649, P00747, P01031	MMSE, P02649 , P05090 , Q14118 , P00747 , P35858, P01024, AGE, P04004, P36222, Q8TEU8	Q969P0 , P12109, MMSE	MMSE	MMSE

Table 8.7: Classification performance (mean $\pm$ SD across 3 outer folds) for the LC cohort (EML-TW formulation). **Bold** values indicate the best result per task and metric across all models and feature sets. MCC: Matthews Correlation Coefficient; AUC: Area Under the ROC Curve; Sens.: Sensitivity; Spec.: Specificity.

Model	Task	Feature Set	MCC	AUC	Sens.	Spec.
RF	Multiclass	clinical	-0.105 ± 0.120	0.450 ± 0.068	0.272 ± 0.100	0.632 ± 0.039
		c-CSF	-0.084 ± 0.059	0.406 ± 0.062	0.264 ± 0.047	0.643 ± 0.018
		proteomic	0.038 ± 0.110	0.546 ± 0.049	0.339 ± 0.078	0.686 ± 0.037
		all_features	0.186 ± 0.050	0.560 ± 0.028	0.435 ± 0.026	0.731 ± 0.018
	Mid OVA	clinical	-0.243 ± 0.261	0.334 ± 0.135	0.206 ± 0.180	0.556 ± 0.096
		c-CSF	-0.040 ± 0.104	0.488 ± 0.063	0.206 ± 0.180	0.750 ± 0.250
		proteomic	-0.219 ± 0.093	0.430 ± 0.014	0.143 ± 0.143	0.667 ± 0.083
		all_features	-0.170 ± 0.039	0.417 ± 0.012	0.143 ± 0.143	0.722 ± 0.173
SVM	Multiclass	clinical	-0.146 ± 0.168	0.546 ± 0.007	0.226 ± 0.128	0.623 ± 0.048
		c-CSF	-0.096 ± 0.091	0.523 ± 0.095	0.269 ± 0.106	0.637 ± 0.025
		proteomic	0.055 ± 0.096	0.587 ± 0.092	0.326 ± 0.040	0.694 ± 0.034
		all_features	0.076 ± 0.104	0.589 ± 0.088	0.332 ± 0.051	0.704 ± 0.039
	Mid OVA	clinical	-0.048 ± 0.048	0.578 ± 0.071	0.619 ± 0.360	0.333 ± 0.333
		c-CSF	-0.096 ± 0.166	0.604 ± 0.080	0.429 ± 0.515	0.472 ± 0.502
		proteomic	-0.081 ± 0.050	0.482 ± 0.117	0.310 ± 0.180	0.611 ± 0.210
		all_features	-0.071 ± 0.095	0.444 ± 0.113	0.294 ± 0.131	0.639 ± 0.048

Table 8.8: Classification performance (mean $\pm$ SD across 3 outer folds) for the ADNI-A+ cohort (EML formulation). **Bold** values indicate the best result per task and metric across all models and feature sets. MCC: Matthews Correlation Coefficient; AUC: Area Under the ROC Curve; Sens.: Sensitivity; Spec.: Specificity.

Model	Task	Feature Set	MCC	AUC	Sens.	Spec.
RF	Multiclass	clinical	0.106 ± 0.018	0.599 ± 0.002	0.383 ± 0.030	0.710 ± 0.005
		c-CSF	0.129 ± 0.139	0.570 ± 0.050	0.384 ± 0.072	0.708 ± 0.051
		proteomic	0.120 ± 0.157	0.588 ± 0.109	0.354 ± 0.042	0.684 ± 0.021
		all_features	0.134 ± 0.096	0.607 ± 0.046	0.370 ± 0.008	0.697 ± 0.015
	Mid OVA	clinical	-0.046 ± 0.223	0.504 ± 0.141	0.222 ± 0.294	0.731 ± 0.013
		c-CSF	0.011 ± 0.126	0.467 ± 0.054	0.111 ± 0.111	0.906 ± 0.018
		proteomic	-0.031 ± 0.054	0.597 ± 0.162	0.000 ± 0.000	0.981 ± 0.033
		all_features	-0.041 ± 0.070	0.524 ± 0.135	0.000 ± 0.000	0.969 ± 0.054
SVM	Multiclass	clinical	0.175 ± 0.135	0.601 ± 0.014	0.438 ± 0.111	0.738 ± 0.051
		c-CSF	0.179 ± 0.036	0.629 ± 0.038	0.434 ± 0.013	0.725 ± 0.014
		proteomic	0.103 ± 0.127	0.569 ± 0.100	0.408 ± 0.092	0.694 ± 0.043
		all_features	0.141 ± 0.068	0.618 ± 0.035	0.402 ± 0.076	0.723 ± 0.016
	Mid OVA	clinical	-0.028 ± 0.025	0.498 ± 0.115	0.296 ± 0.280	0.664 ± 0.310
		c-CSF	-0.031 ± 0.068	0.554 ± 0.089	0.593 ± 0.170	0.363 ± 0.085
		proteomic	0.027 ± 0.166	0.514 ± 0.128	0.222 ± 0.222	0.825 ± 0.132
		all_features	-0.070 ± 0.035	0.435 ± 0.038	0.185 ± 0.170	0.745 ± 0.162

Table 8.9: Complete list of 51 proteins statistically significant in the Cox proportional hazards survival analysis on the ADNI-A+ cohort ($p < 0.05$), ordered by increasing p -value. $\log(\text{HR})$ values are per one standard deviation change in normalised protein relative abundance. A positive $\log(\text{HR})$ indicates an association with faster conversion (risk); a negative $\log(\text{HR})$ indicates an association with slower conversion (protective).

UniProt	Gene	$\log(\text{HR})$	p -value	Direction
Q14515	SPARCL1	-0.238	0.0017	Protective
P61981	YWHAG	+0.218	0.0019	Risk
Q96MU8	KREMEN1	-0.206	0.0023	Protective
P60510	PPP4C	-0.219	0.0035	Protective
P10092	CALCB	-0.232	0.0036	Protective
O75023	LILRB5	-0.215	0.0037	Protective
O60462	NRP2	-0.207	0.0044	Protective
Q13253	NOG	-0.199	0.0047	Protective
P62258	YWHAE	+0.216	0.0059	Risk
P10747	CD28	-0.195	0.0062	Protective
P02751	FN1	-0.220	0.0066	Protective
P31946	YWHAB	+0.197	0.0078	Risk
P63104	YWHAZ	+0.196	0.0118	Risk
Q96JB5	CDK5RAP3	-0.164	0.0122	Protective
Q9H320	VCX	-0.173	0.0126	Protective
P02751	FN1	-0.208	0.0136	Protective
P06276	BCHE	-0.174	0.0151	Protective
Q6UXG2	ELAPOR1	-0.181	0.0153	Protective
Q92859	NEO1	-0.168	0.0154	Protective
Q8NFY4	SEMA6D	-0.172	0.0171	Protective
P14778	IL1R1	-0.194	0.0176	Protective
O75882	ATRN	-0.158	0.0178	Protective
Q13308	PTK7	-0.166	0.0179	Protective
Q92185	ST8SIA1	-0.175	0.0180	Protective
O14737	PDCD5	+0.169	0.0181	Risk
Q9NYL9	TMOD3	+0.183	0.0187	Risk
Q9Y646	CPQ	-0.179	0.0197	Protective
P41217	CD200	-0.153	0.0209	Protective
P06744	GPI	+0.185	0.0212	Risk
Q12904	AIMP1	-0.157	0.0213	Protective
P06396	GSN	-0.171	0.0236	Protective
P31946	YWHAB	+0.167	0.0241	Risk
Q96PB8	LRRC3B	-0.155	0.0245	Protective
Q8N2S1	LTBP4	-0.176	0.0250	Protective
Q13444	ADAM15	-0.159	0.0252	Protective
Q9NPH3	IL1RAP	-0.164	0.0330	Protective
P18075	BMP7	-0.150	0.0330	Protective
Q8IWV2	CNTN4	-0.151	0.0341	Protective
P39059	COL15A1	-0.156	0.0352	Protective
P41235	HNF4A	-0.164	0.0353	Protective
Q8TEU8	WFIKKN2	-0.155	0.0359	Protective
P00742	F10	-0.162	0.0369	Protective
P07996	THBS1	-0.172	0.0390	Protective
Q9Y2I2	NTNG1	-0.142	0.0403	Protective
Q15063	POSTN	-0.165	0.0403	Protective
P19827	ITIH1	-0.154	0.0437	Protective
Q9NV23	OLAH	-0.145	0.0458	Protective
P35247	SFTPD	+0.144	0.0468	Risk
P00740	F9	-0.153	0.0483	Protective
Q969P0	IGSF8	-0.128	0.0497	Protective

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